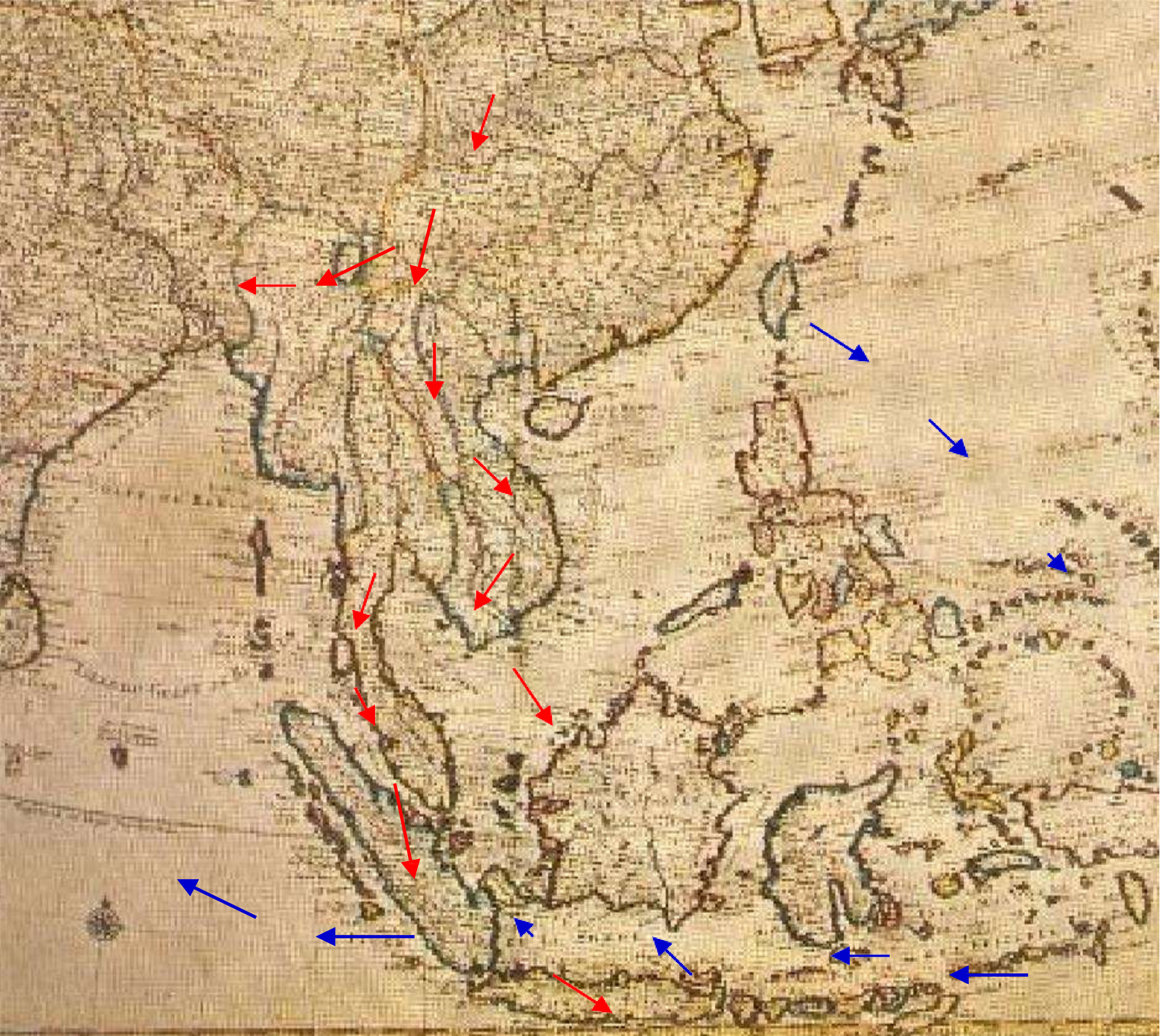
# Malaysian Housing Evolution

## Then and Now


Section 1: Early History of Malay Civilisation
Section 2: Malacca Straits Settlements
Section 3: Kuala Lumpur Origin
Section 4: Malaysia Mass Housing

Section 5: Work experience
Section 6: Research In Japan



a. **First Theory :**
   Related to Asian Continent/ Mainland –Dongsan Civilisation

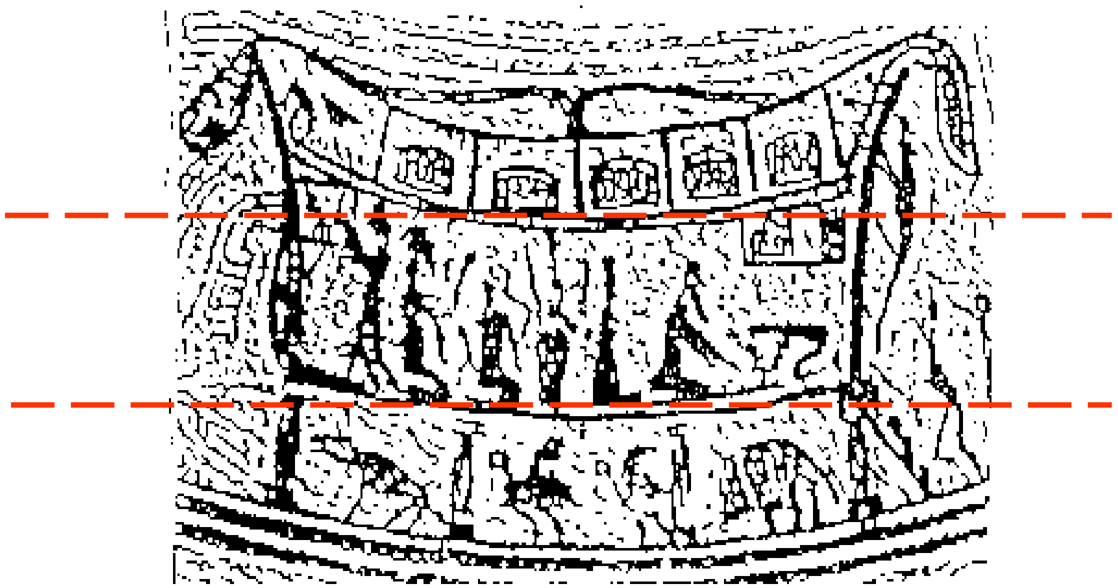
b. **Second Theory:**
   Related to Pacific Ocean- Seafarers Civilization (oceanic).

"The early settlement by the river, at deltas and by the sea front appeared to be an amphibian (water) habitat"

Sumet Jumsai, NAGA (1988)

"The range of architectural styles which have developed in the communities of Southeast Asia reveal some startling similarities, which strongly suggestive of perhaps distant but common origin."

Roxana Waterson, LIVING HOUSE (1990)

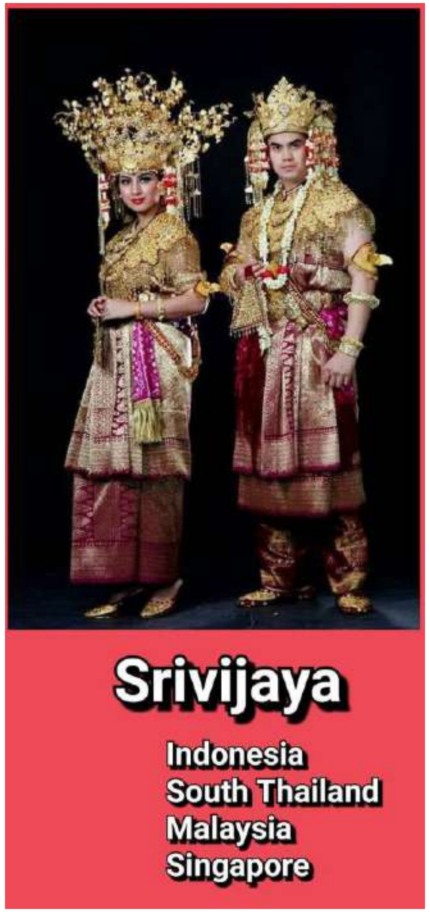


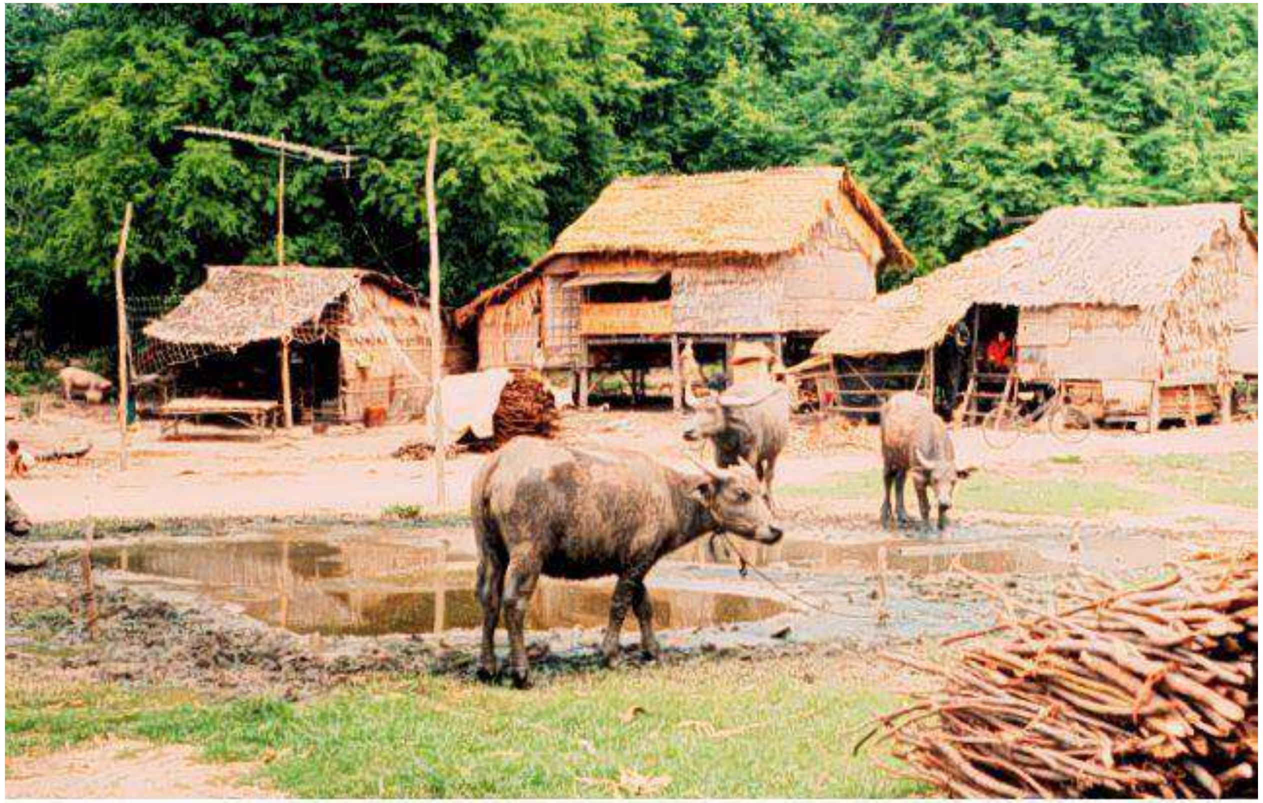
“The Sangeang Drum “ dwelling-occupied by people wearing what appear to be Chinese Han Dynasty costumes. The structure of the house with its under floor level occupied by animal (a pig, two chicken and a dog) the main level by humans and partitioned attic containing what appear shows three-tiered division of inhabited space which is so typical of dwellings in many present-day Indonesia Society”

Roxana Waterson, LIVING HOUSE (1990)


There is no definite evidence which dates the first Indian voyages across the Bay of Bengal but conservative estimates place the earliest arrivals on Malay shores at least 2,000 years ago. The discovery of jetty remains, iron smelting sites, and a clay brick monument dating back to 110 CE in the Bujang Valley, shows that a maritime trading route with South Indian Tamil kingdoms was already established since the second century.[Devan,2010]

Srivijaya Empire: 7th-13th Cent
Majapahit Empire : 14th Cent
Malacca Empire: 15th Cent
Colonalisation: 16th Cent

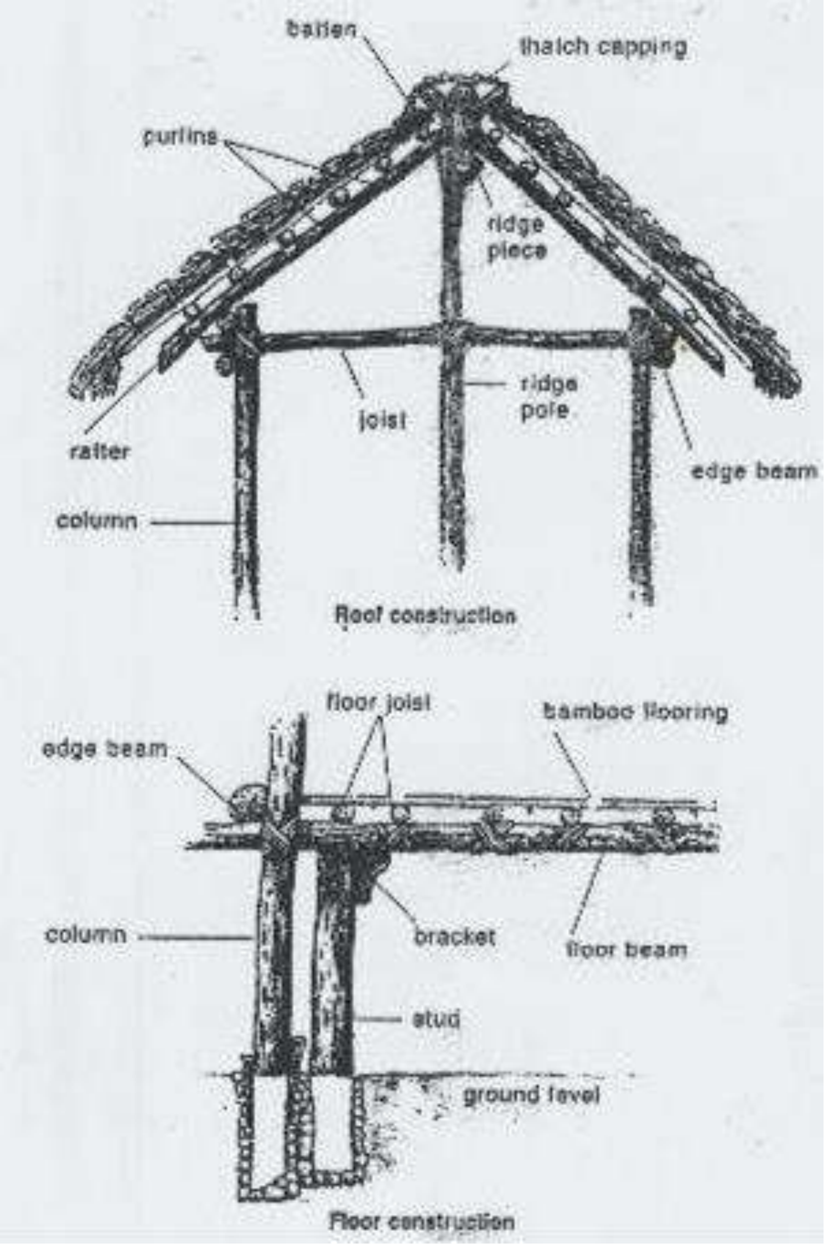



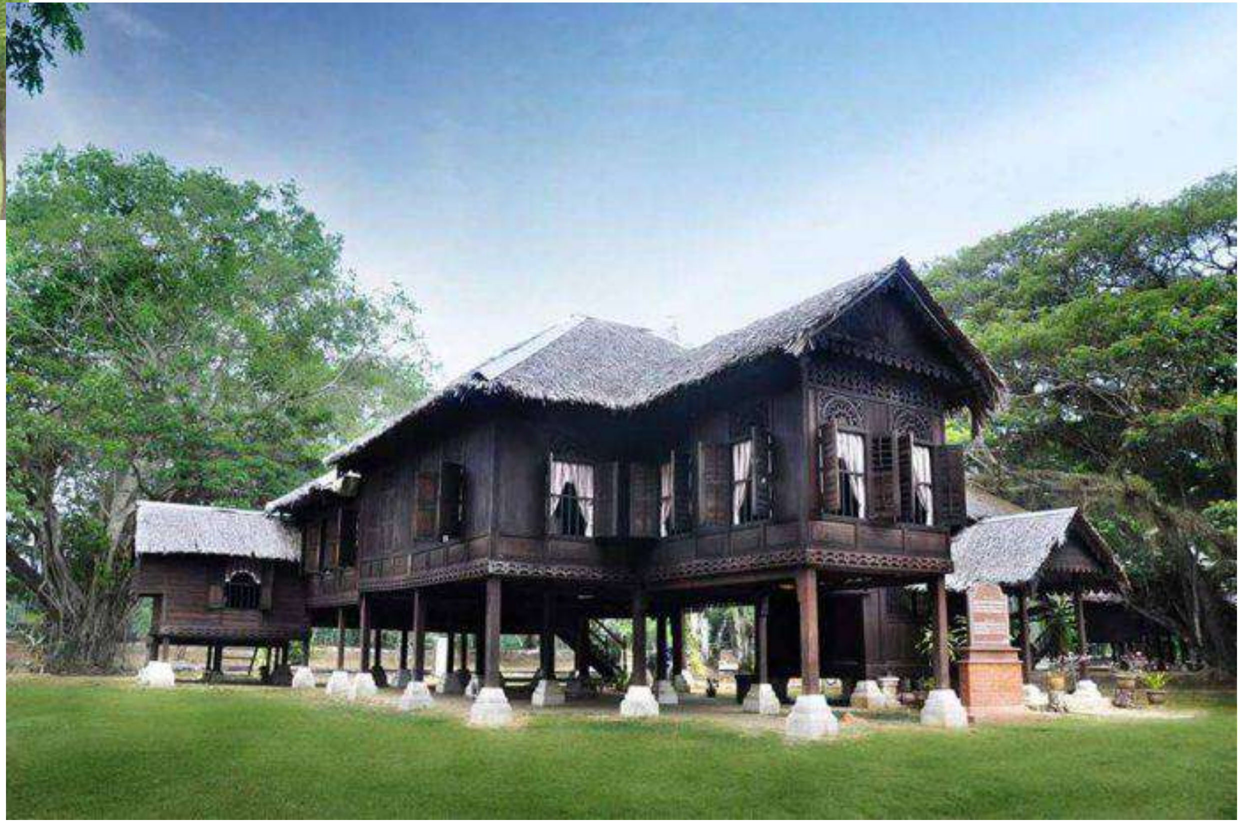


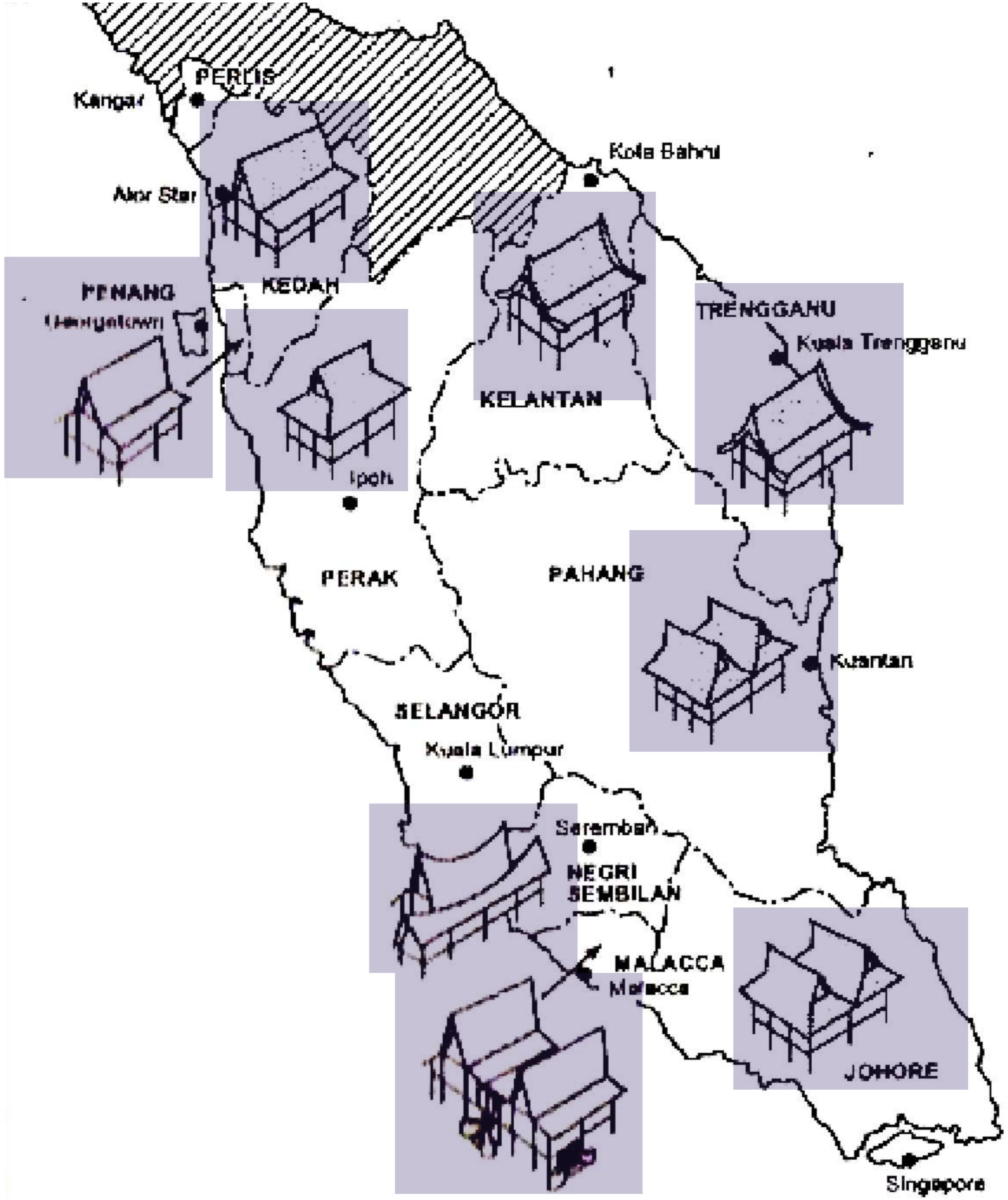
Housing near Phnom Da

Raised on stilts :

to be safe from animals and flood, space under the house is used as a storage area, for rearing animals or as a working area.






PERLIS
Kangar
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Alor Star
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Georgetown
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Ipoh
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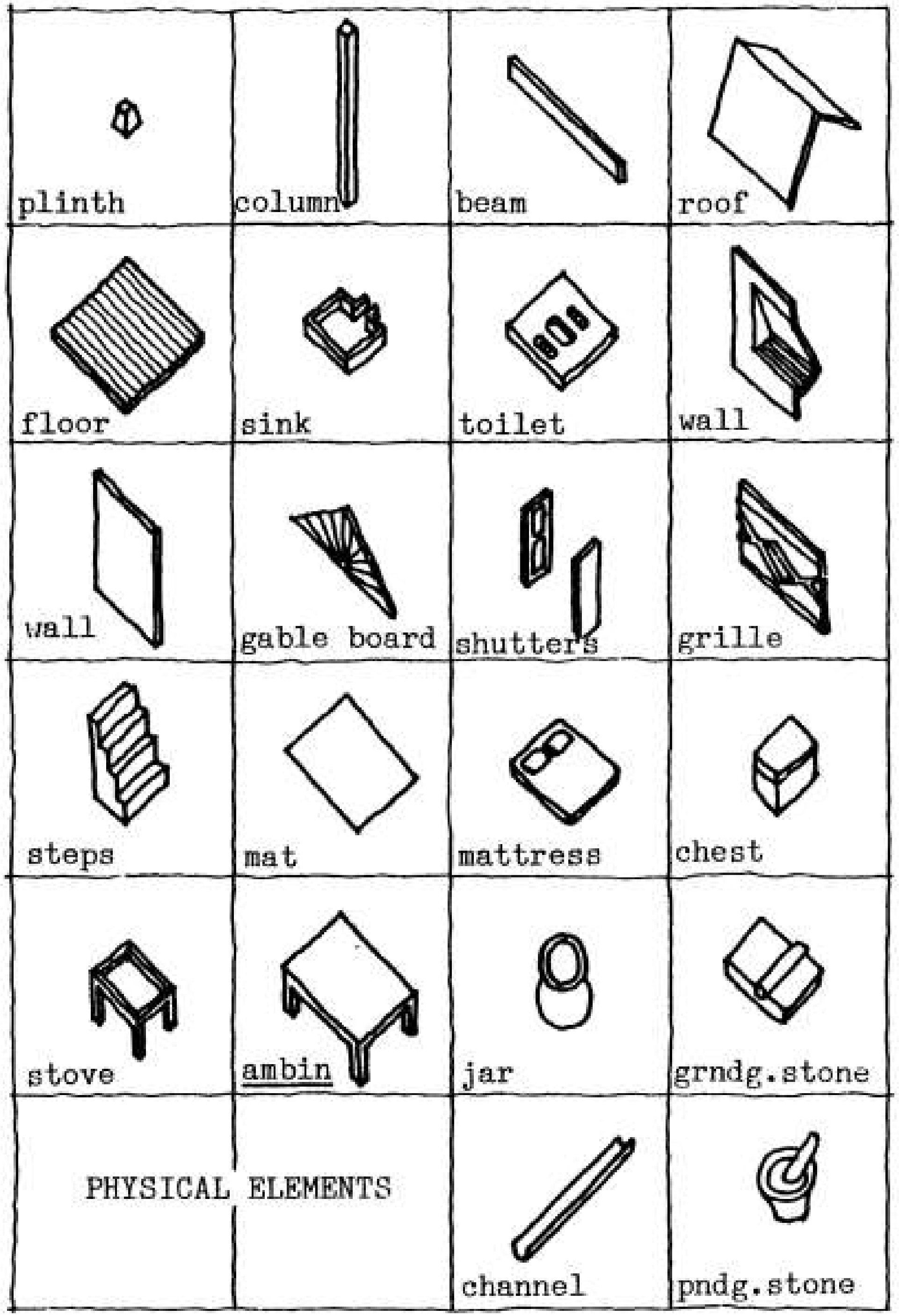
Kuantan
SELANGOR
Kuala Lumpur
Seremban
NEGRI SEMBILAN
MALACCA
Malacca
JOHORE
Singapore


plinth
column
beam
roof
floor
sink
toilet
wall
wall
gable board
shutters
grille
steps
mat
mattress
chest
stove
ambin
jar
grndg.stone
PHYSICAL ELEMENTS
channel
pndg.stone

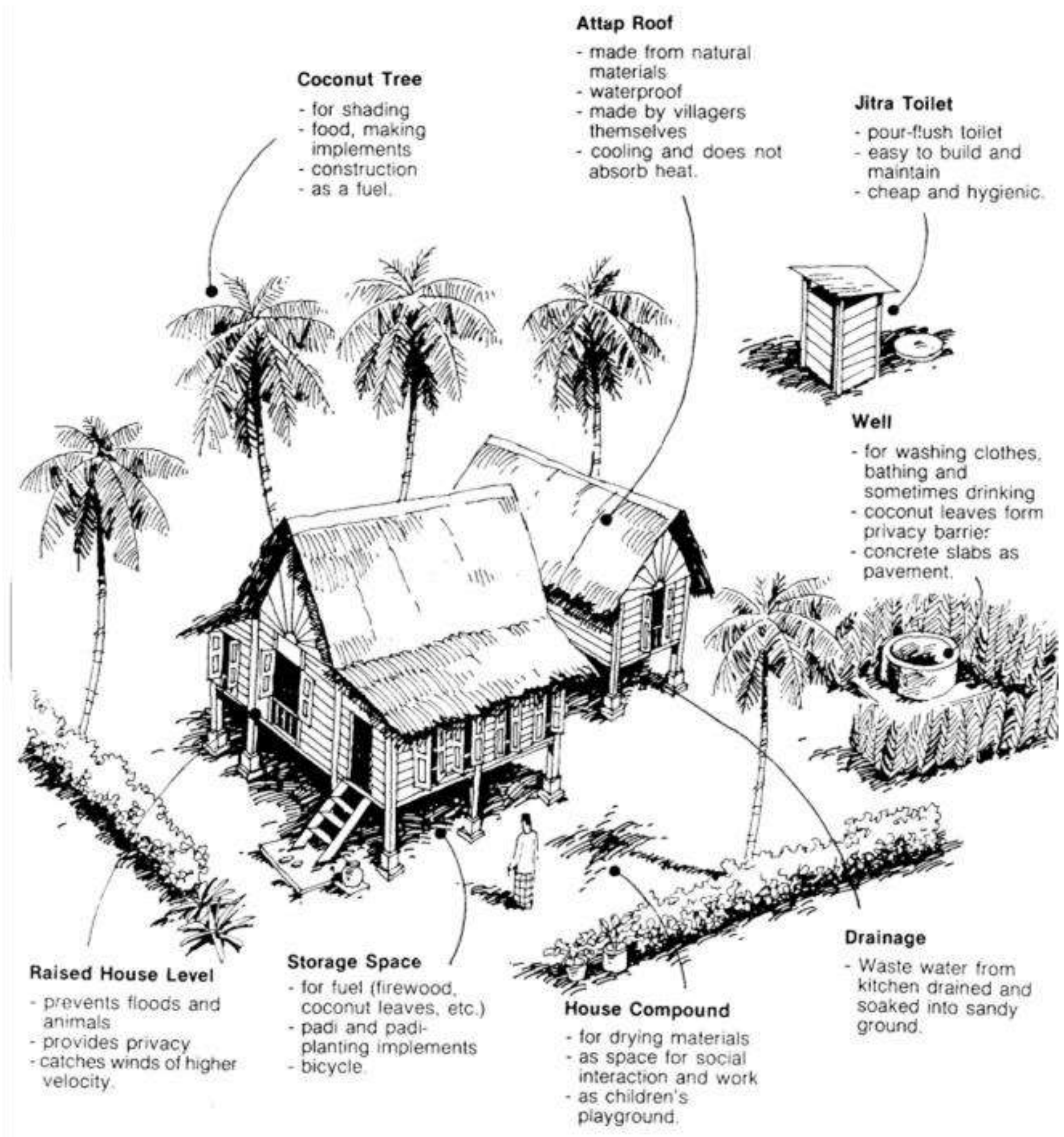



**Figure 2**: The external environment of the traditional Malay house

Setting and orientation :

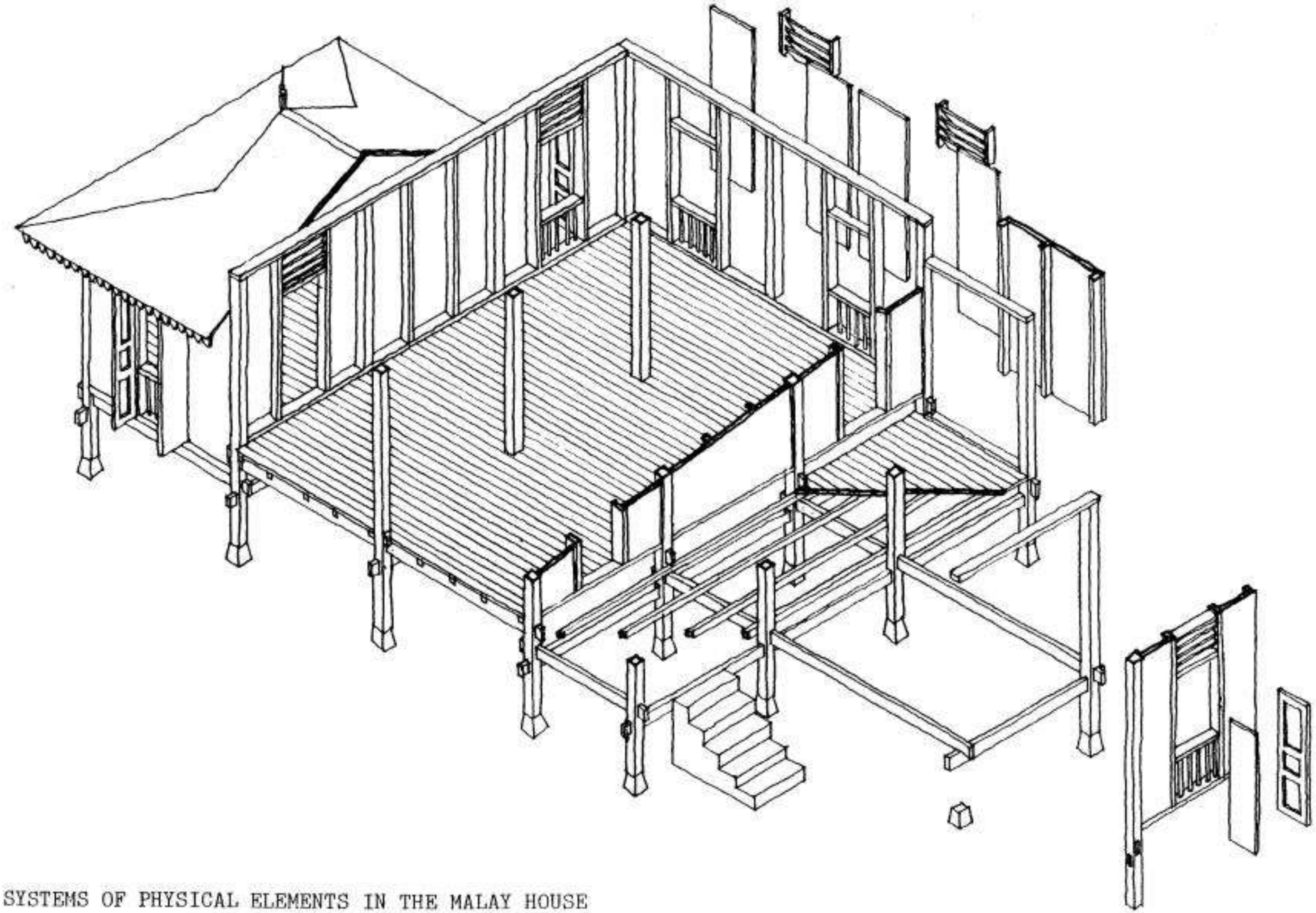
built on a big plot of land, freestanding, care was taken in not clearing the whole site Once the house is built, a few fruit and herb trees are planted around it.

Raised on stilts :

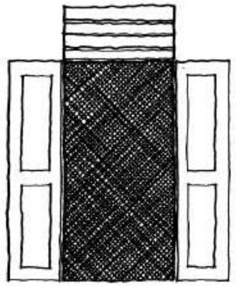
to be safe from animals and flood, space under the house is used as a storage area, for rearing animals or as a working area.

Source: Nasir 1985

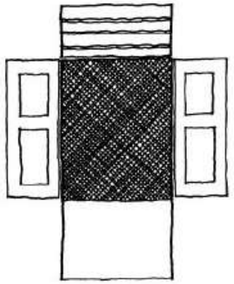
**Figure 1**: The basic design of the traditional Malay house



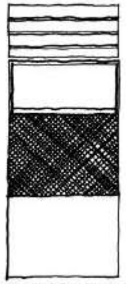
SYSTEMS OF PHYSICAL ELEMENTS IN THE MALAY HOUSE



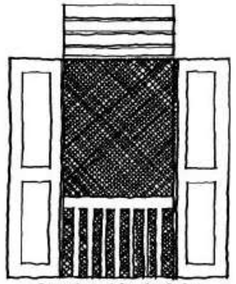
Unblocked Opening (door)
Long side-hung shutters



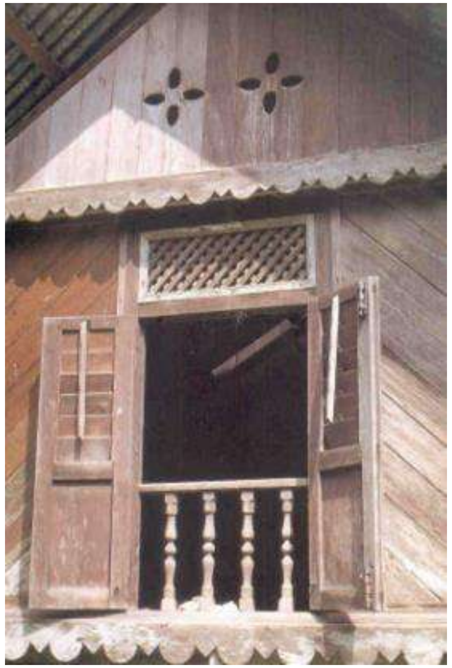
Opening blocked by
a solid wall (window)
Short side-hung shutters

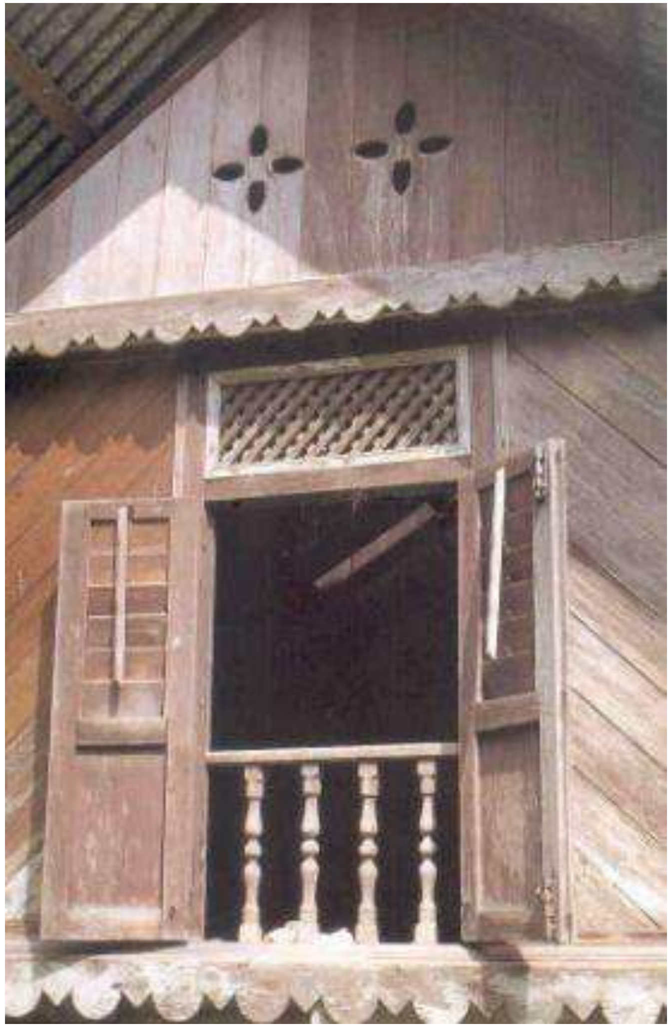


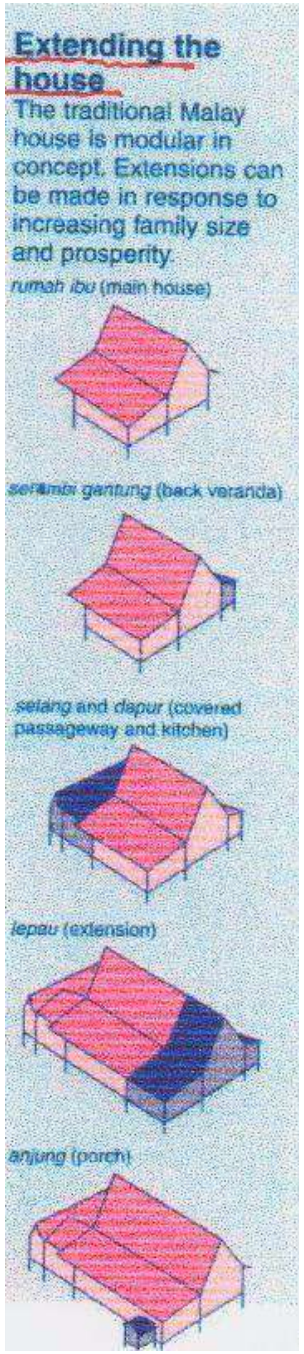
Opening blocked by
a solid wall (window)
Top hung shutters



Opening blocked by
an open wall (window)
Long side-hung shutters





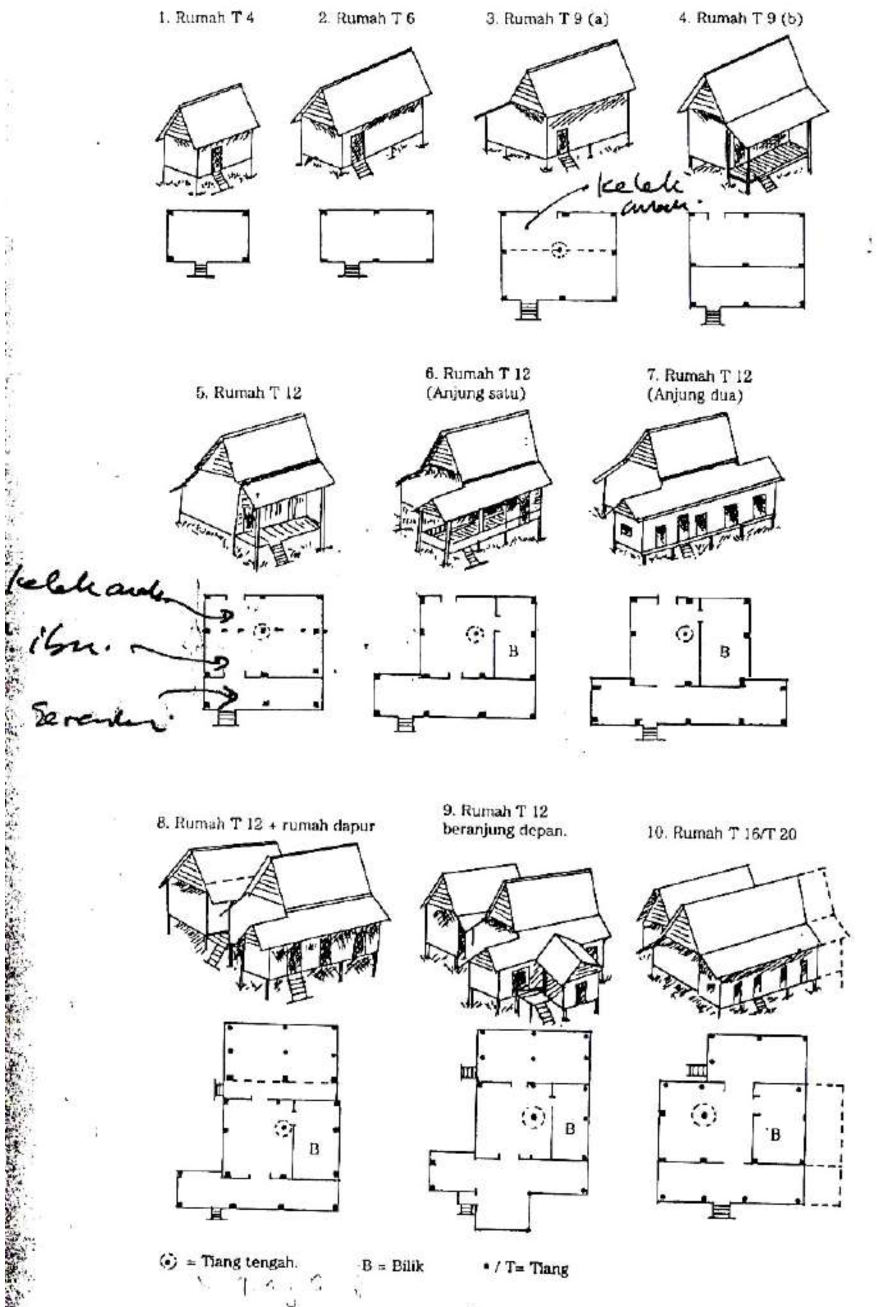




**Four-pillared**
structure denotes basic shelter

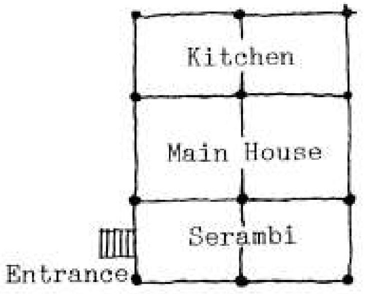
**Six -pillared house**
= rumah bujang = ibu rumah
(A living unit with private and public spaces)

**Nine-pillared house**
centre pillar = tiang seri
additional space at the back of the house ‘kelek anak’
(originally for cooking place, but later on as female gathering area)

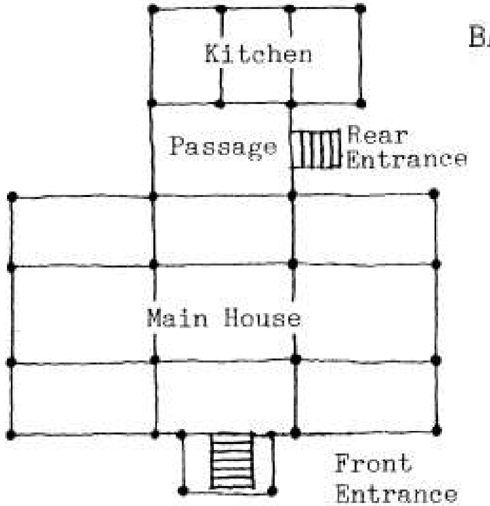
**Twelve-pillared house**
.additional space in front of the house ‘serambi’ = ‘selasar’
(for male guests)
.common housing type



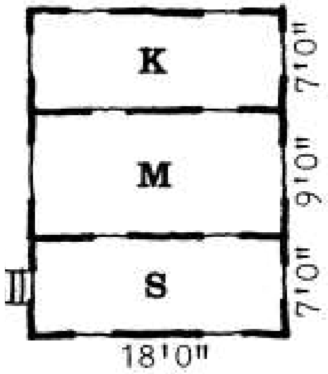

**Rajah 3.2** Konsep ruang dan perkembangannya



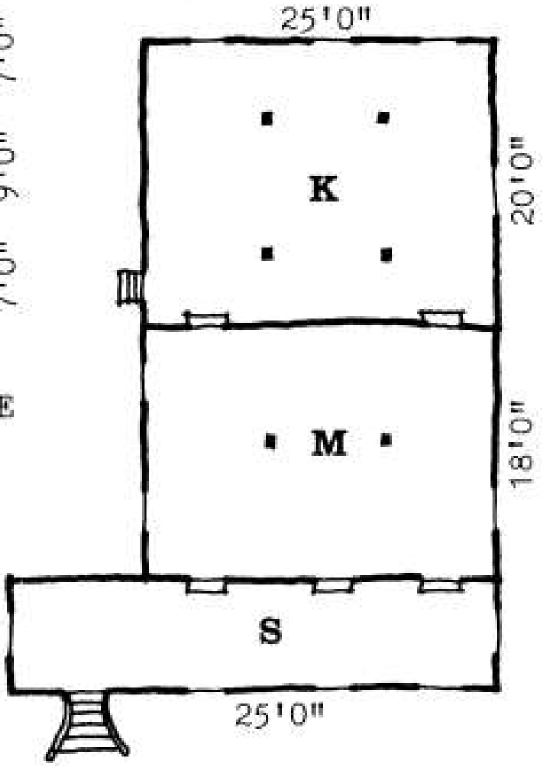

Basic Malay House



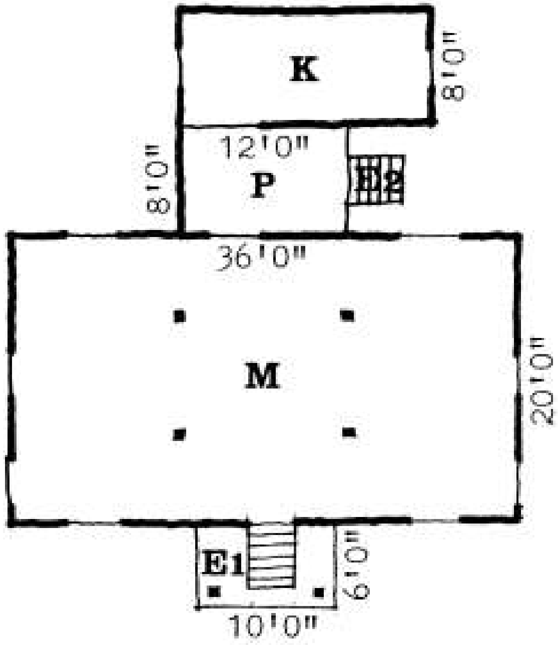

House at Bota Kiri



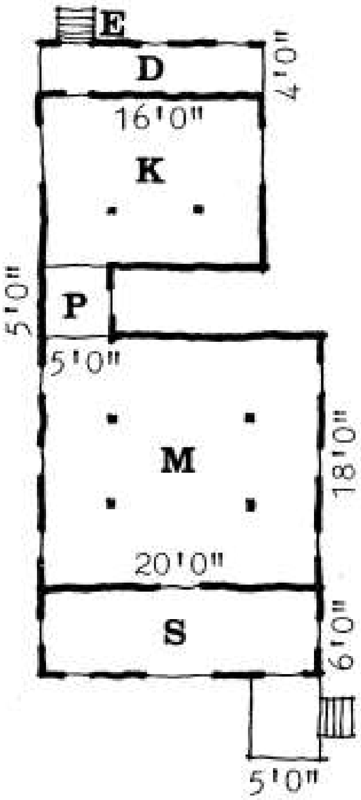

BASIC MALAY HOUSE
(Hilton)



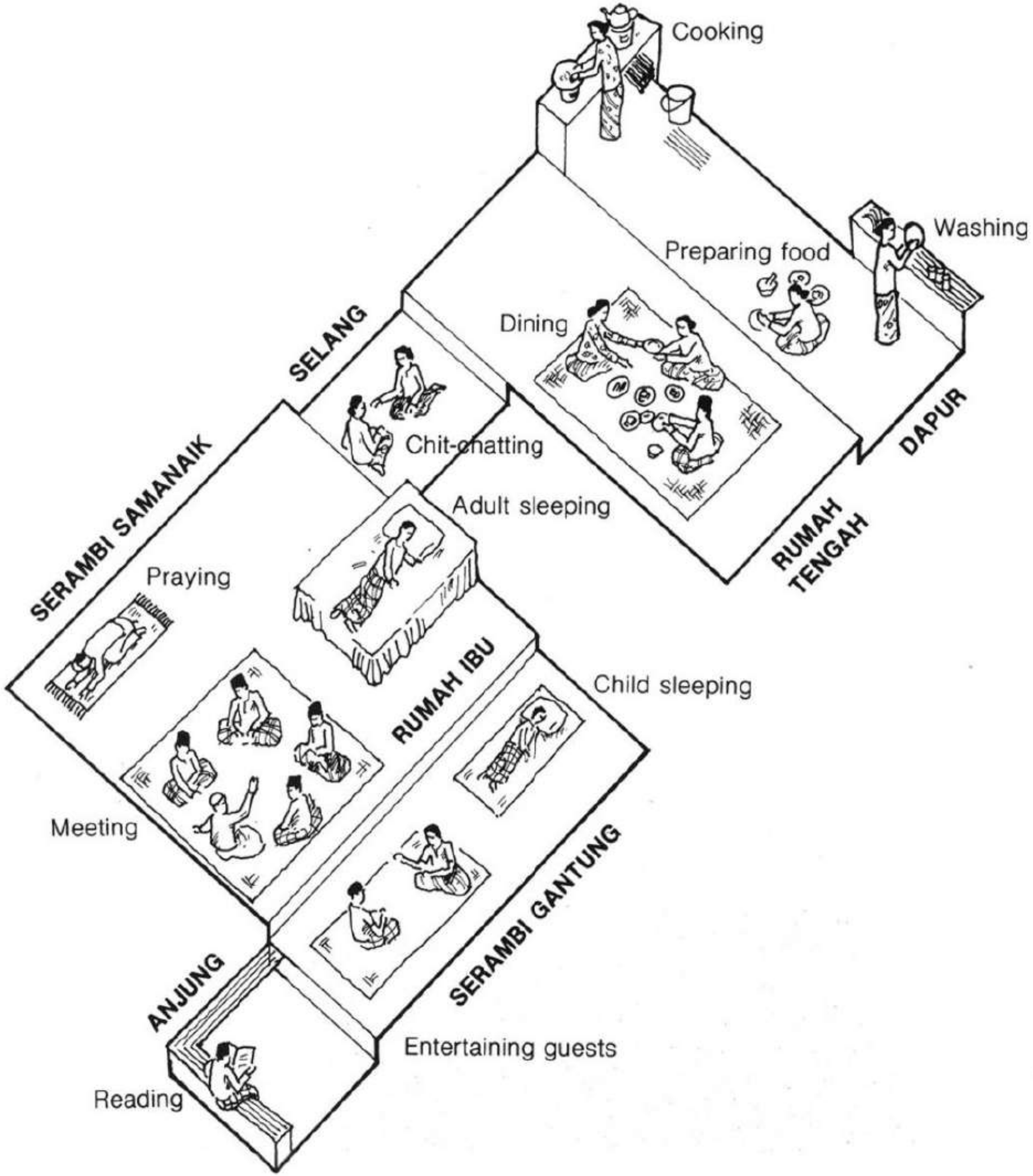

MALACCA HOUSE
(Morley)

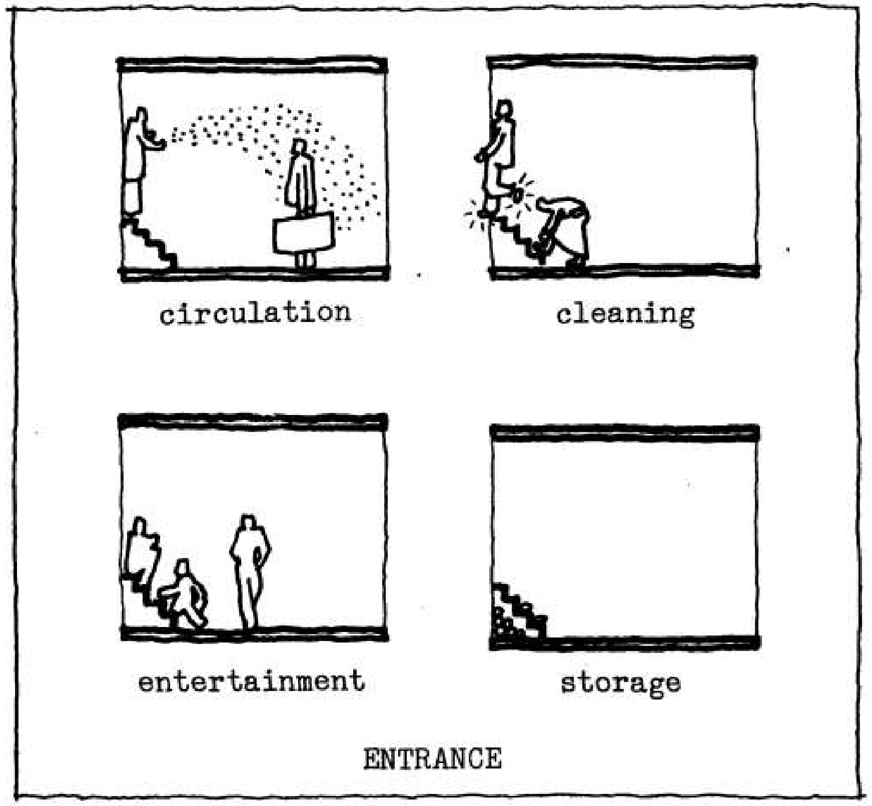


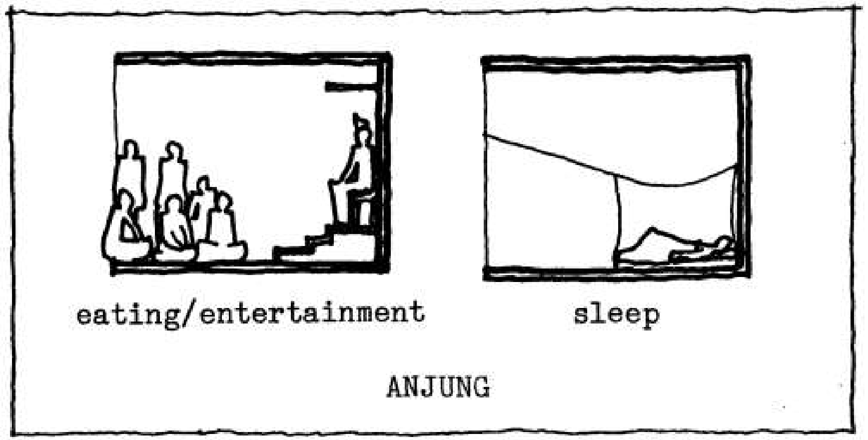

HOUSE AT BOTA KIRI
(Morley)

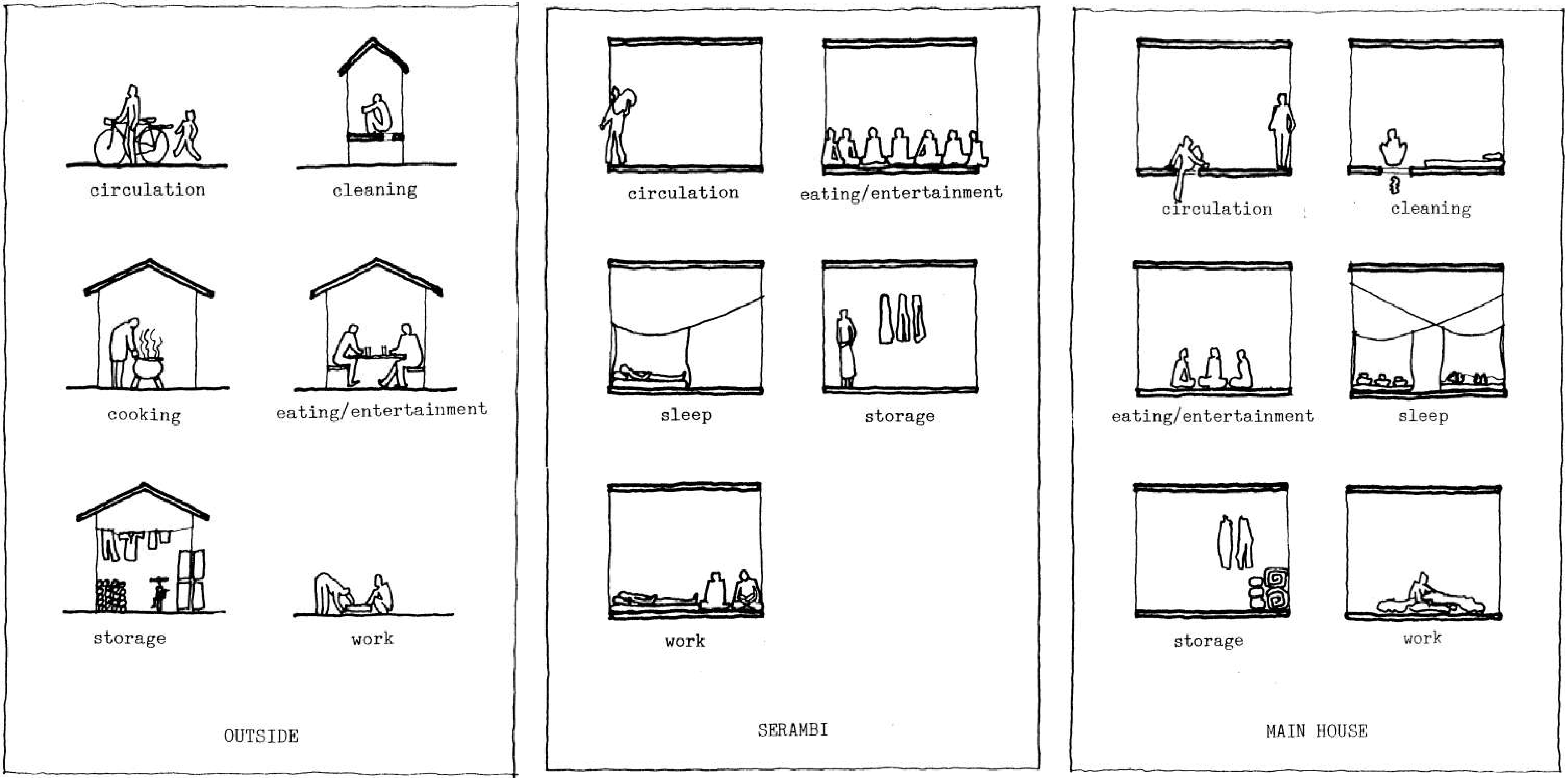


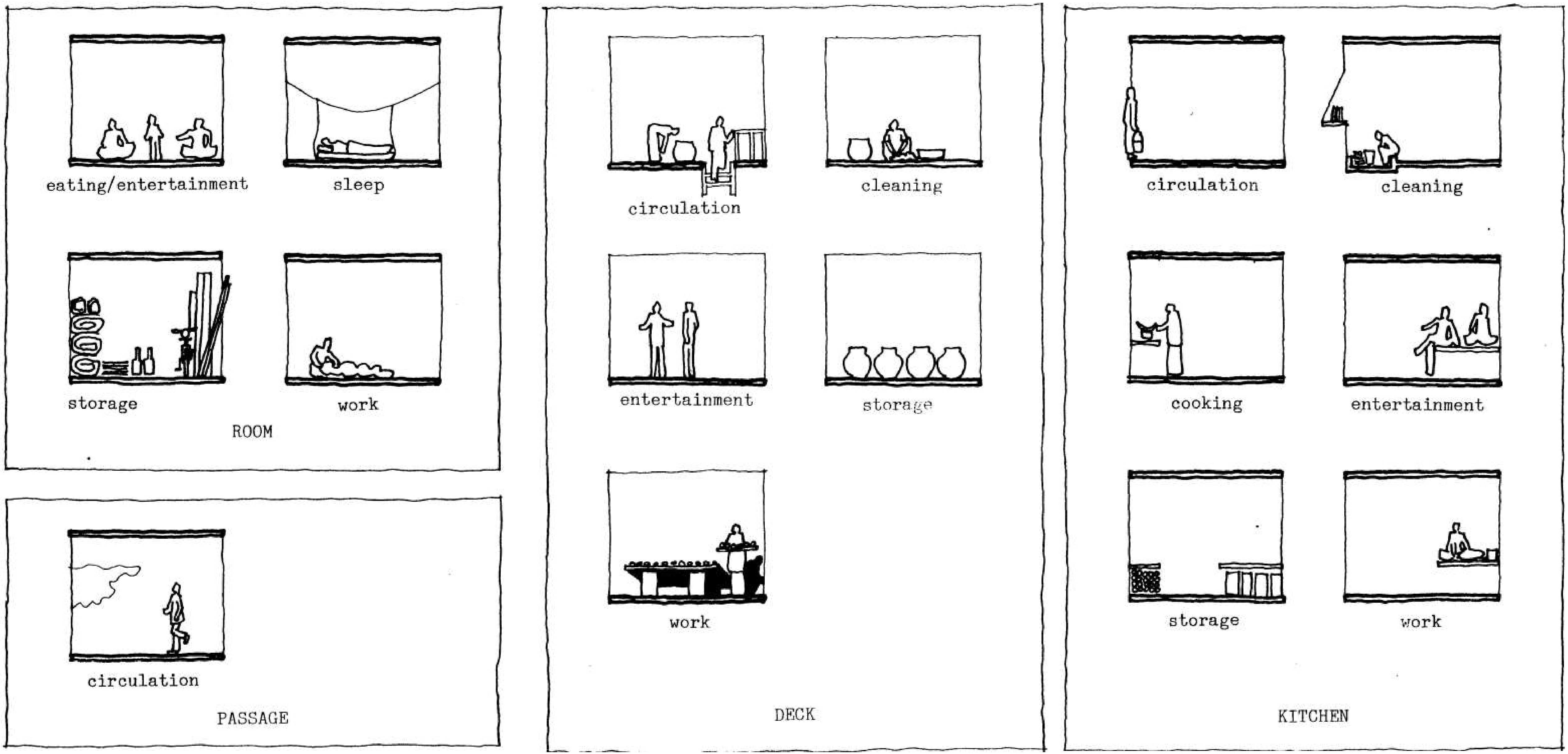

MALAY HOUSE
(C.W.Lim)


Cooking
Washing
Preparing food
Dining
SELANG
DAPUR
Chit-chatting
Adult sleeping
RUMAH TENGAH
SERAMBI SAMANAIK
Praying
RUMAH IBU
Child sleeping
Meeting
SERAMBI GANTUNG
ANJUNG
Entertaining guests
Reading

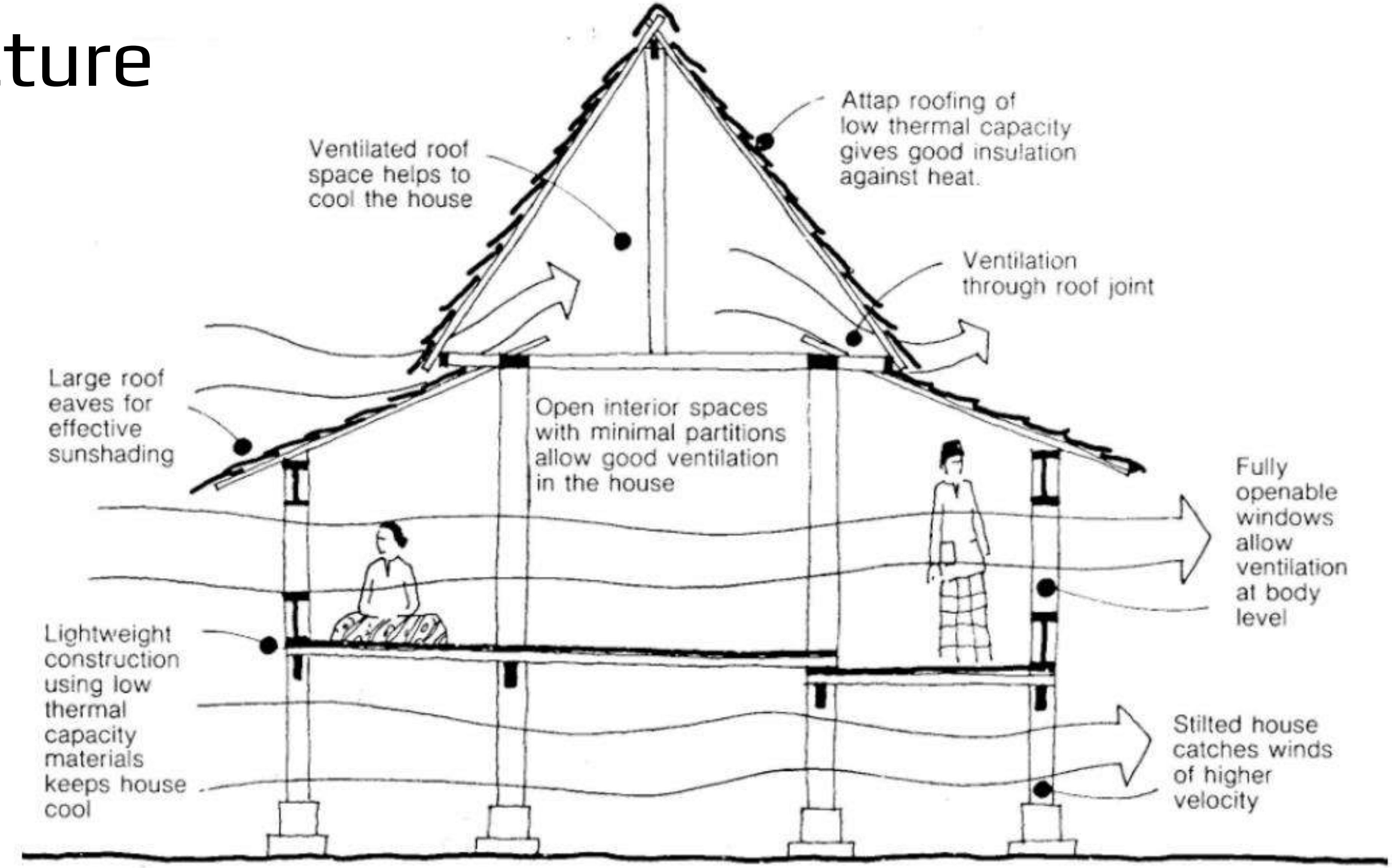

circulation
cleaning
entertainment
storage
ENTRANCE


eating/entertainment
sleep
ANJUNG


circulation
cleaning
cooking
eating/entertainment
storage
work
OUTSIDE
circulation
eating/entertainment
sleep
storage
work
SERAMBI
circulation
cleaning
eating/entertainment
sleep
storage
work
MAIN HOUSE

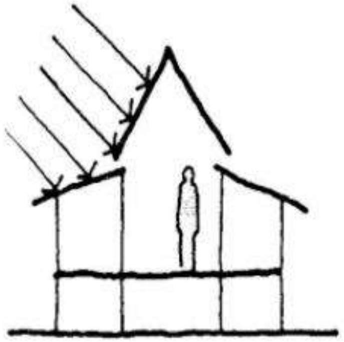

eating/entertainment
sleep
storage
work
ROOM
circulation
PASSAGE
circulation
cleaning
entertainment
storage
work
DECK
circulation
cleaning
cooking
entertainment
storage
work
KITCHEN

# Traditional Architecture



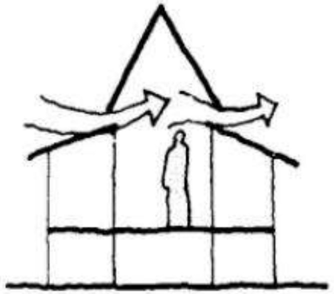

Source: Yuan 1987

**Figure 3**: Climatic design of the traditional Malay house



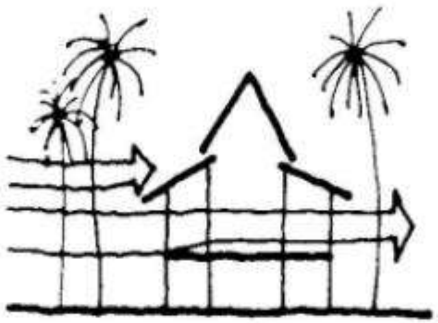
**Building Materials**

◄ Traditional Malay houses use lightweight construction of wood and other natural materials. The lightweight construction of low thermal capacity holds little heat and cools adequately at night. The attap roof is an excellent thermal insulator. Glazed areas are seldom found in the traditional Malay house.



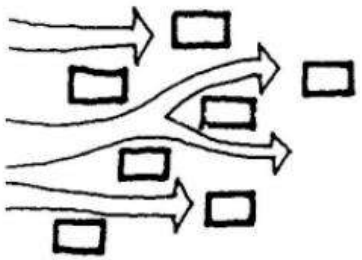
**Ventilation of Roof Spaces**

◄ Roof spaces in the traditional Malay house are properly ventilated by the provision of ventilation joints and panels in the roof construction.



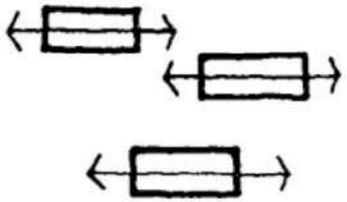
**Vegetation**

◄ The use of coconut trees and other tall trees in the kampong not only provides good shade but also does not block the passage of winds at the house level.

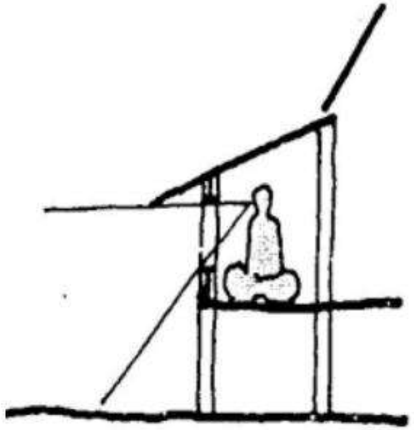


**Layout**

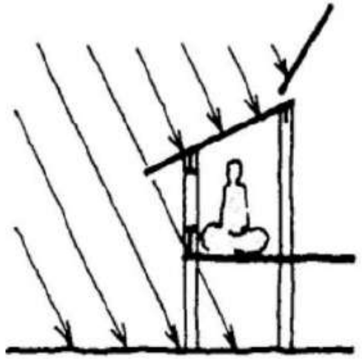
◄ Traditional Malay houses are randomly arranged. This ensures that wind velocity in the houses in the latter path of the wind will not be substantially reduced.



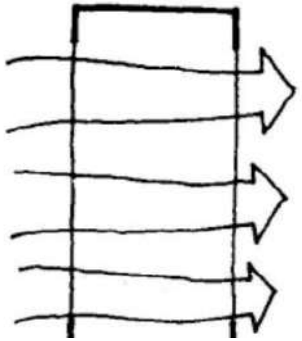
◄ Traditional Malay houses are often oriented to face Mecca (i.e., in an east-west direction) for religious reasons. The east-west orientation minimizes areas exposed to solar radiation.



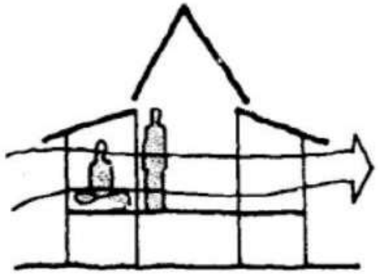
◄ Glare in the traditional Malay house is controlled by large roof overhangs and low windows which exclude the open skies from the visual field. Glare is also lessened by the less reflective natural ground covers and wooden walls of neighbouring houses.



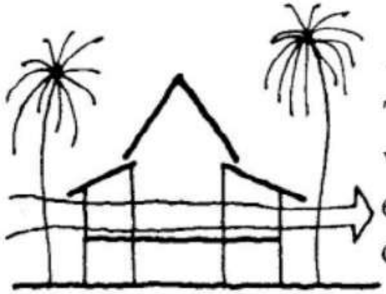
◄ Large overhangs and the low exposed vertical areas (windows and walls) in the traditional Malay house provide good protection against driving rain, provide good shading, and allow the windows to be left open most of the time for ventilation.



◄ The elongated open plans of the traditional Malay house allow easy passage of air and good cross ventilation. There are minimal interior partitions in the Malay house which restrict air movement in the house.



◄ The body level is the most vital area for ventilation for comfort. The traditional Malay house allows ventilation at the body level by having many full-length fully openable windows and doors at body level.



◄ The velocity of wind increases with altitude. The traditional Malay house on stilts capture winds of higher velocity at a higher level. This is especially vital in areas where there are plant cover on the ground which restricts air movement.

The area beneath the house is comfortable for children to play and various daily activities.

The raised floor presented solution to social responsibilities such as bathing the physically challenged elderly and also bathing the deceased.

The raised floor of the traditional Malay house is an effective counter measures from flood, wild animals and etc. It allows easy passage of air into and through the house.

Gaps between the floorboards solved problems of house cleaning, ventilation and also lighting.

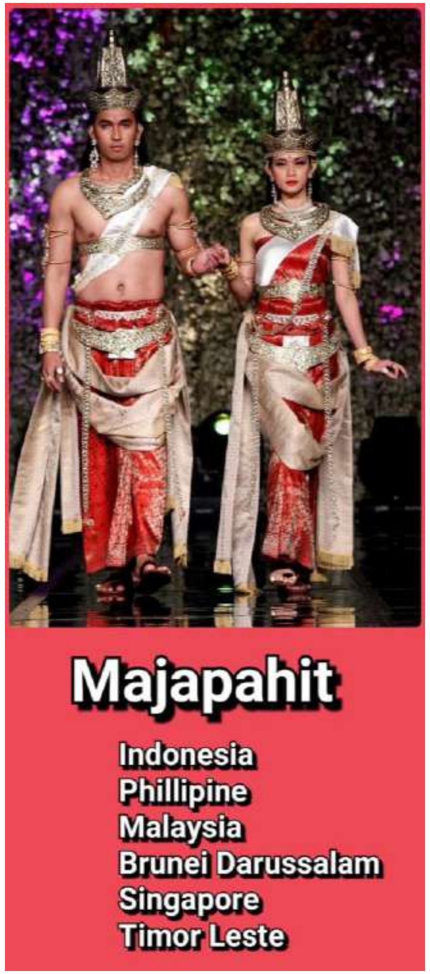


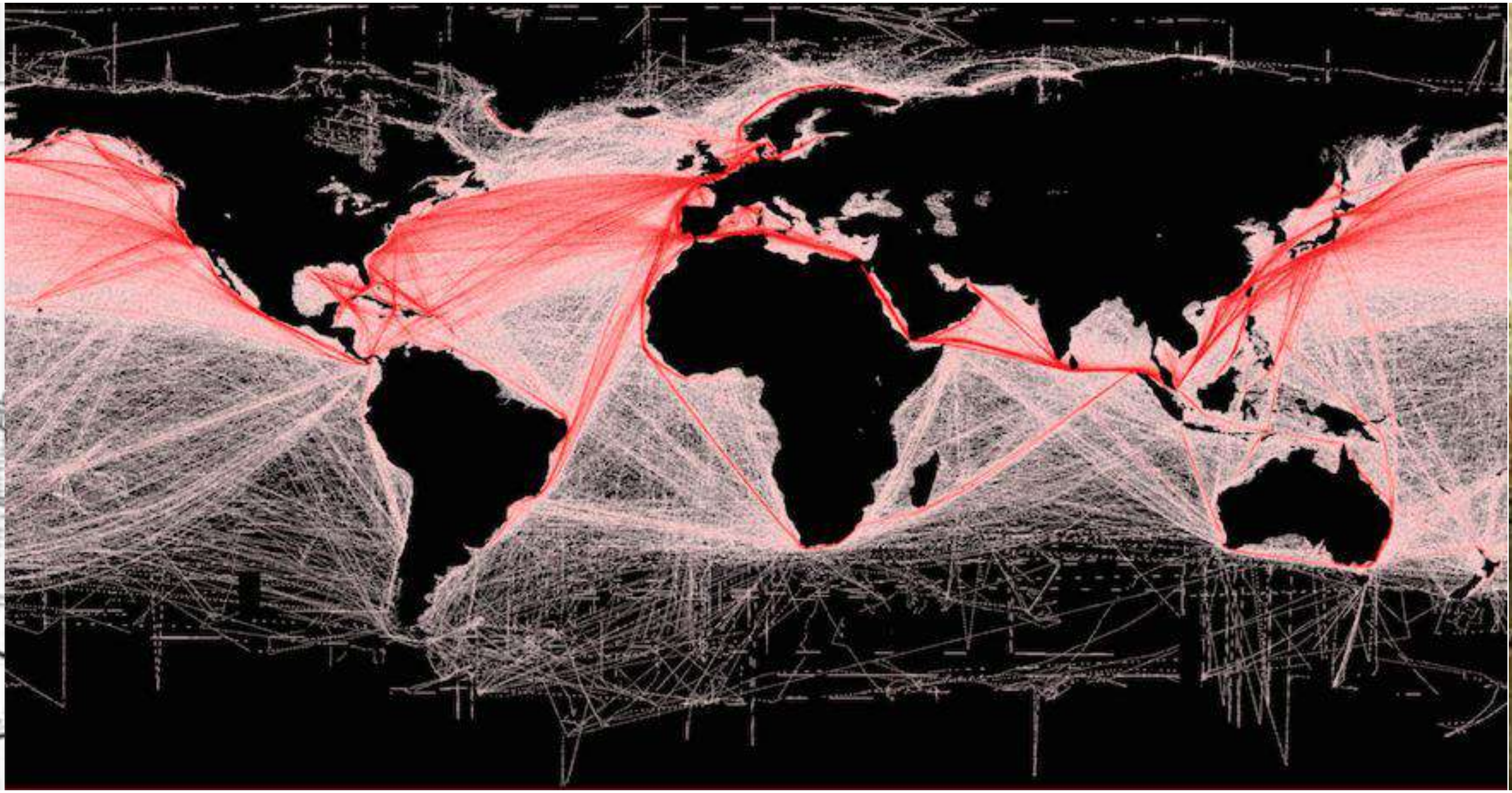


atap
atap.co

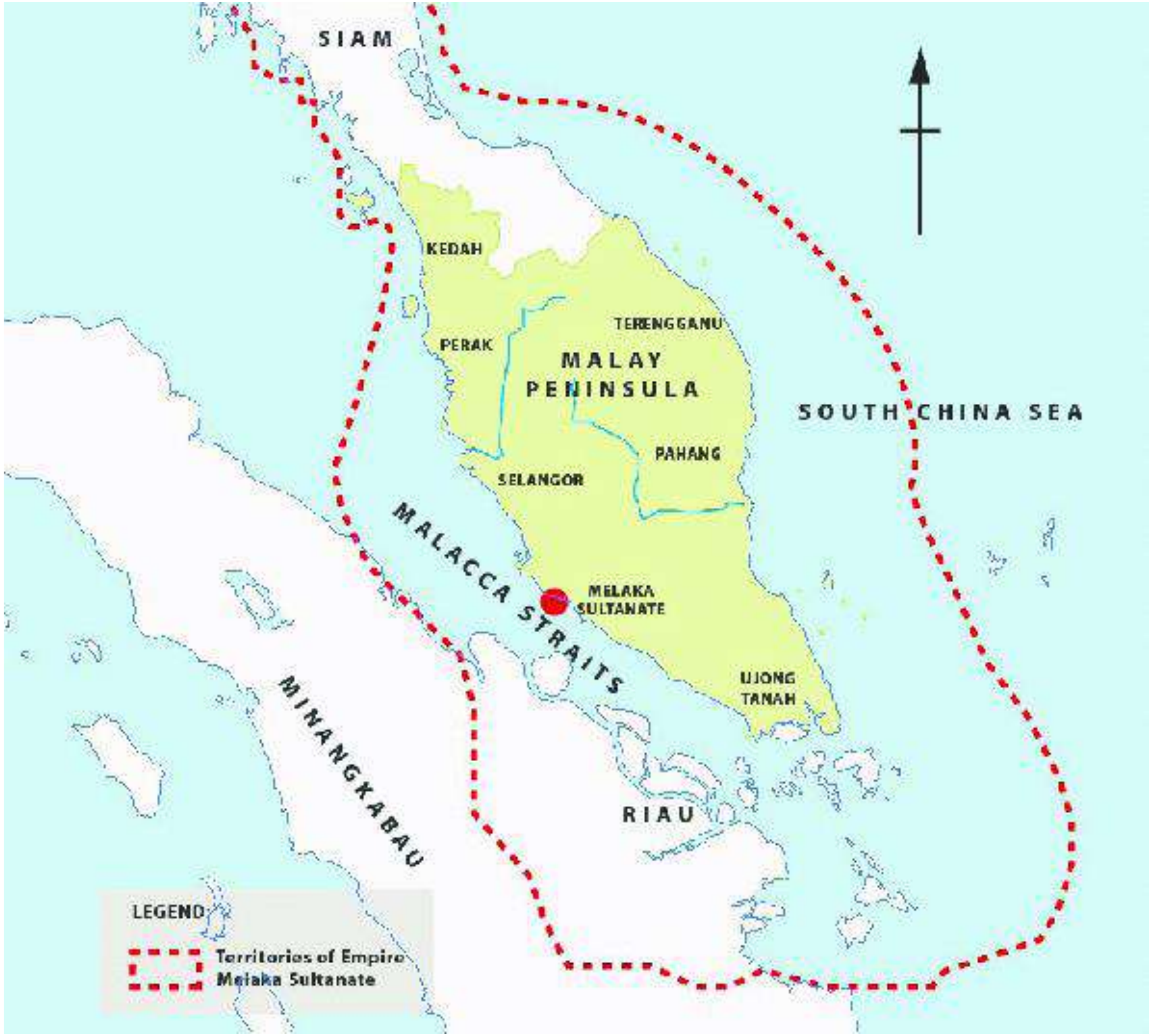


Sang (Nila) Utama is a reference to a 13th-century Palembang prince from the Srivijaya ruling house. His link to the pre-1819 history of Singapore is his founding of a settlement called Singapura on the island Temasek sometime in 1299. Sang Utama's descendants ruled Singapore until the reign of Iskandar Shah (Parameswara), the fifth ruler of Singapore, who was driven out by the Majapahit (Javanese) forces and later founded the kingdom of Malacca (1400s). Thus the lineage of Malacca kings could be traced to Sang Nila Utama and all the way to the powerful Srivijaya.




In his journey to the West, **Admiral Cheng Ho** had led 62 ships and docked in Melaka in **1405** or according to some historians in 1409 when **King Parameswara** discarded his loyalty to the deteriorating Majapahit Empayar in Jawa. Admiral Cheng Ho then received protection from China by paying tribute. By that, **Melaka stayed as one of the oldest settlement of Chinese in Malaya** where a continuous record was found spanning 500 years ago.




The city was founded about 1400, when Paramesvara, the ruler of Tumasik (now Singapore), fled from the forces of the Javanese kingdom of Majapahit and found refuge at the site, then a small fishing village.

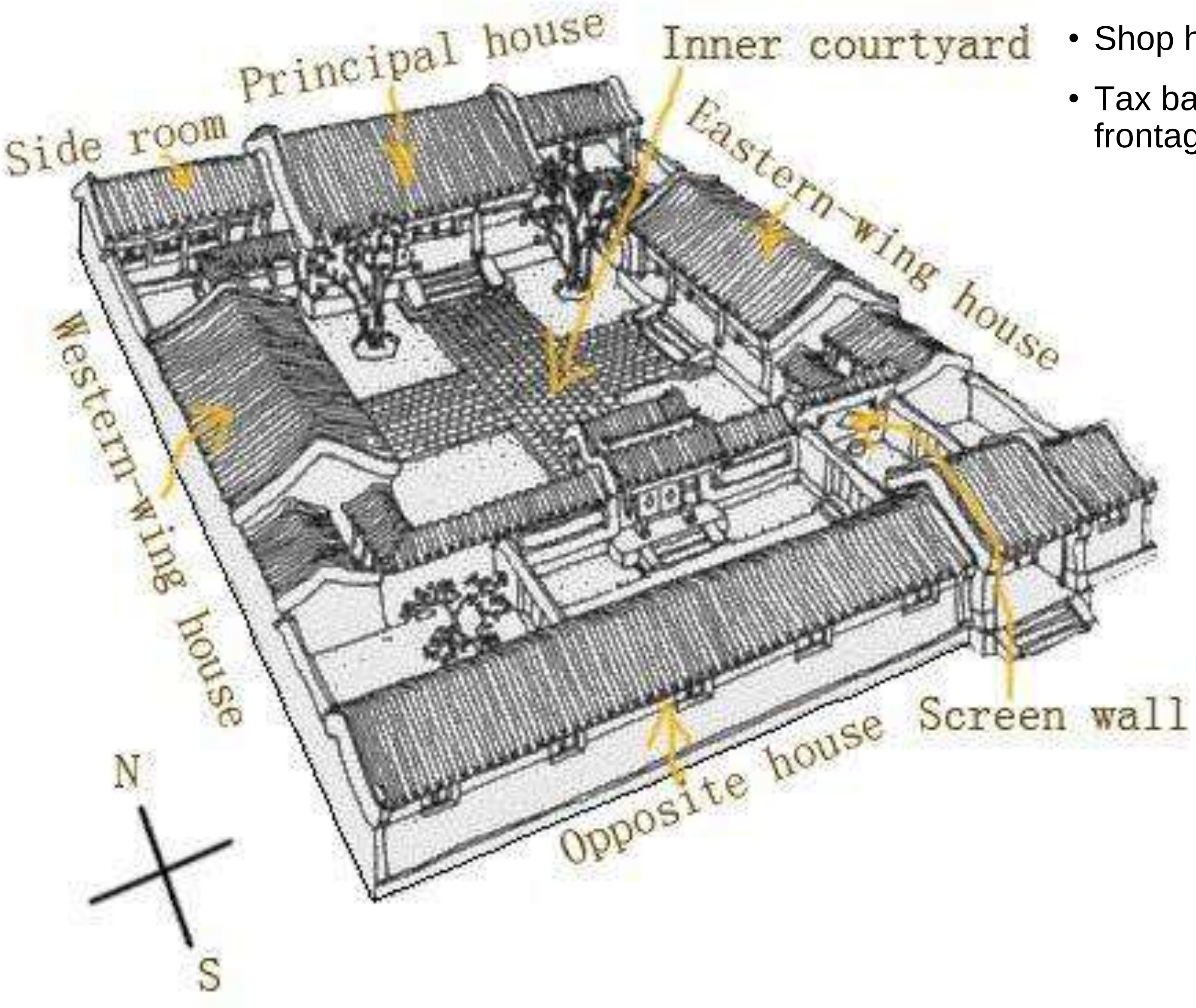
There he founded a Malay kingdom, the kings of which—aided by the Chinese—extended their power over the peninsula. The port became a major stopping place for traders to replenish their food supplies and obtain fresh water from the hill springs.

Malay rule ended in 1511

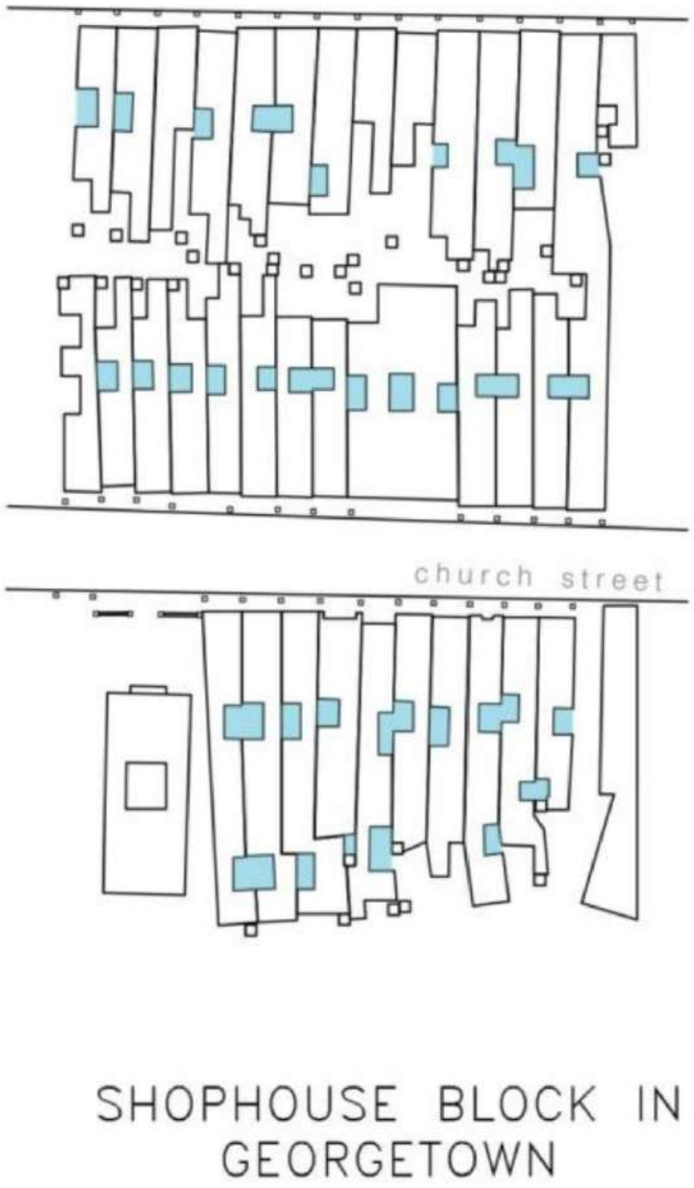




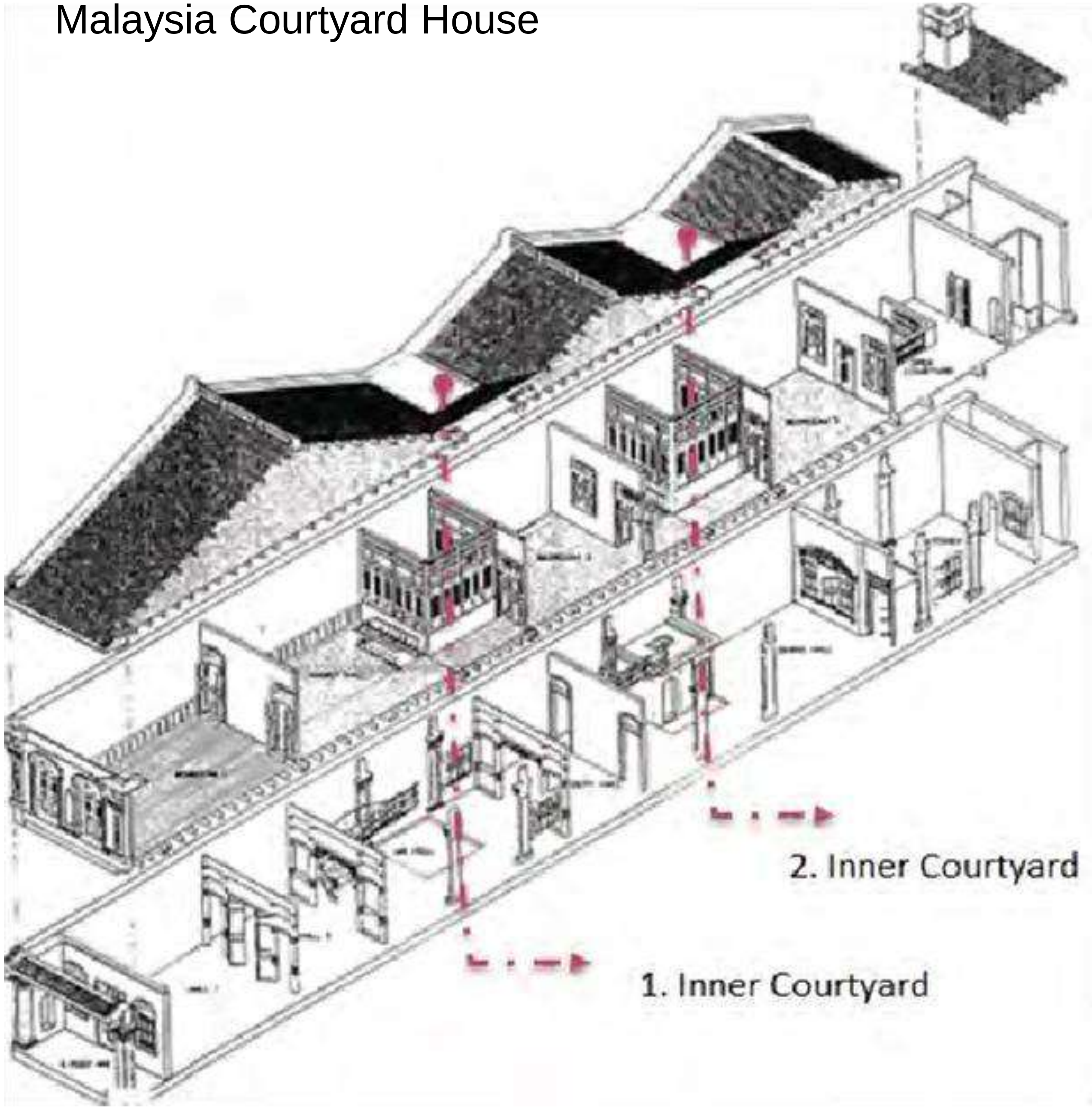
# Chinese Courtyard House



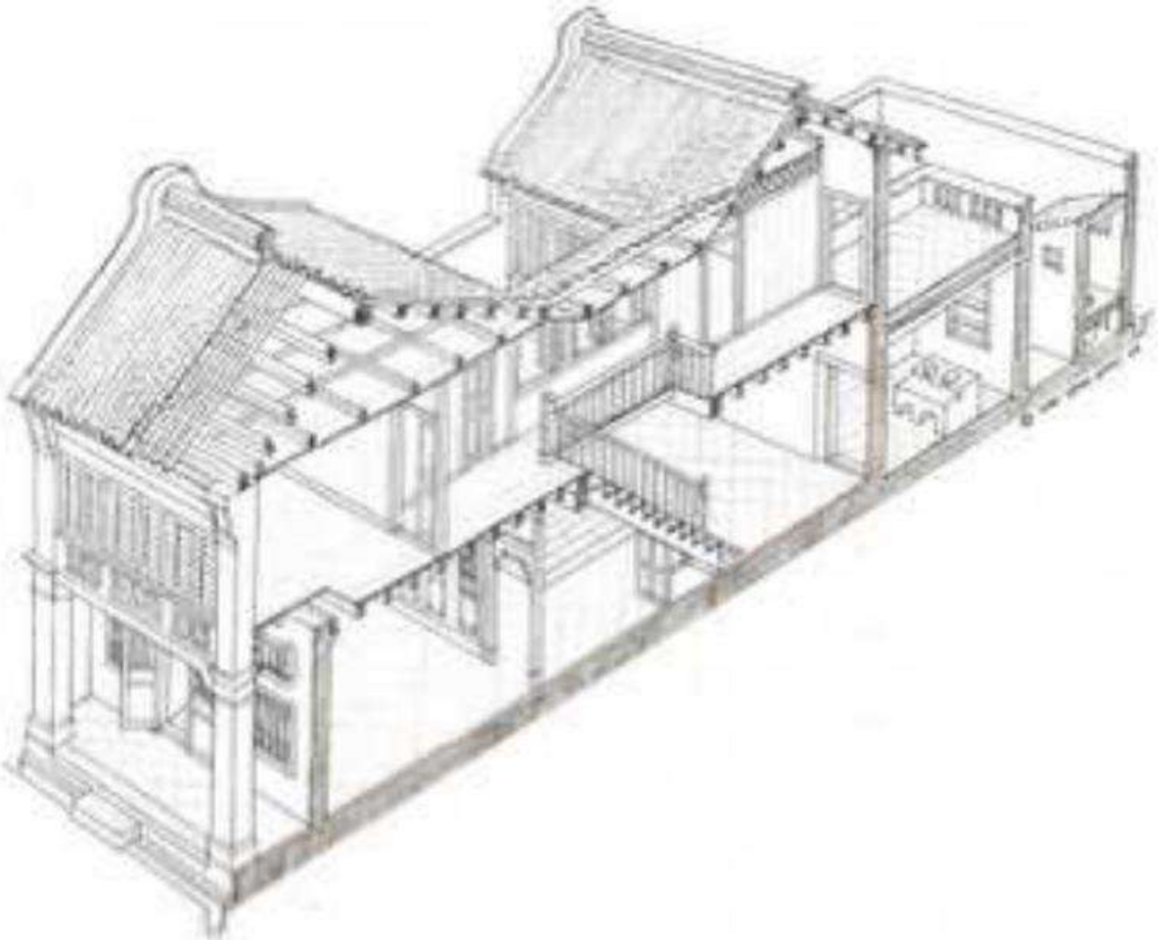

- From South China
- Courtyard house
- Shop house
- Tax based on road frontage



# Malaysia Courtyard House




- Shop house
- Tax based on road frontage

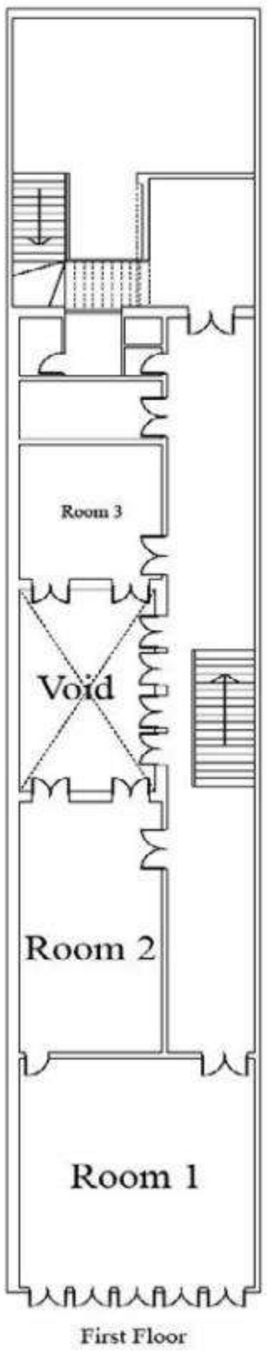


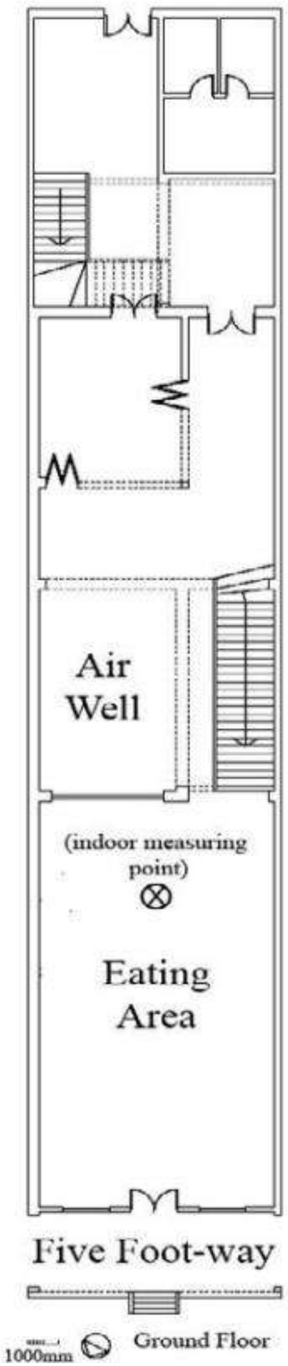
| | Early "Penang" Style 1790s-1850s | "Southern Chinese" Eclectic Style 1840s-1900s | Early "Straits" Eclectic Style 1890s-1910s | Late "Straits" Eclectic Style 1910s-1940s | Art Deco Style 1930s-1960s | Early Modernism Style 1950s-1970s |
|---|---|---|---|---|---|---|
| Description | **Physical Appearance**<br>- 1 storey<br>- simple facade<br>- colour in fashion: light indigo blue, ocre, white<br><br>**Building Materials**<br>- Timber<br>- Clay bricks<br>- Lime | **Physical Appearance**<br>- 1-2 storey<br>- simple facade<br>- colour in fashion: light indigo blue, ocre, white<br><br>**Building Materials**<br>- Timber<br>- Clay bricks<br>- Lime | **Physical Appearance**<br>- 2-3 storey<br>- decorative facade mixture of Malay, Chinese & European ornaments<br>- colour in fashion: light indigo blue, ocre, white<br><br>**Building Materials**<br>- Timber<br>- Clay bricks<br>- Lime | **Physical Appearance**<br>- 2-3 storey<br>- decorative facade mixture of Malay, Chinese & European ornaments<br>- colour in fashion: light indigo blue, ocre, white, opel green<br><br>**Building Materials**<br>- Timber<br>- Clay bricks<br>- Lime | **Physical Appearance**<br>- 2-3 storey<br>- decorative facade geometrical designs<br>- colour in fashion: grey of Shanghai plaster walls<br><br>**Building Materials**<br>- Shanghai plaster<br>- clay bricks<br>- reinforced concrete<br>- glass | **Physical Appearance**<br>- 2-3 storey<br>- simple and clean facade designs<br>- colour in fashion: white<br><br>**Building Materials**<br>- clay bricks<br>- reinforced concrete glass |

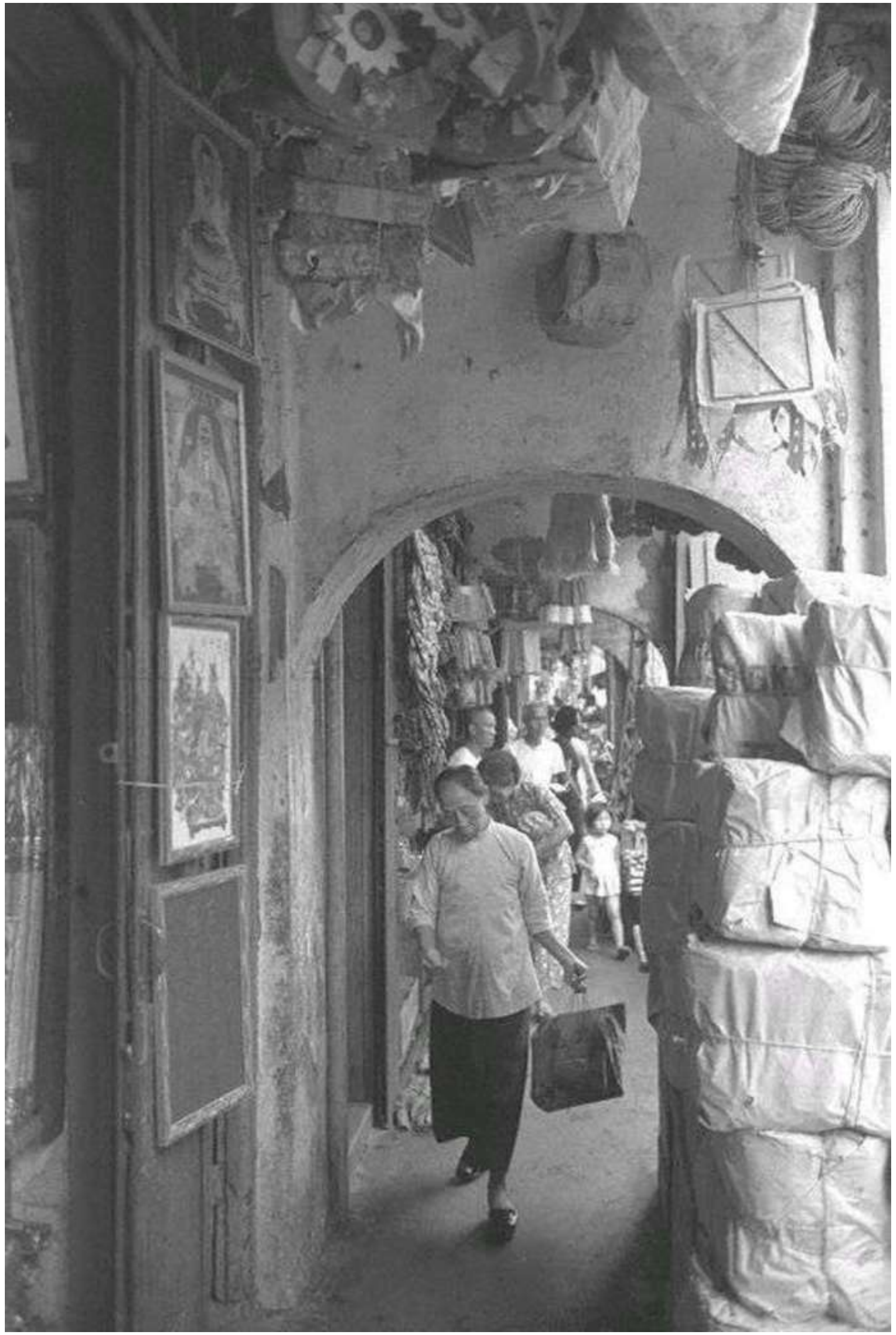

1700
1800
1840
1860
1890
1900
1920
1930
1960

*Figure 2. The interior of the shophouse of Peranakan reused as a coffee shop.*

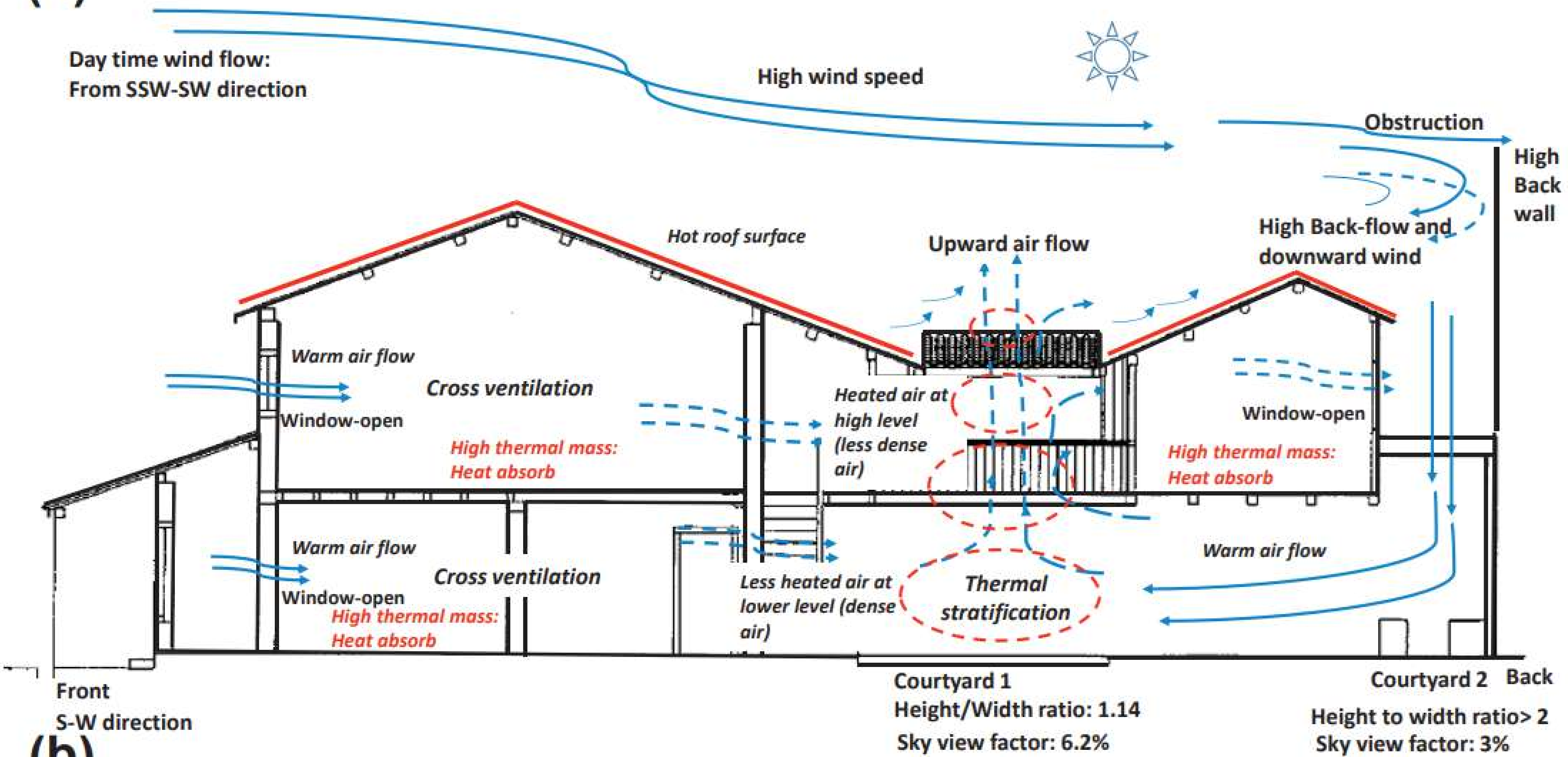
*Figure 5. The original Vs contemporary Peranakan tiles (captured by the researcher)*


Room 3
Void
Room 2
Room 1
First Floor


Air Well
(indoor measuring point)
Eating Area
Five Foot-way
1000mm
Ground Floor



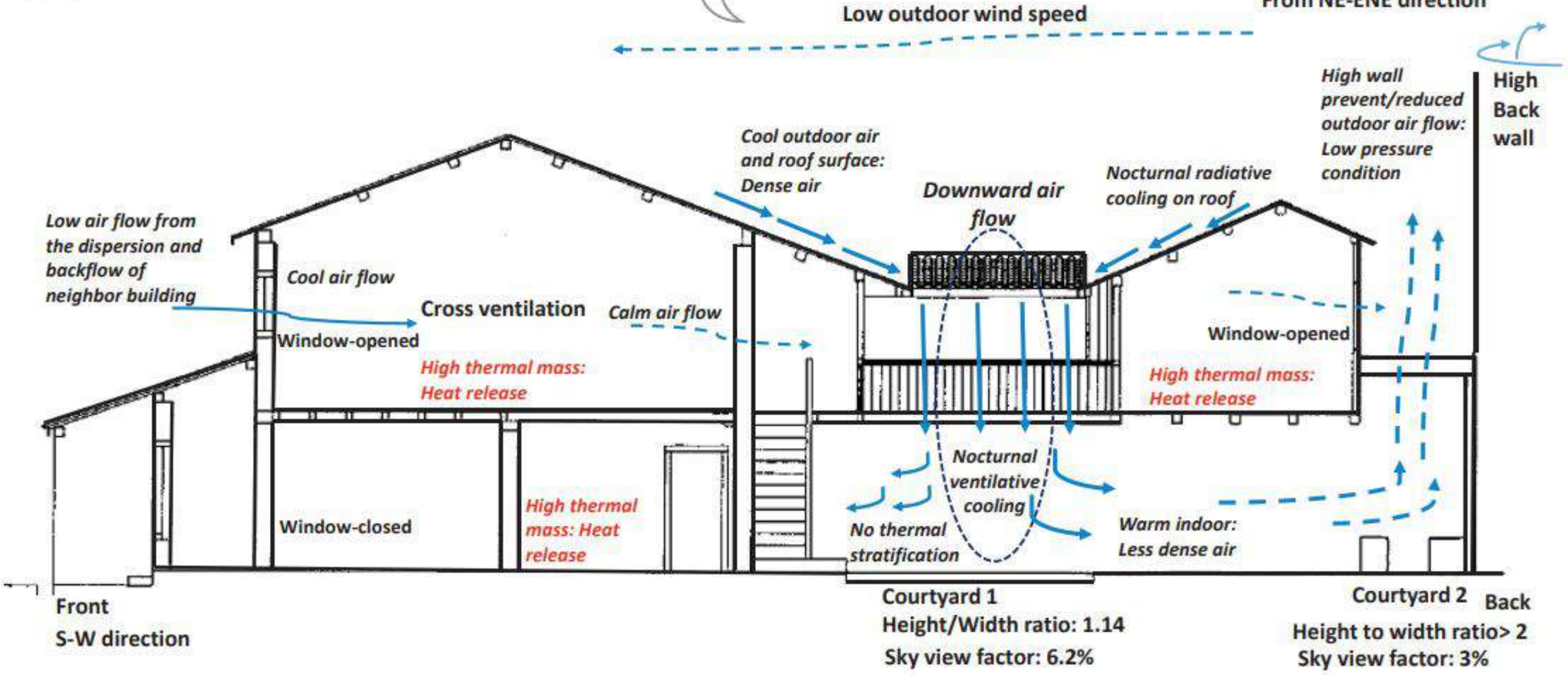
(a)



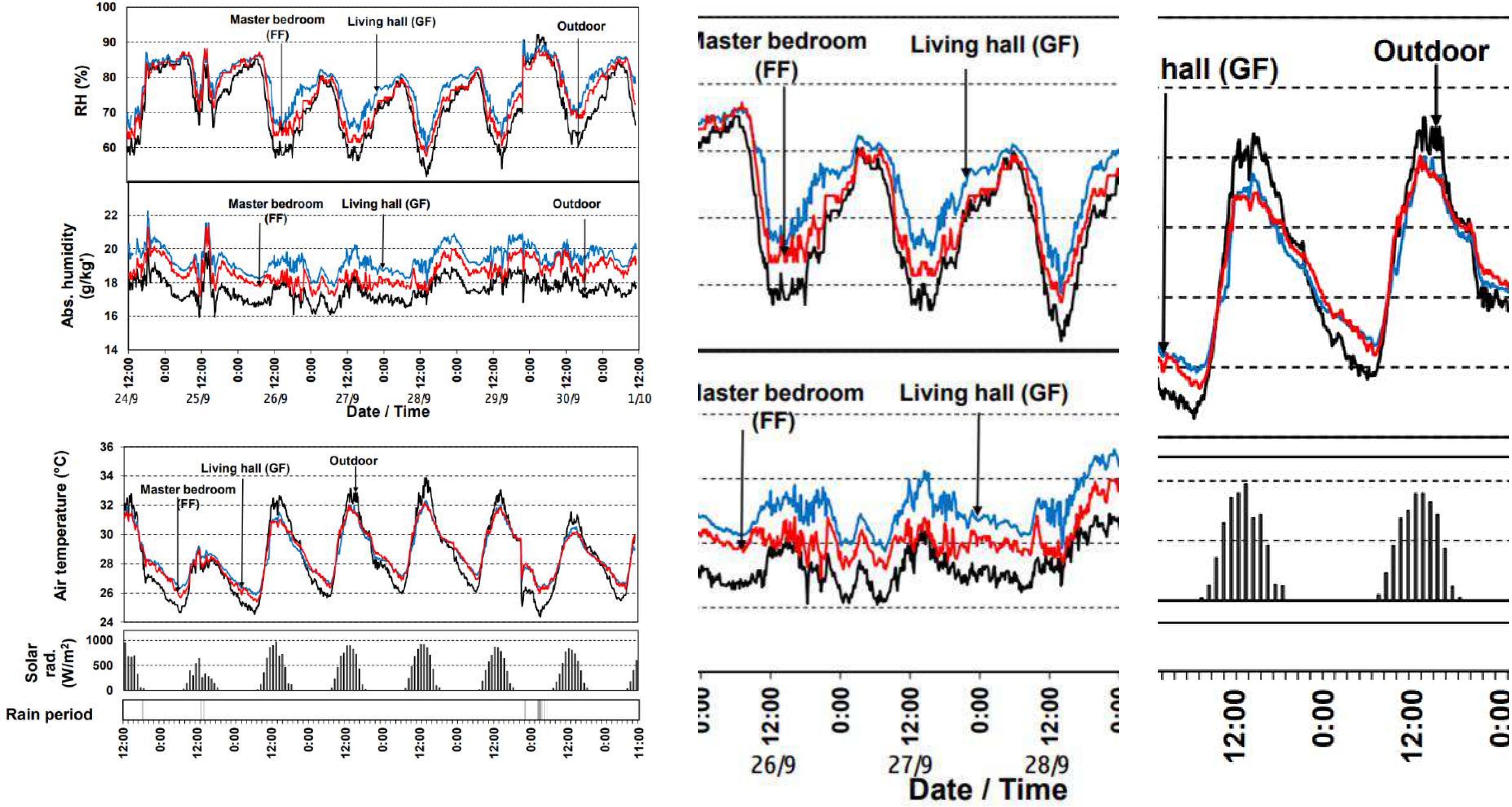
Day time wind flow:
From SSW-SW direction
High wind speed
Obstruction
High
Back
wall
High Back-flow and
downward wind
Hot roof surface
Upward air flow
Warm air flow
Cross ventilation
Window-open
High thermal mass:
Heat absorb
Heated air at
high level
(less dense
air)
Window-open
High thermal mass:
Heat absorb
Warm air flow
Cross ventilation
Window-open
High thermal mass:
Heat absorb
Less heated air at
lower level (dense
air)
Thermal
stratification
Warm air flow
Front
S-W direction
Courtyard 1
Height/Width ratio: 1.14
Sky view factor: 6.2%
Courtyard 2
Back
Height to width ratio> 2
Sky view factor: 3%


(b)

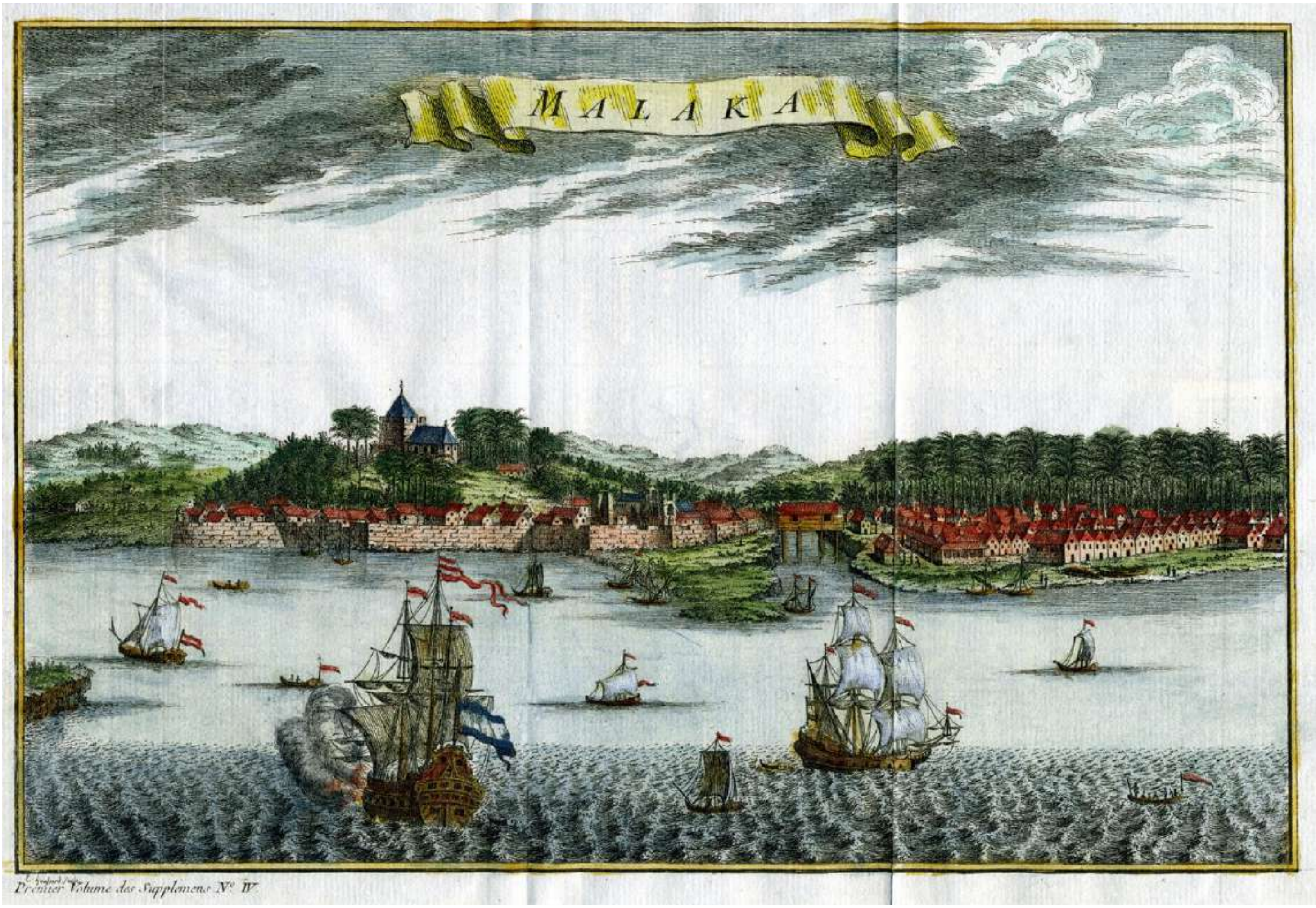
(c)



Night time wind flow:
From NE-ENE direction
Low outdoor wind speed
High wall prevent/reduced outdoor air flow: Low pressure condition
High Back wall
Cool outdoor air and roof surface: Dense air
Downward air flow
Nocturnal radiative cooling on roof
Low air flow from the dispersion and backflow of neighbor building
Cool air flow
Cross ventilation
Calm air flow
Window-opened
High thermal mass: Heat release
Window-opened
High thermal mass: Heat release
Window-closed
High thermal mass: Heat release
Nocturnal ventilative cooling
No thermal stratification
Warm indoor: Less dense air
Front
S-W direction
Courtyard 1
Height/Width ratio: 1.14
Sky view factor: 6.2%
Courtyard 2
Back
Height to width ratio> 2
Sky view factor: 3%


RH (%)
Master bedroom (FF)
Living hall (GF)
Outdoor
Abs. humidity (g/kg')
Date / Time
24/9
25/9
26/9
27/9
28/9
29/9
30/9
1/10
Air temperature (°C)
Solar rad. (W/m2)
Rain period

Dutch Malacca, c. 1750




| | |
|---|---|
| Malacca Sultanate | 15th century |
| Portuguese control[3] | 24 August 1511 |
| Dutch control[4][5] | 14 January 1641 |
| British control[4][5][6][7] | 17 March 1824 |
| Japanese occupation[8][9] | 11 January 1942 |
| Accession into the Malayan Union[10] | 1 April 1946 |
| Accession into the Federation of Malaya[11] | 1 February 1948 |
| Independence as part of the Federation of Malaya[12] | 31 August 1957 |


1630 map of the Portuguese fort and the city of Malacca

## The Straits Settlements

The Straits Settlements, comprising Penang, Malacca and Singapore, was an administrative unit of the East India Company (1826–1867) and later the British Colonial Office (1867–1946).



The first settlement was the Penang territory, in 1786.

Singapore became the site of a British trading post in 1819 after its founder, Stamford Raffles, successfully involved the East India Company in a dynastic struggle for the throne of Johor.

In 1824, the Dutch conceded any rights they had to the island in the Anglo-Dutch Treaty of 1824, and from 1832, Singapore was the seat of government of the Straits Settlements for 114 years until its dissolution in 1946.

Malacca River 1907, Church of St. Francis Xavier in the background

The Port of Penang in George Town during the 1910s

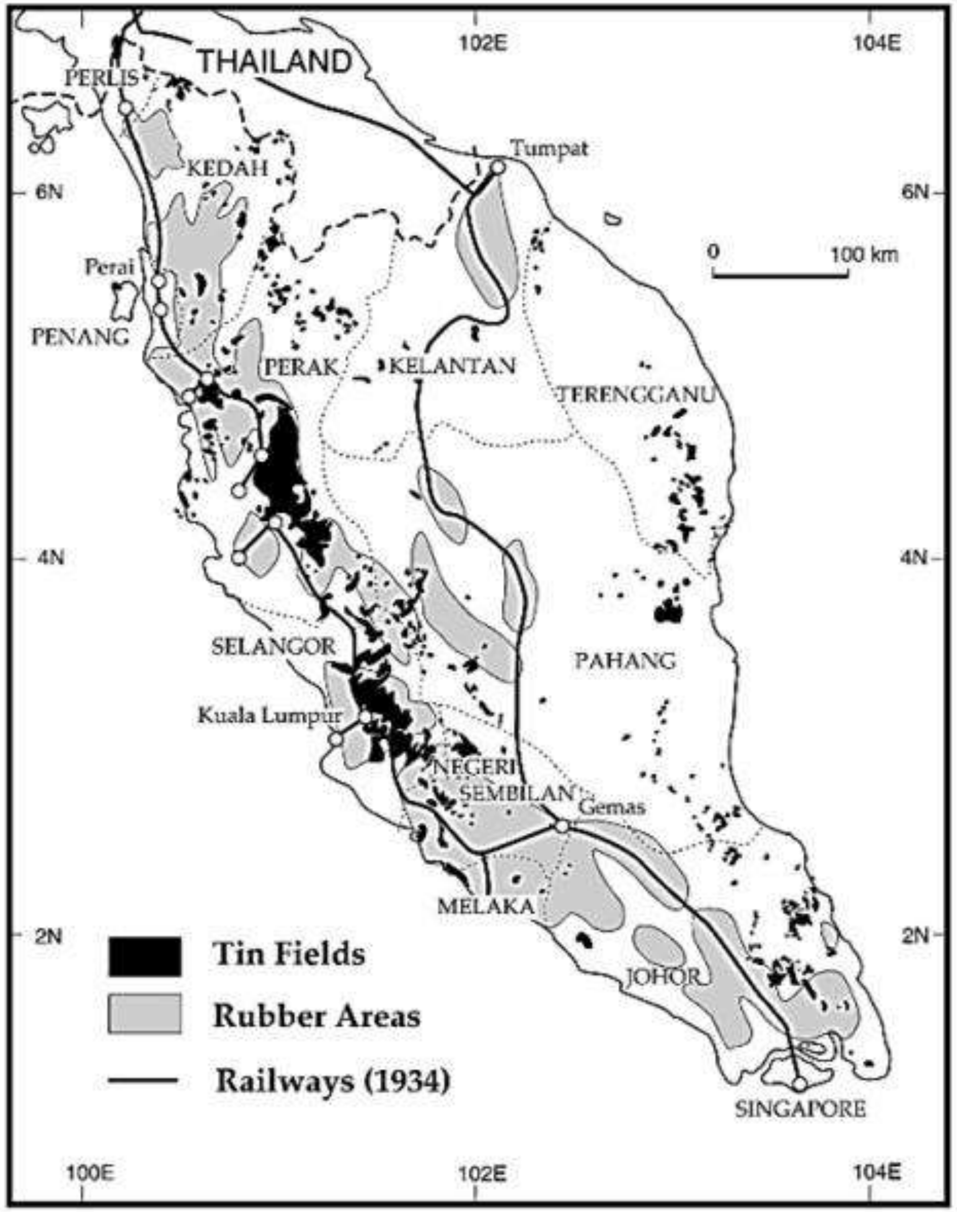
Singapore

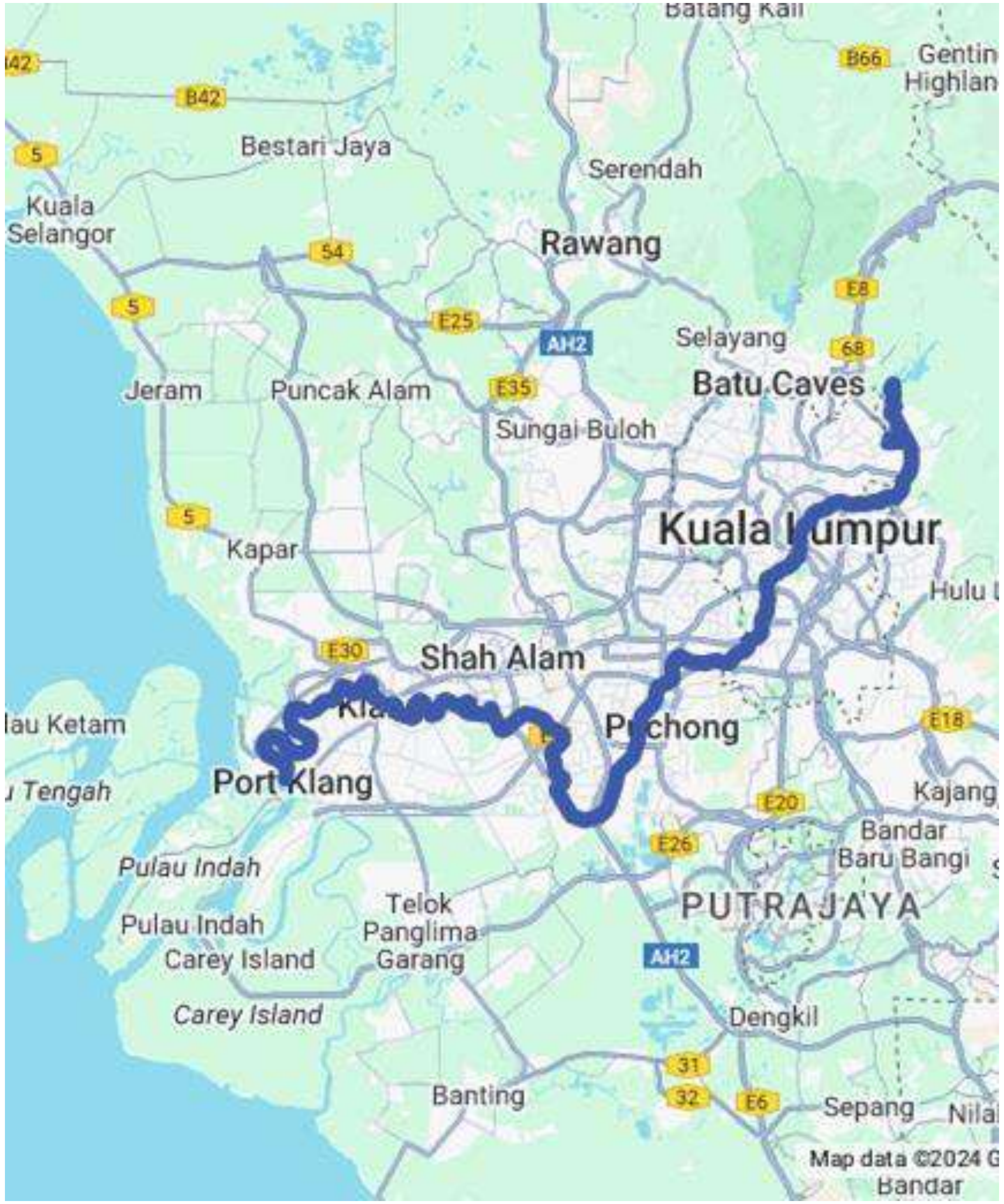
# History of Kuala Lumpur

Kuala Lumpur was founded *ca.* 1857 at the confluence of the Gombak and Klang rivers. In English, the name Kuala Lumpur literally means "muddy confluence".

Selangor royal family hired tin prospectors to open tin mines in the Klang Valley.

Kuala Lumpur was the furthest point up the Klang River to which supplies could conveniently be brought by boat; it therefore became a collection and dispersal point serving the tin mines.



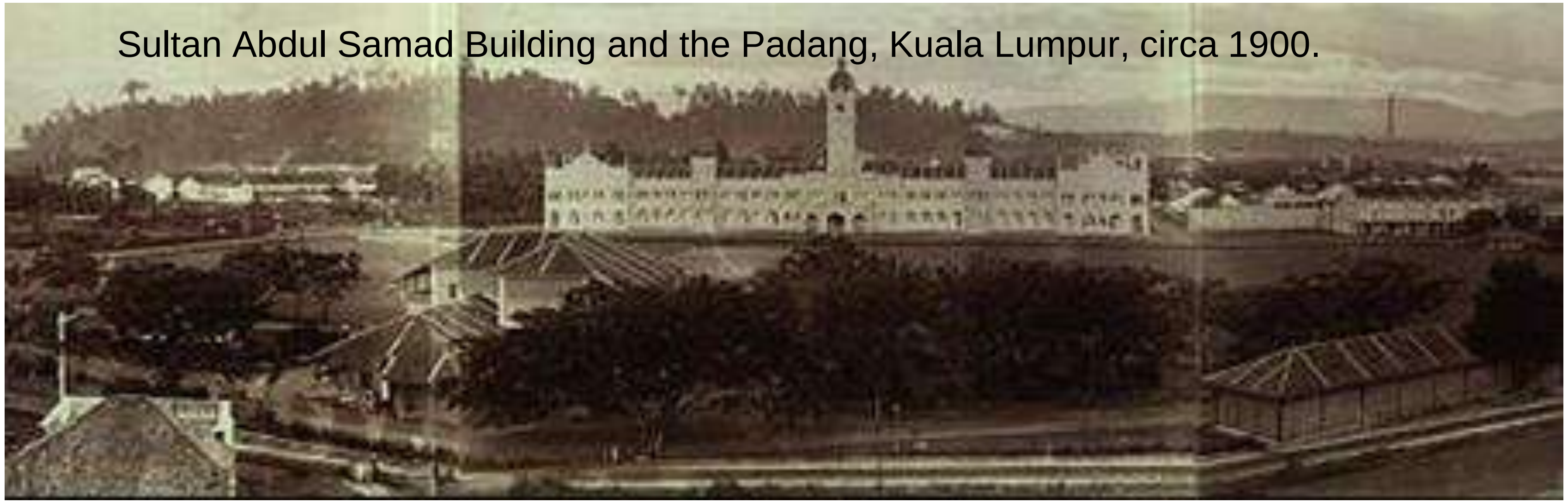



Kuala Lumpur, circa 1884.

On 4 January 1881 the entire town was burnt down, and later the same year, the town was severely flooded.

KITLV Universiteit Leiden 3990

**A street in Kuala Lumpur in the 1880s**



Sultan Abdul Samad Building and the Padang, Kuala Lumpur, circa 1900.

Frank Swettenham, on becoming the British Resident, began improving the town by cleaning up the streets. He also stipulated that buildings should be constructed of brick and tile so that they would be less flammable.[15][16] He ordered that Kuala Lumpur be rebuilt with wider streets, and the houses to be replaced with buildings in brick and tile street by street. The rebuilding program lasted about five years.[16]

KL Old Market Square circa 1920

Kuala Lumpur

| | | |
|---|---|---|
| 1895 | 0.65 $km^2$ | |
| 1903 | 20 $km^2$ | |
| 1948 | 93 $km^2$ | became a municipality |
| 1974 | 243 $km^2$ | Federal Territory. |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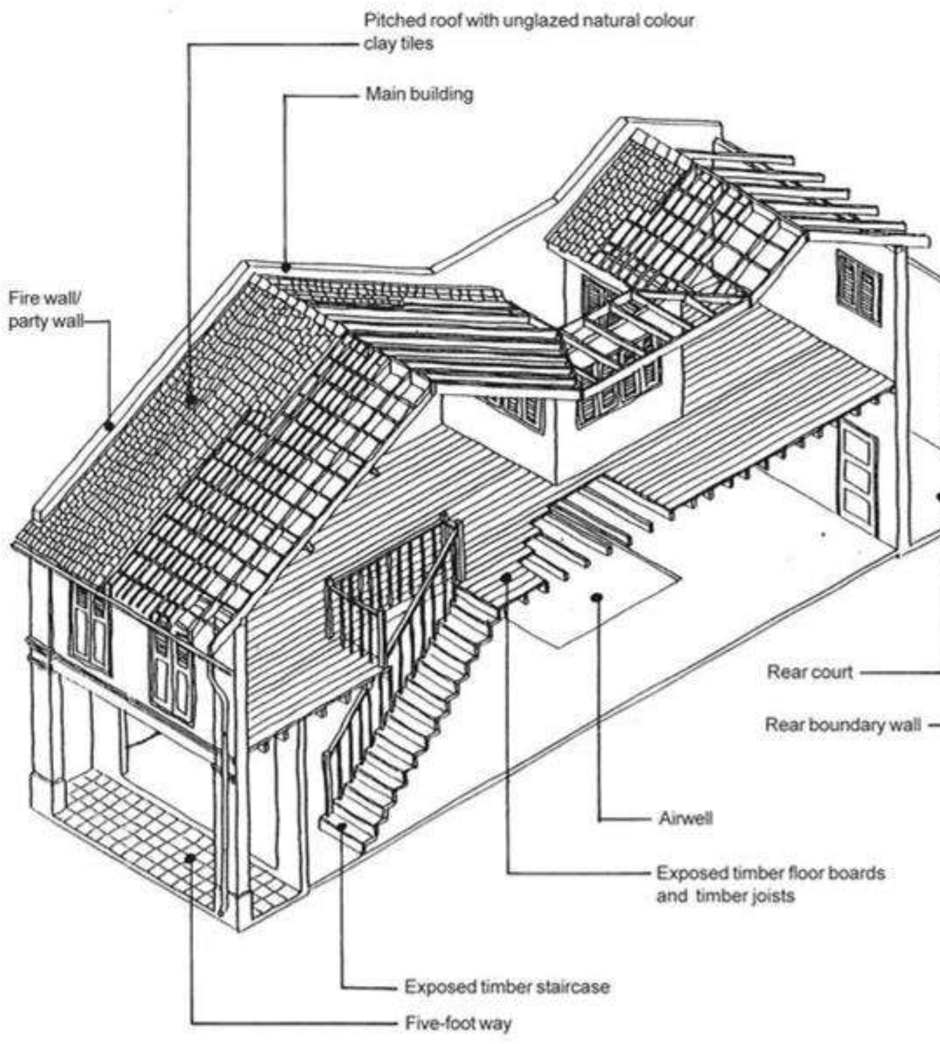
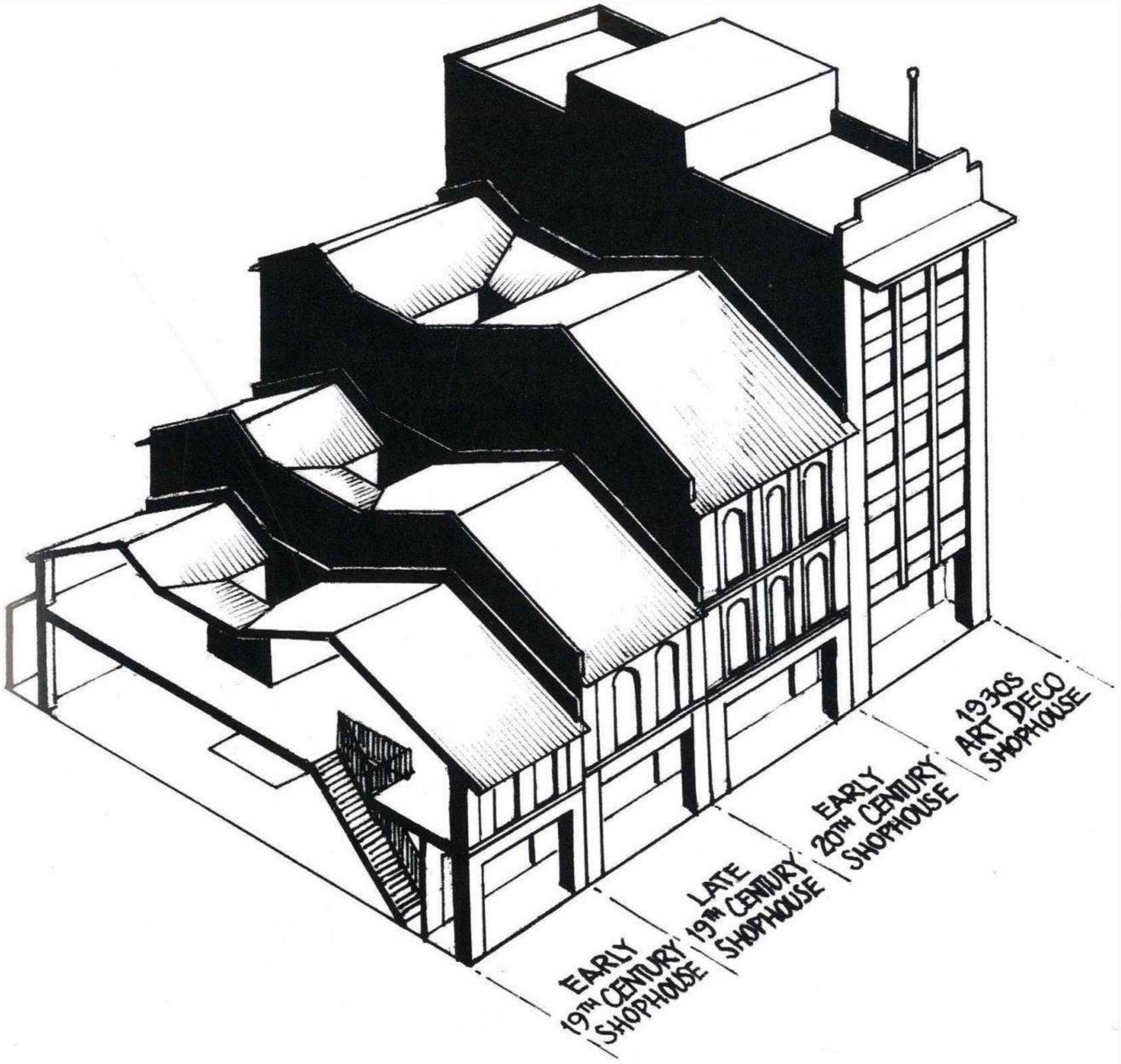
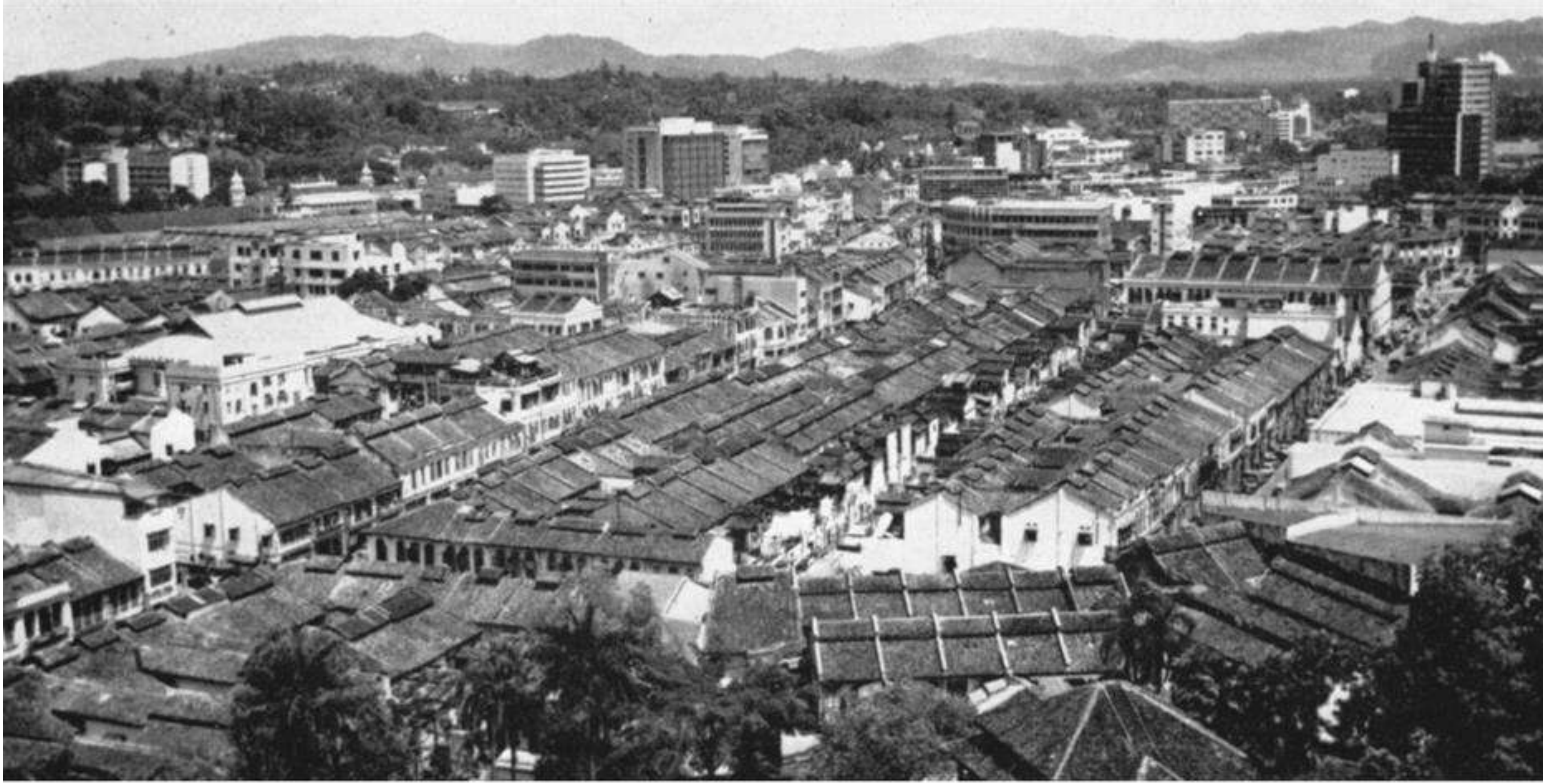
**Old Market Square in the 1910s**

**Medan Pasar bus stop in 2008**




KEY ELEMENTS OF A TYPICAL SHOPHOUSE





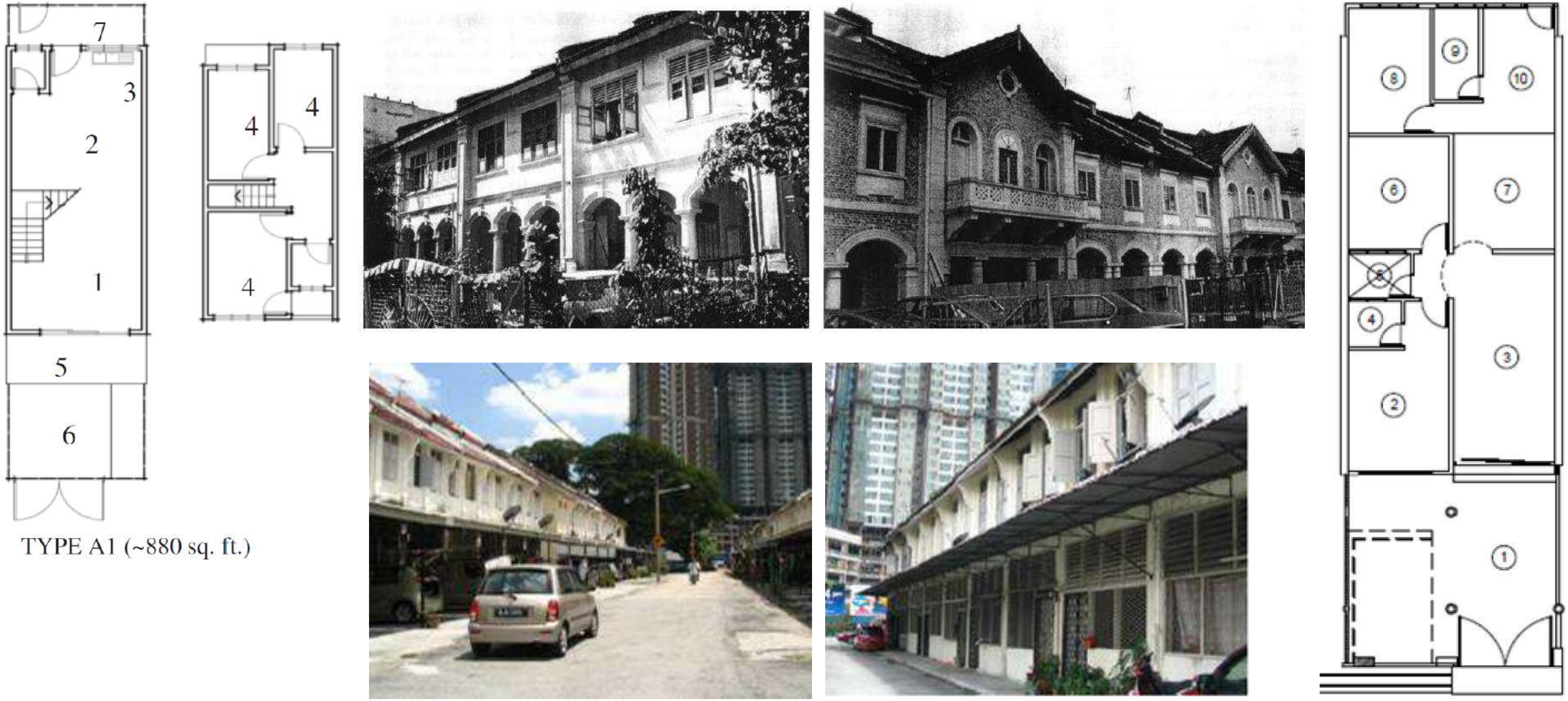
**Source:** Hamzah Sendut (1965, p. 133).

After industrial revolution,
the world saw a different kind of housing development in the urban areas in the form of dense shelters serving mainly as night time refuge for factory workers who had migrated from the countryside to these new industrial towns in search for better income.

Sir Frank Swettenham (born March 28, 1850, Belper, Derbyshire, Eng.—died June 11, 1946, London) was a British colonial official in Malaya who was highly influential in shaping British policy and the structure of British administration in the

In 1871 Swettenham was sent to Singapore as a cadet in the civil service of the Straits Settlements

In 1882 he was appointed resident (adviser) to the Malay state of Selangor.




Before Modification

**Figure 2,** Terrace house for middle class at Jalan Campbell, Doraisamy, Link Houses at Brickfields, (100 quarters) today, Source: S Vlatseas (1990)

# Petaling Jaya - First New Township


Introduction to motor age brought in newer issues. Automobiles became the mainstay in urban life. For the first time in history, walking was relegated to secondary as cities became wider and wider,
travel distances became longer, and there was no alternate for motorized vehicles.

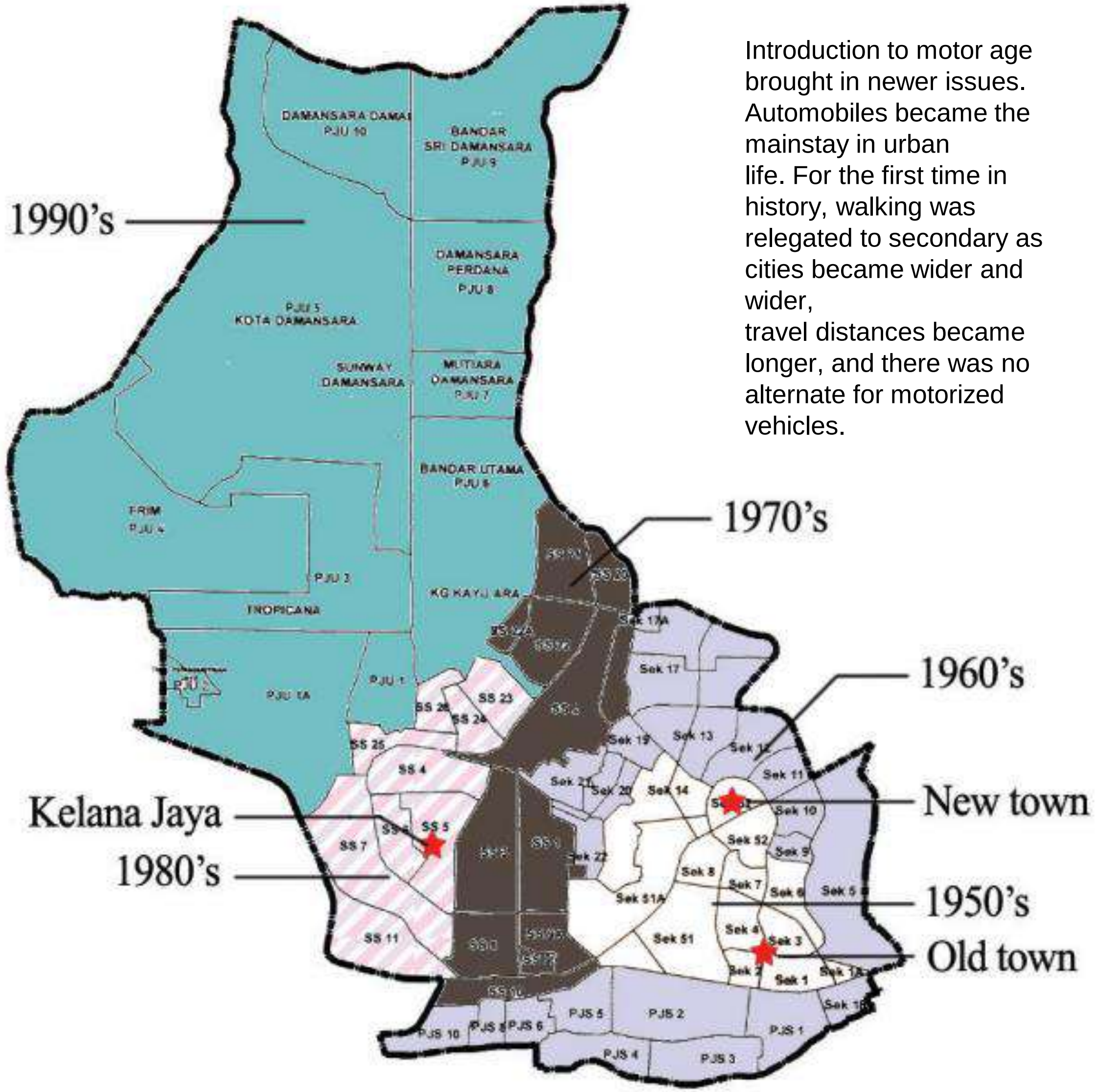



Fig.2. Development of Petaling Jaya

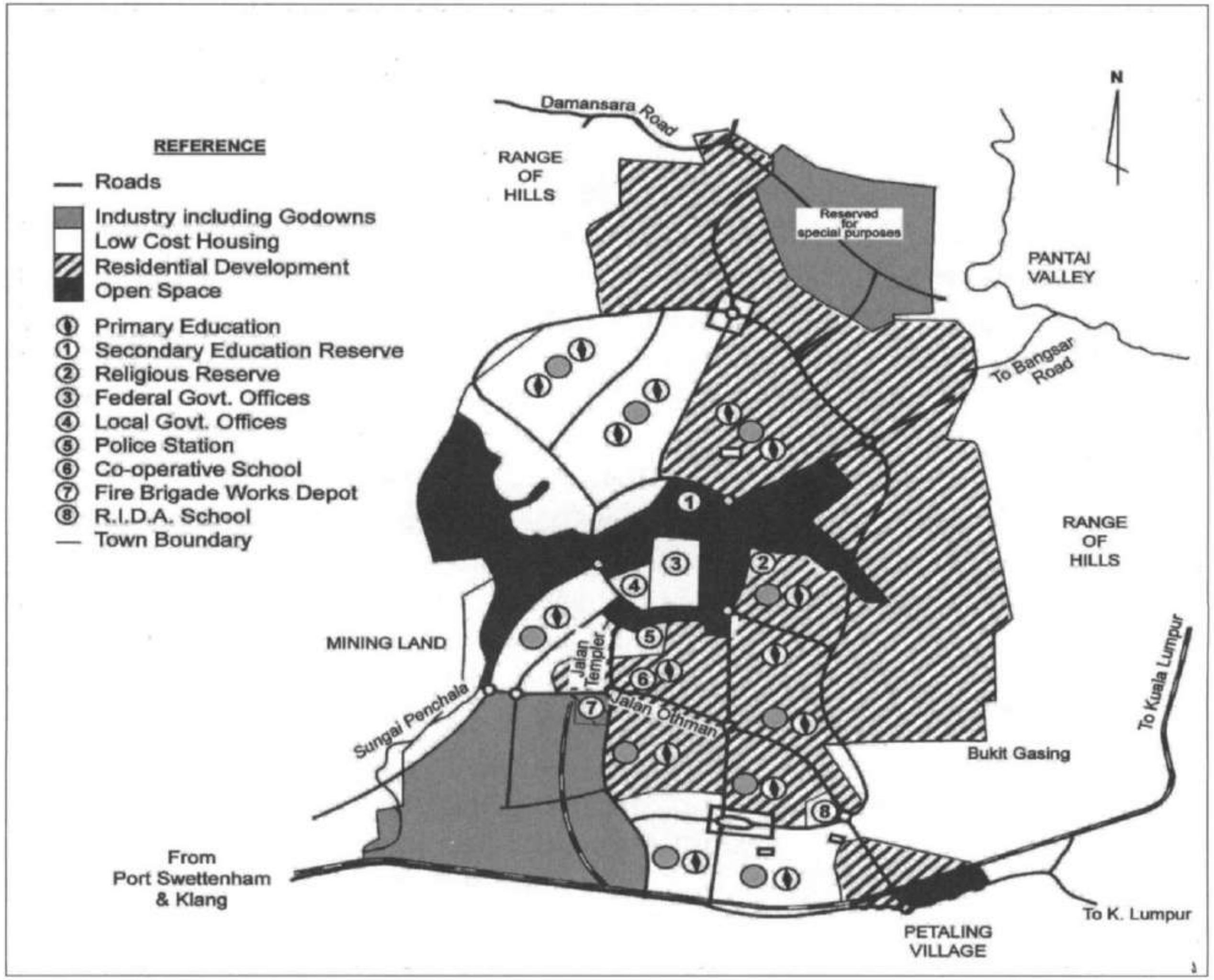



FIG. 5 The more detailed plan for the entire area of Petaling Jaya in 1955.

FIG. 7 The newly opened Assunta Hospital in 1961 and the modern bungalows of the 'Ministers' Row' in the background.

FIG. 4 An aerial photograph mosaic of early Petaling Jaya at the end of 1954. (*Source:* Mosaic from 1:4000 air photos taken by 81 Squadron RAF. British Library Map No. Y839)

**Figure 5 Kuala Lumpur terraced housing, 1975**

**Closely spaced terraced housing at Bungsar Baru, Kuala Lumpur. The houses have almost no front yards, and at the back open directly onto narrow lanes. The white scars on the hills in the distance are housing development areas in southeastern Kuala Lumpur**

**Source:** Aiken and Leigh (1975, p. 561).

Kuala Lumpur Municipal Council formed in 1957

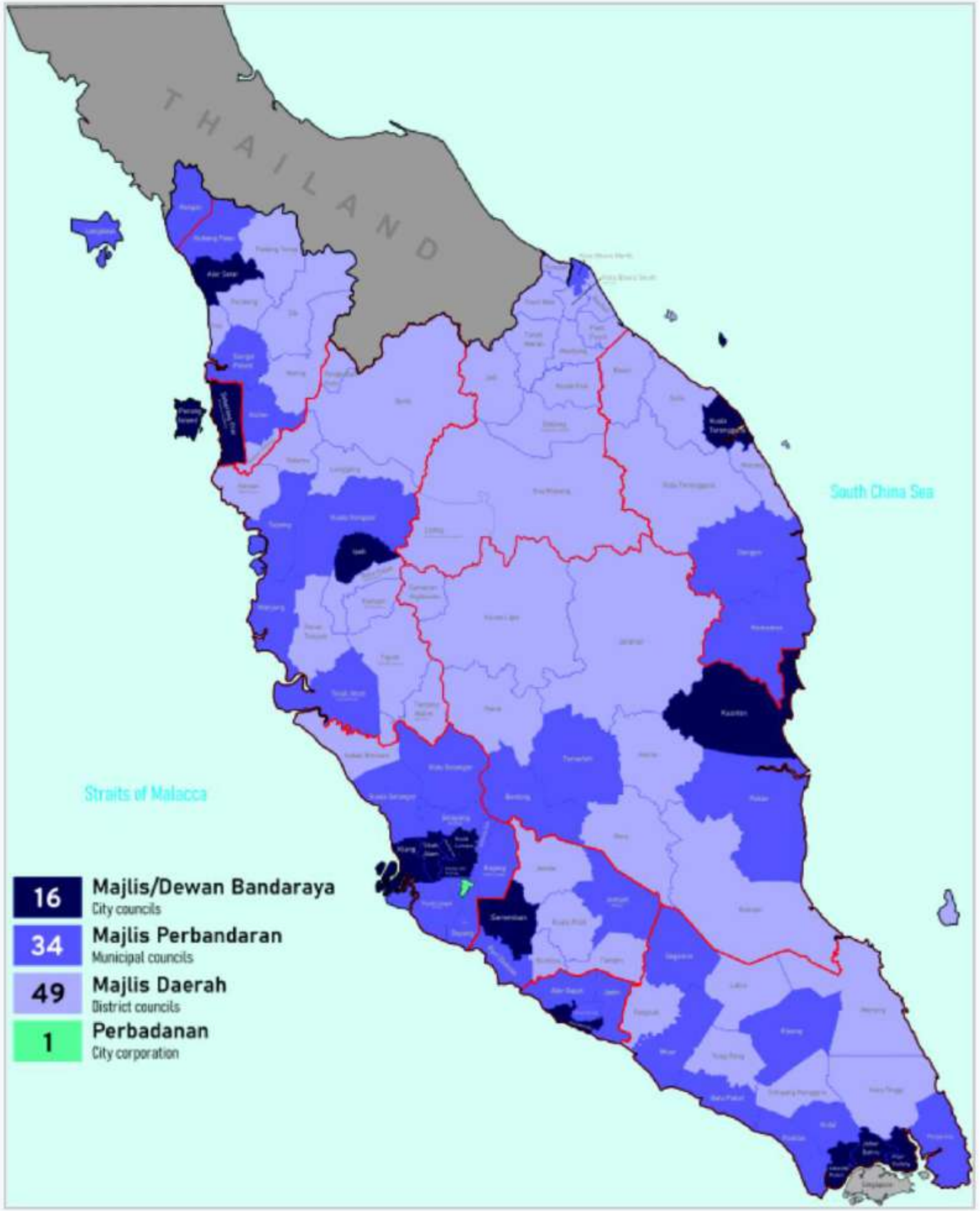
**Category of Local Authority :**

**14 states = 151 local authority**

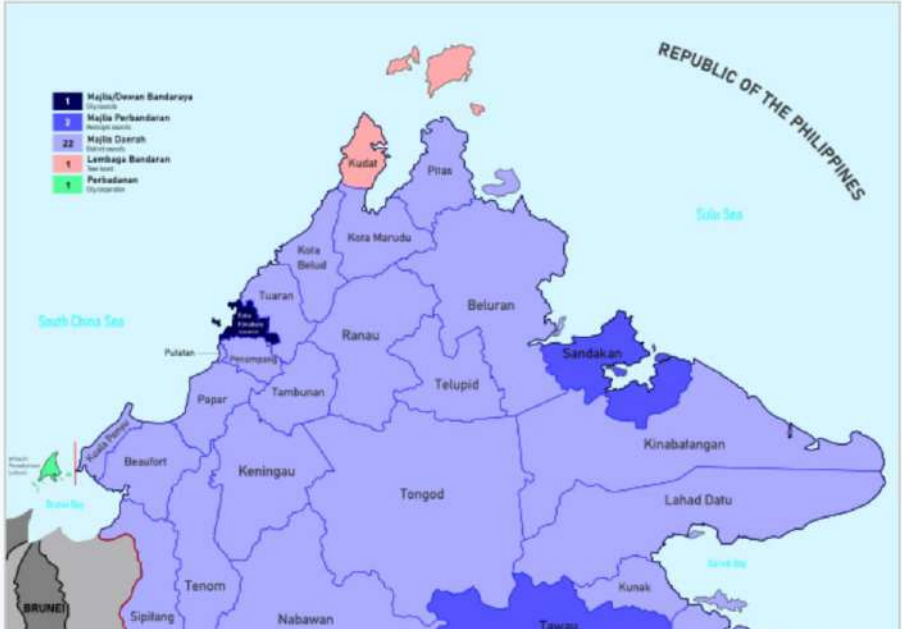
A) City Council
Have people not less than 500,000

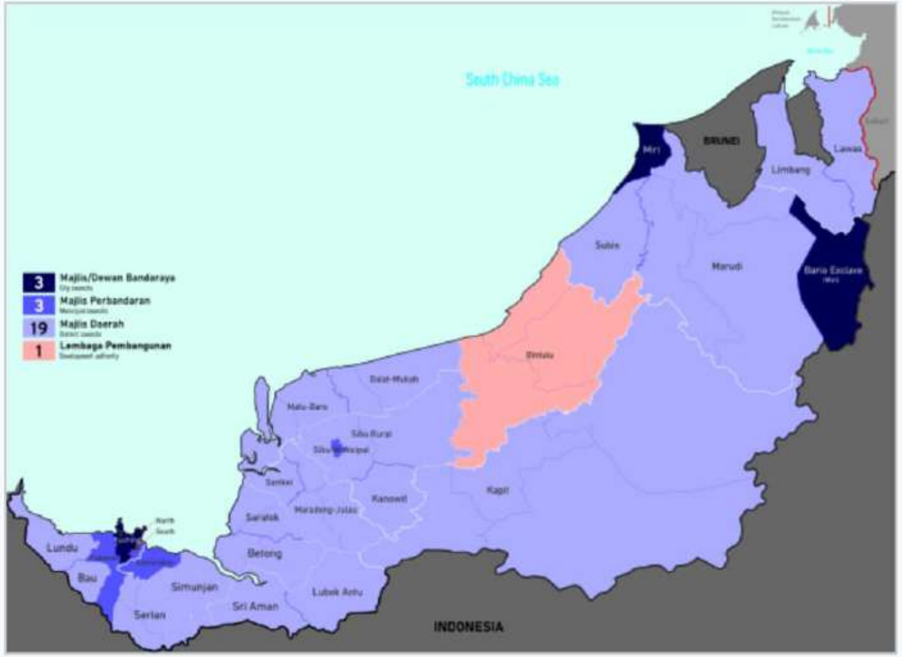
B) Municipal
Have people not less than 150,000

C) District Council
Inhabitants less than 150,000 people




Local governments in Peninsular Malaysia



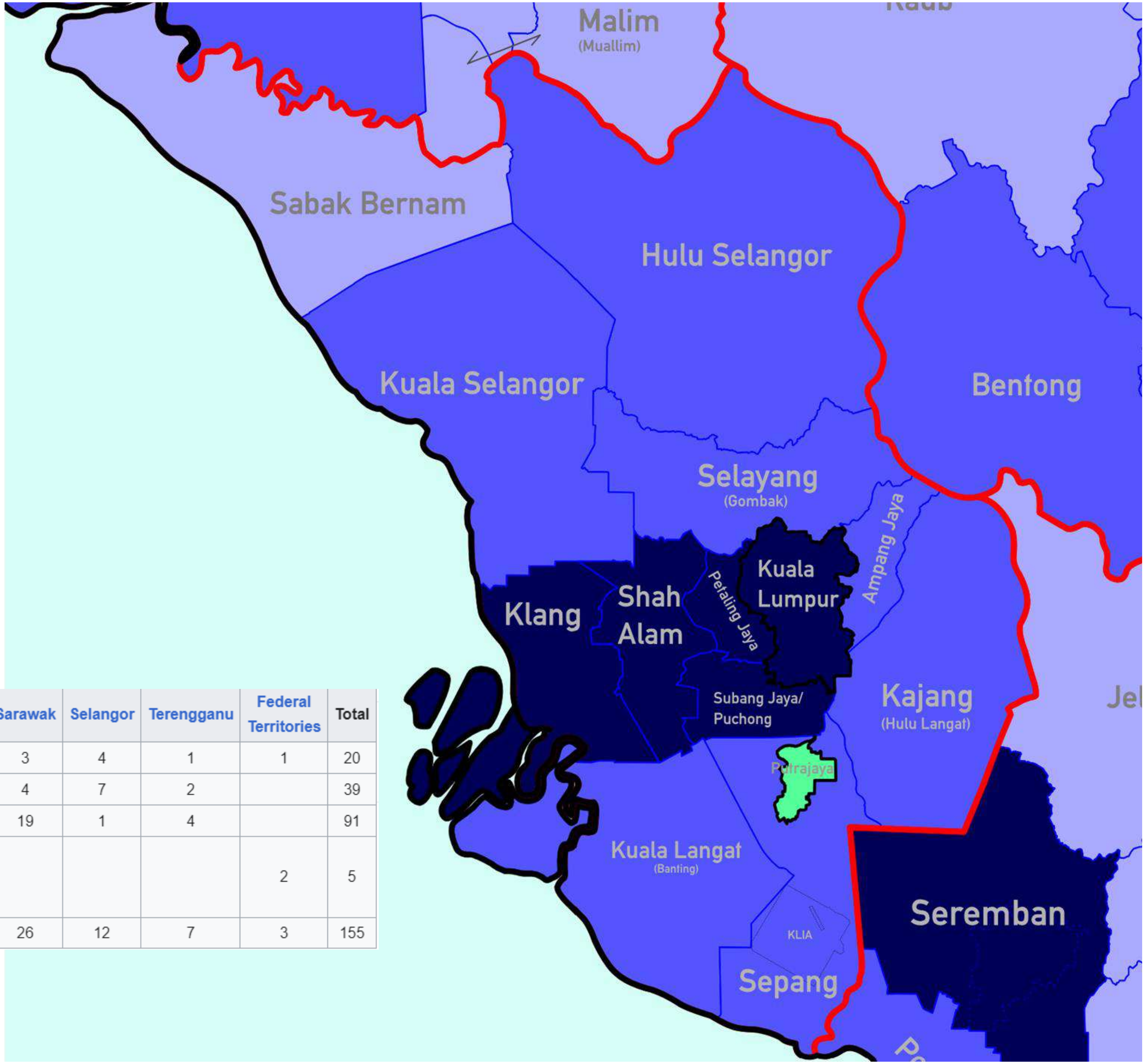




Local governments in Sarawak

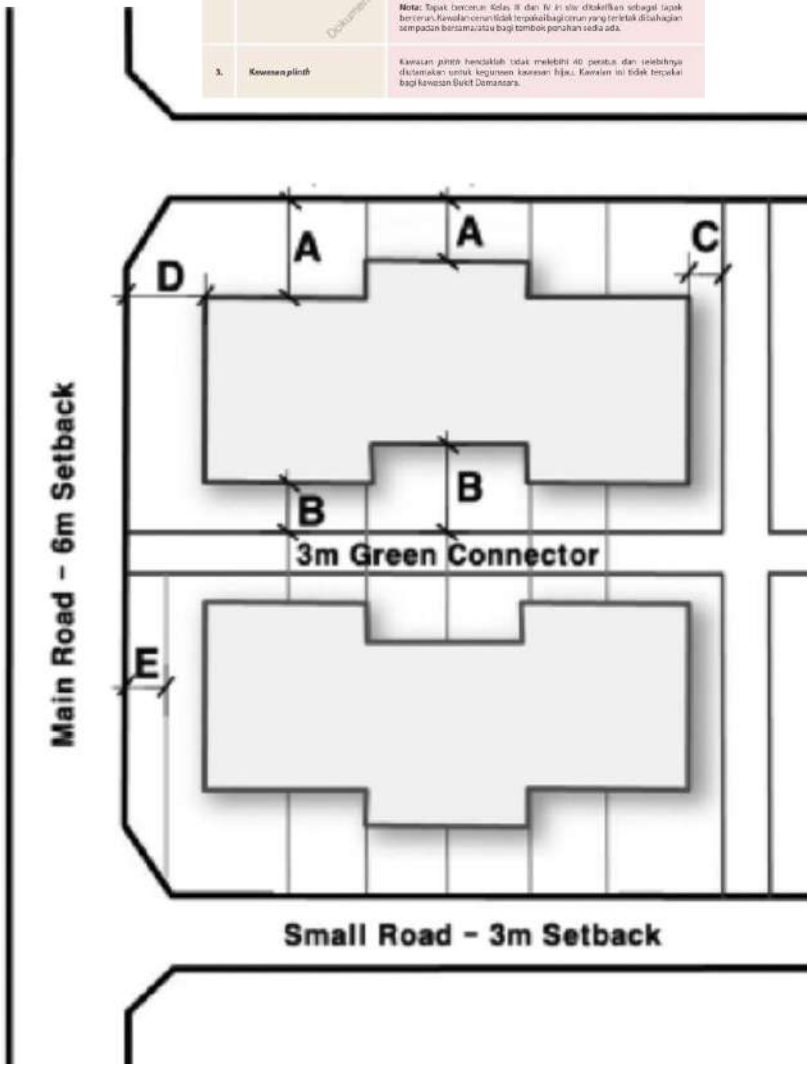
**Building Authority**

Building Department of the local authority
Engineering Department of the local authority
Landscape Department of the local authority
Health Department of the local authority
State Town and Country Planning Department (JPBD)
Land and Mine Office / District Land Administrator (PTD)
Public Works Department (JKR)
Department of Irrigation and Drainage (JPS)
Environmental Department
Water Supply Department (SYABAS)
Tenaga Nasional Berhad (TNB)
Fire and Rescue Department Malaysia (JBPM)
Telekom Malaysia Berhad (Telekom)
Sewerage Services Department (WK)
Department of Minerals and Geosciences (JMG)

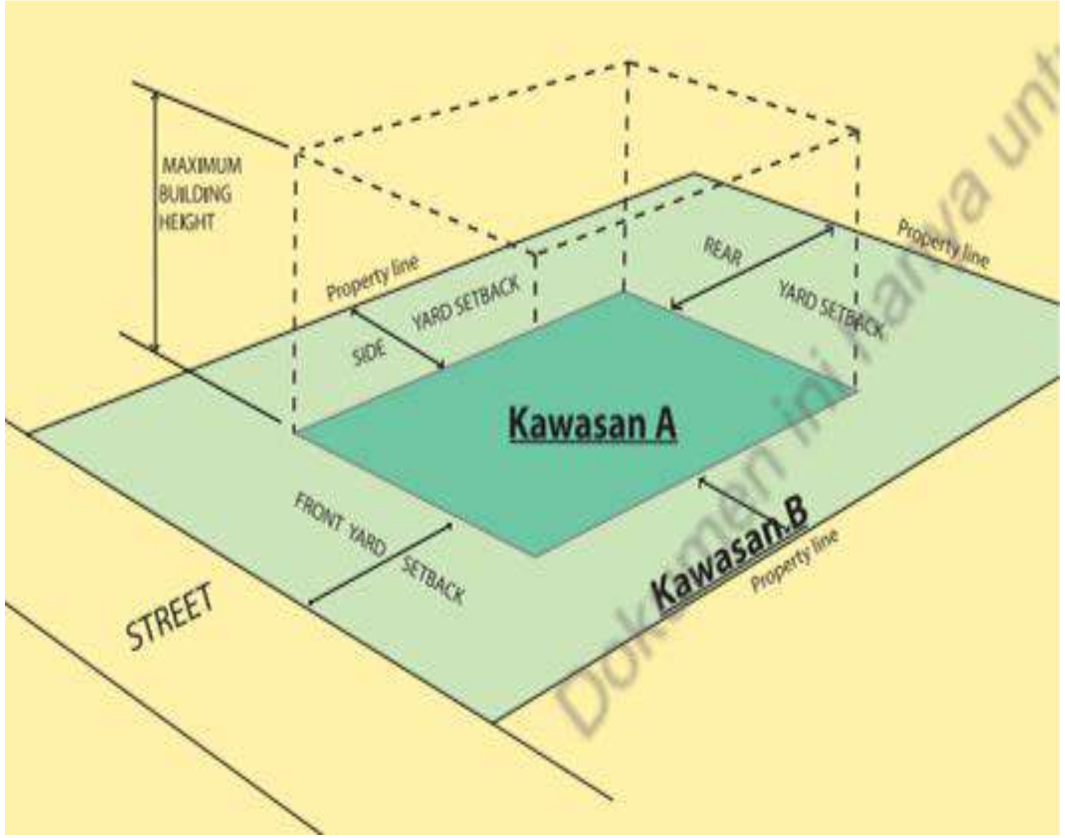
| Type | Johor | Kedah | Kelantan | Malacca | Negeri Sembilan | Pahang | Penang | Perak | Perlis | Sabah | Sarawak | Selangor | Terengganu | Federal Territories | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Cities | 3 | 1 | | 1 | 1 | 1 | 2 | 1 | | 1 | 3 | 4 | 1 | 1 | 20 |
| Municipalities | 7 | 4 | 1 | 3 | 2 | 2 | | 4 | 1 | 2 | 4 | 7 | 2 | | 39 |
| Districts | 6 | 6 | 11 | | 4 | 8 | | 10 | | 22 | 19 | 1 | 4 | | 91 |
| Special or modified local councils | 1 | 1 | | | | 1 | | | | | | | | 2 | 5 |
| Total | 17 | 12 | 12 | 4 | 7 | 12 | 2 | 15 | 1 | 25 | 26 | 12 | 7 | 3 | 155 |

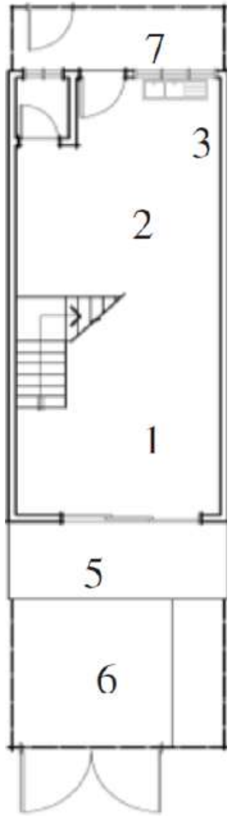


| Bil. | Kriteria | Garis Panduan |
|---|---|---|
| 1. | **Permohonan bagi tujuan pecah sempadan lot** | Hanya lot berkeluasan 1 ekar dan lebih sahaja yang layak memohon untuk pecah sempadan lot. |
| 2. | **Keluasan minimum lot rumah sesebuah** | Cadangan pecahan tanah perlu memastikan keluasan minimum lot rumah sesebuah seperti yang berikut:<br>a. Bukit Damansara<br>• minimum 8,000 kaki persegi<br>b. Bukit Tunku, Taman Duta dan Bukit Persekutuan<br>- minimum 15,000 kaki persegi bagi tapak yang rata<br>- minimum 20,000 kaki persegi bagi tapak yang bercerun<br>**Nota:** Tapak bercerun Kelas III dan IV *in situ* ditakrifkan sebagai tapak bercerun. Kawalan cerun tidak terpakai bagi cerun yang terletak di bahagian sempadan bersama/atau bagi tembok penahan sedia ada. |
| 3. | **Kawasan *plinth*** | Kawasan *plinth* hendaklah tidak melebihi 40 peratus dan selebihnya diutamakan untuk kegunaan kawasan hijau. Kawalan ini tidak terpakai bagi kawasan Bukit Damansara. |



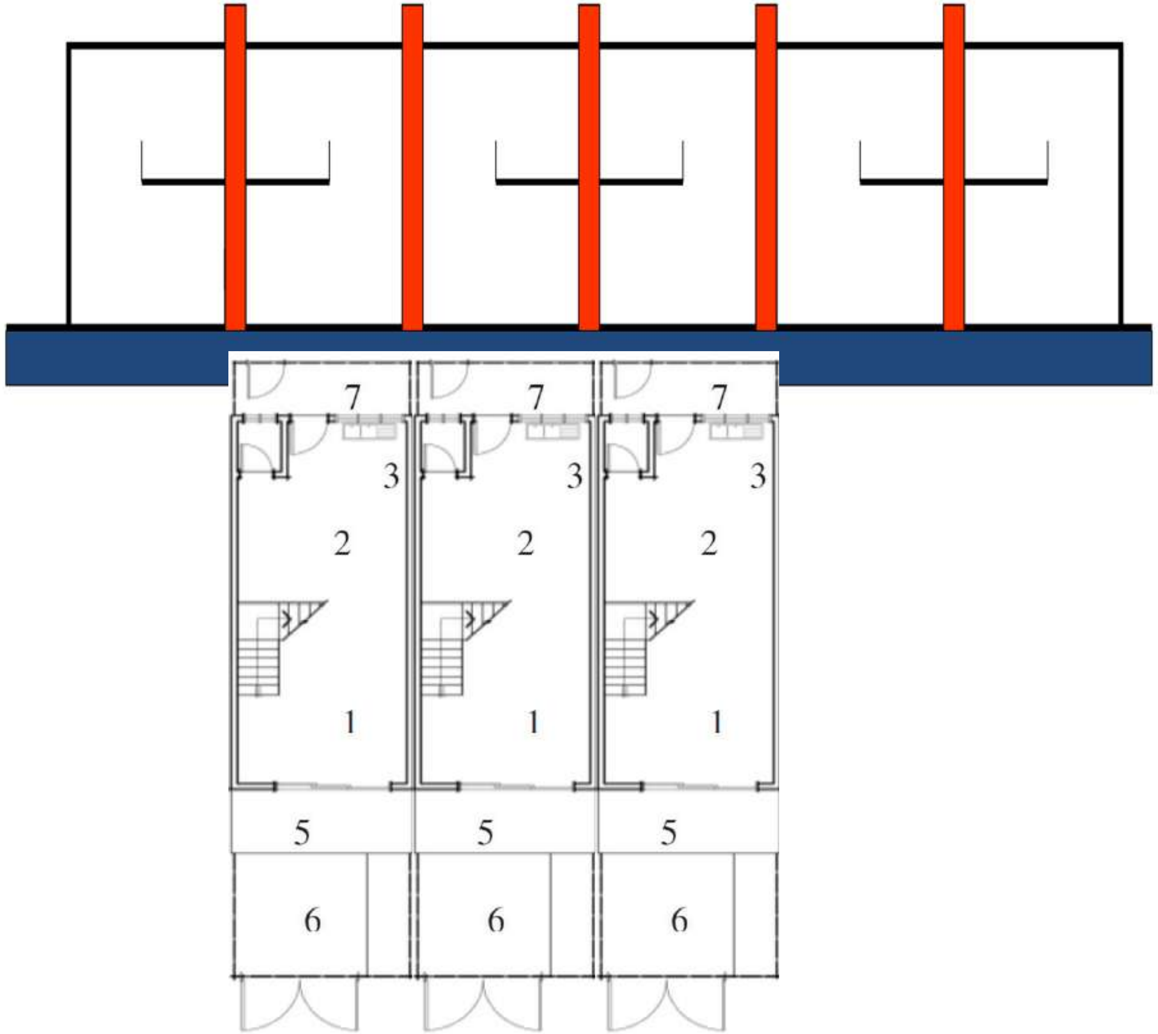

| Tingkat | Hadapan | Sisi | Belakang |
|---|---|---|---|
| **Tingkat Bawah** | | | |
| Dinding Bangunan | 6.0 meter | 3.0 meter | 3.0 meter |



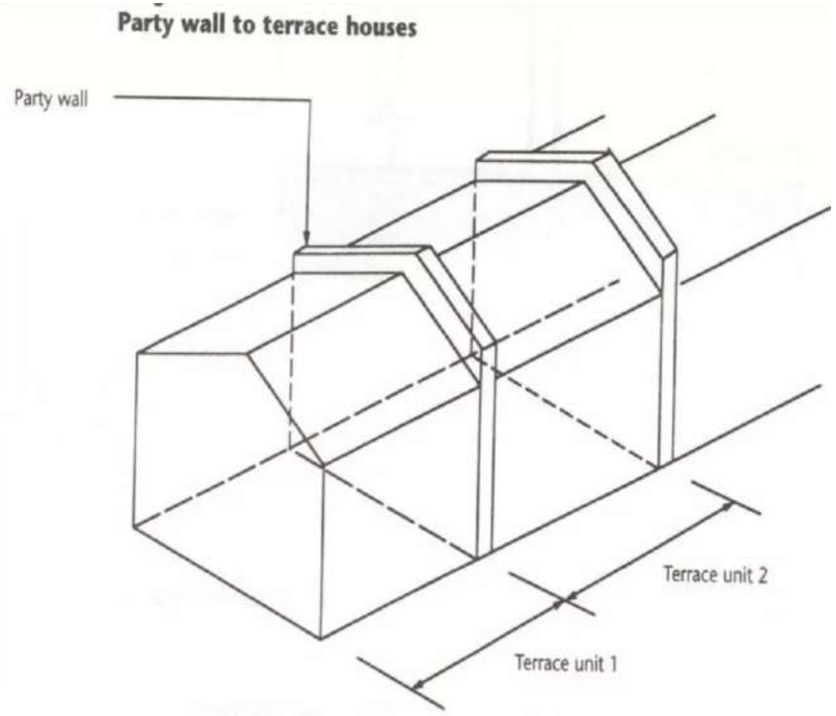




TYPE A1 (~

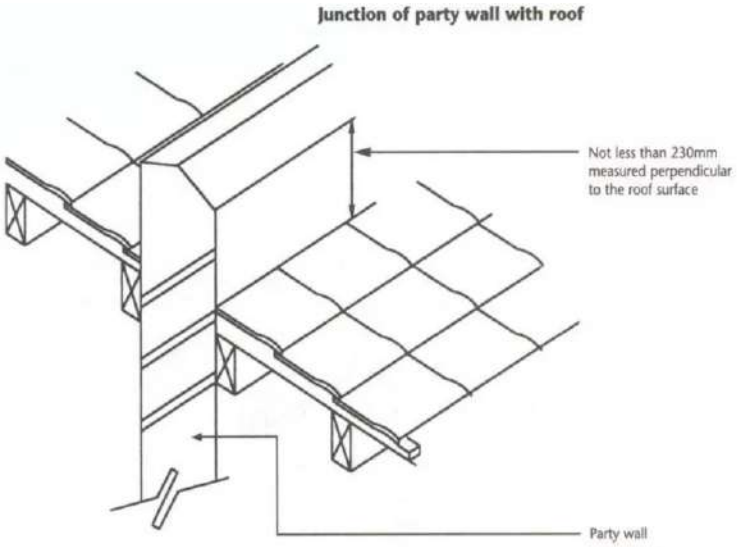
# UBBL 2012 : SEPARATING WALL : TERRACE HOUSES




All party walls shall generally be of not less than 200 millimetres total thickness of solid masonry or in situ concrete which may be made up of two separate skins each of not less than 100 millimetres thickness if constructed at different times:




All party walls shall be carried above the upper surface of the roof to a distance of not less than 230 millimetres at right angles to such upper surface



# Developer

| Detached House (Bungalow) | Terrace House (Link House) | Semi-Detached Housing |
|---|---|---|



The private housing developers, who are now the main provider of residential units in the country, have been able to influence and shape the urban housing. This has been made possible through the strategies adopted in overcoming the problems and constraints imposed on them, particularly in dealing with the application of planning permission. The imposition of planning controls towards housing developers in Malaysia has its implication not only in creating a good urban living environment, but at the same it has some impacts on the price due to the developers' strategies. Thus, it is the concern of this paper to look into the impacts of the developers' strategies on housing development that have some implication on the formulation of national housing policy.

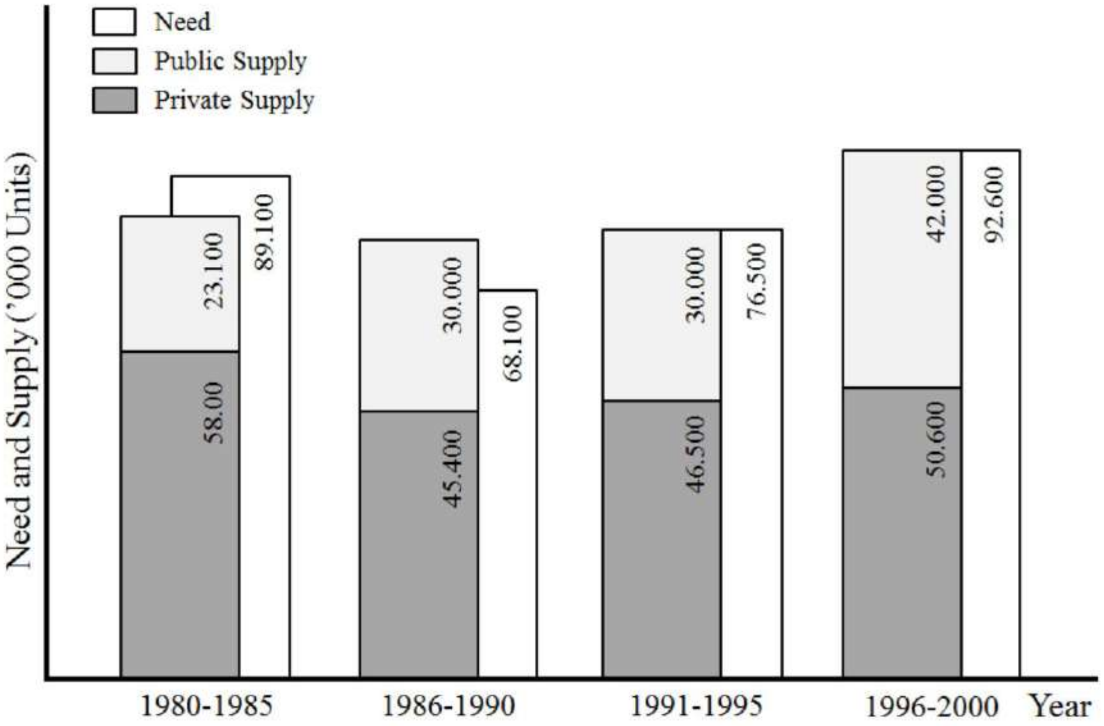



Fig.1. Summary of Housing Need and Supply
(Source: Kuala Lumpur Structure Plan, p. 39)

i. **Kawasan Lapang Awam**
Dikenakan bagi cadangan pembangunan yang melibatkan keluasan tapak kasar melebihi 10 ekar. Spesifikasi kawasan lapang awam adalah seperti berikut:

- 5% daripada keluasan tapak bersih.

ii. **Kawasan Lapang Berpusat**
Dikenakan bagi cadangan pembangunan melibatkan keluasan tapak bersih melebihi 20,000 kaki persegi.

Spesifikasi kawasan lapang berpusat adalah seperti yang berikut:

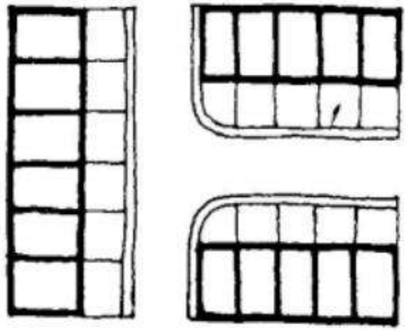
- 10% daripada keluasan tapak bersih.
- Keluasan kawasan lapang berpusat tidak kurang daripada 8% atas tanah boleh dibenarkan dengan syarat menyediakan sekurang-kurangnya 16% (tapak bersih) kawasan lapang di aras podium.

Mandatory minimum of
5% green area for 10 Acres development

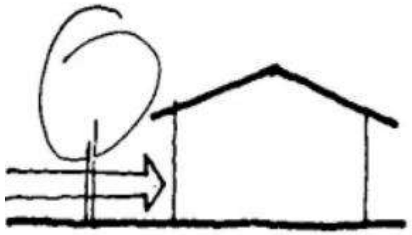
Or
10% for 20k sqft development
Or
8% + 16% podium for 20k sqft development

The concept of connectors mainly comes from fire-fighting regulations. Its job was to break the length of the block in order to give easy access to each house for fire-fighting trucks. The idea was that fire-fighting trucks should reach the back side of each house accessing through the connectors turning to the back lane.

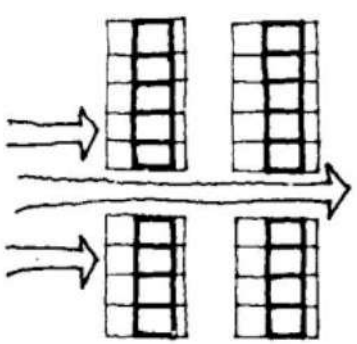
The number of row-houses was limited to a range of eight to sixteen to allow firebreaks. Backlanes provided ample space for sewage system to run. Therefore, a lot of perforation by vehicular streets dominated the neighborhood planning. Moreover, linear face-to-face blocks did not allow much space for developing social activities and natural surveillance. Thereafter, a whole generation of estates comprising terrace houses has generated a kind of car dependent noninteractive unsecure neighborhood, with people isolated from each other, just the way the American and the Europeans faced before the 1960s (Ghazali 2007).



► For profit motives, housing estate houses often disregard orientation for minimizing solar radiation and the orientation of the houses often becomes a jigsaw puzzle of fitting the most units into the site within permissible densities.



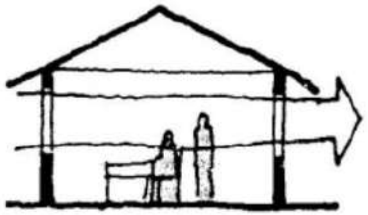
► The housing estate house at ground level receives wind of lower velocity. Hedges and solid fences built around the house to provide privacy often block winds and create steeper wind velocity gradient.



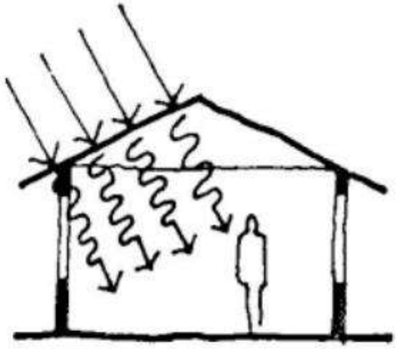
► Rigid patterns in the arrangement of housing estates houses create barriers that block the passage of wind to the houses in the latter path of the wind.

► Plans of housing estate houses are of more complicated shapes, and the partitioning of the house into different rooms and areas restrict air movement and cross ventilation in the house.

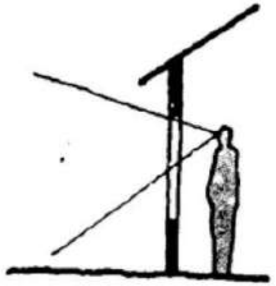
They are arranged probably only one thing in the mind of the developers, that is, how to maximize the number of houses (Ghazali 2007).



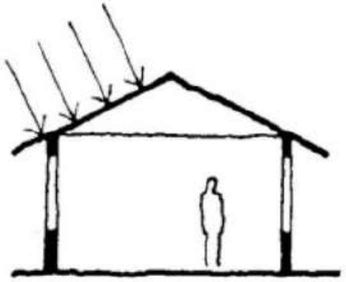
► Ventilation in the housing estate house is often only directed at the upper part of the body because windows and other openings are located at higher levels to provide privacy.



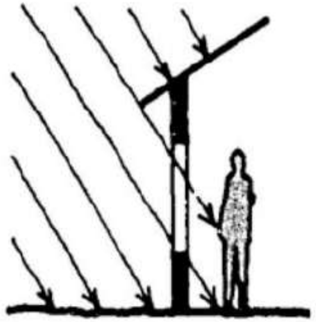
► Modern housing estate houses use bricks, tiles, concrete and other materials of high thermal capacity. These materials store up heat and reradiate it into the house, causing considerable discomfort. Glazed areas are usually abundant in these houses.



► Glare is usually more evident in the housing estate house due to the open skies which are not excluded from the visual field because of the use of bigger and higher unshaded windows. Glare from paved concrete areas and brightly lit exterior walls of other houses also causes considerable discomfort.



► Roof spaces in the housing estate house are insulated by trapped air instead of being ventilated. Such construction requires a high ceiling to be effective.



► The higher and larger exposed vertical areas of the windows in the modern house are often penetrated by direct sunlight and cause considerable discomfort. The walls which act as direct sun-shading devices get heated up and in the evenings reradiate heat into the interior areas.

## highest priority to car dependency

pedestrian movement is relegated to secondary, and often neglected
No distinct pedestrian walkways.
They must walk on the road along with the car
do not offer much natural surveillance.
green field location is not well-planned.
playgrounds are not at walking distance from children's perspective
playgrounds are not accessible without crossing car roads. Therefore it
traffic free route is necessary for community to build up.
no proper pedestrian
Communal facilities are not within walking distance
There are not other alternate than cars
Tropical climate do not allow comfortable walking during daytime, unless they are shaded.

Meanwhile in the rural areas:

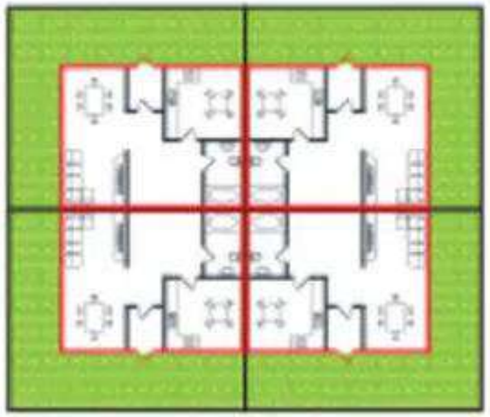
Switch to Brick and Zinc
Less maintenance
Bricks are cheaper than wood

Zinc can last 80 to 100 years when used in roofing, and even longer when applied as a wall system. Other than copper roofing, there is no other material with such a long lifespan. Zinc is fire resistant, insect-proof, and a fungistat, meaning it prohibits the reproduction of mold, mildew and fungus.

Semi Detached

Town House

Detached



Cluster

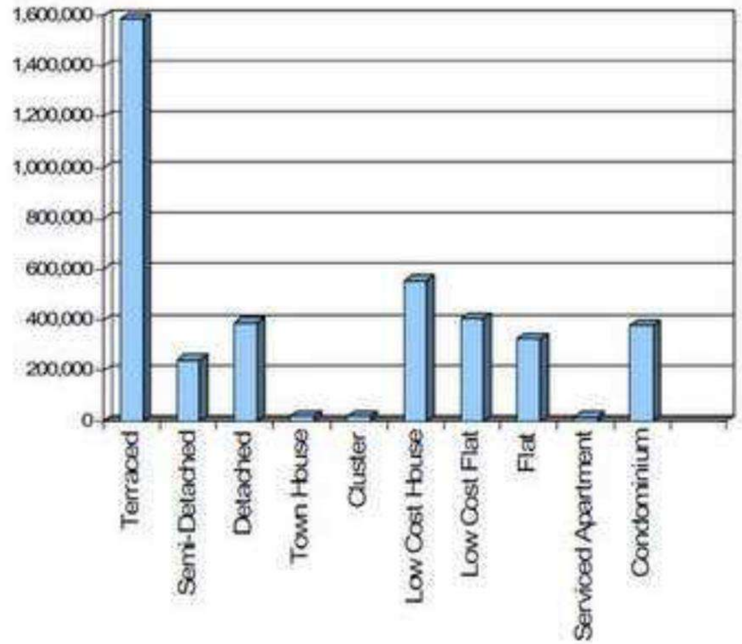
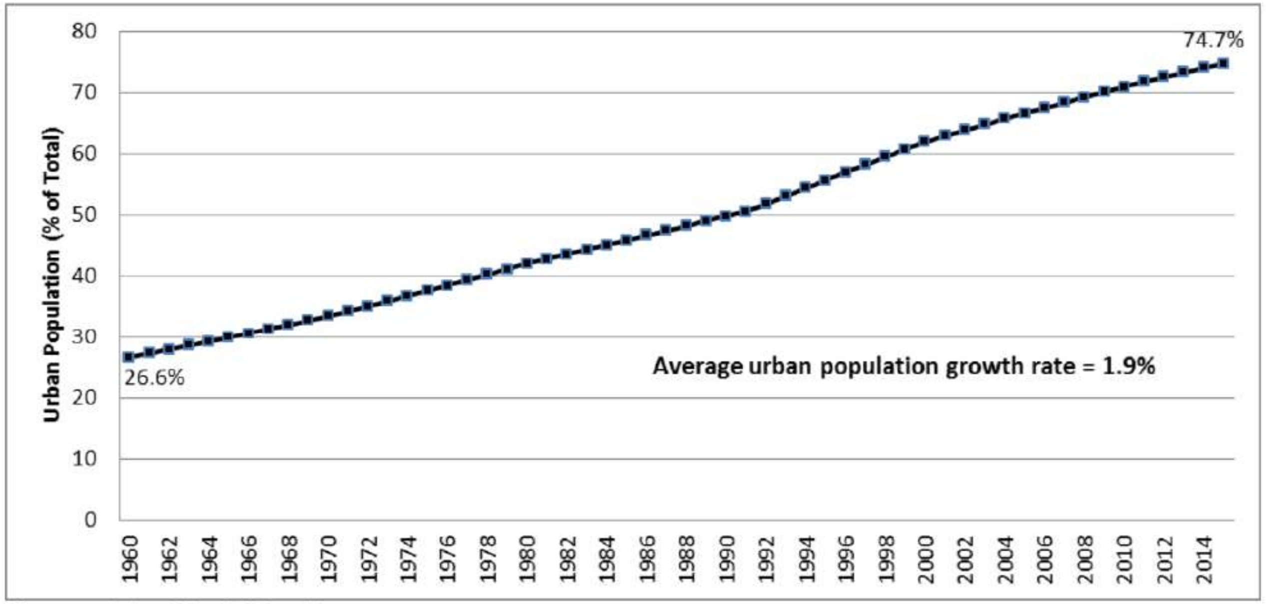
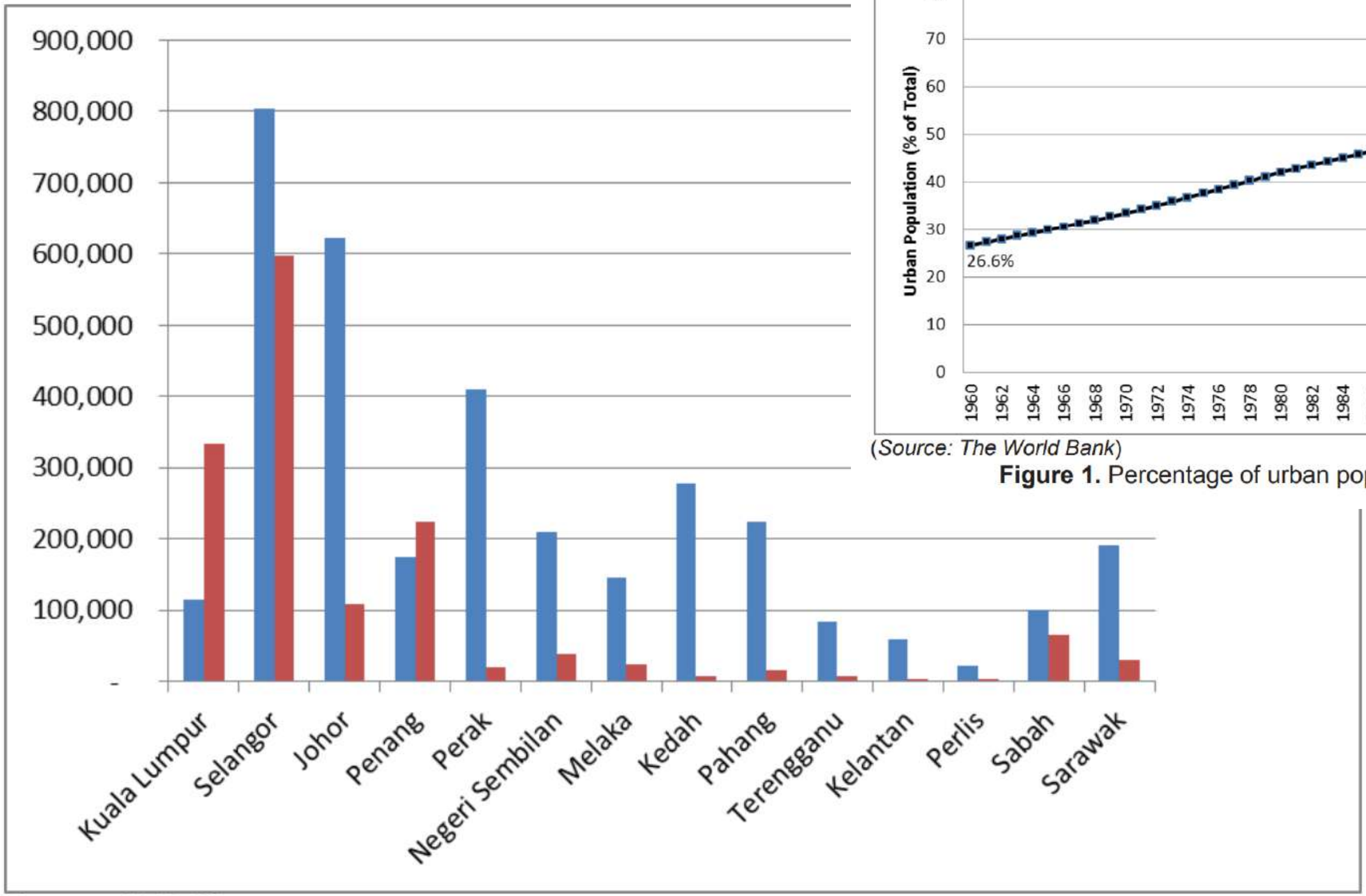



Figure 5: Summary of Supply of Residential Units by Type in Malaysia – Existing Stock NAPIC 2007)




(*Source: The World Bank*)

**Figure 1.** Percentage of urban population in Malaysia, 1960 – 2015



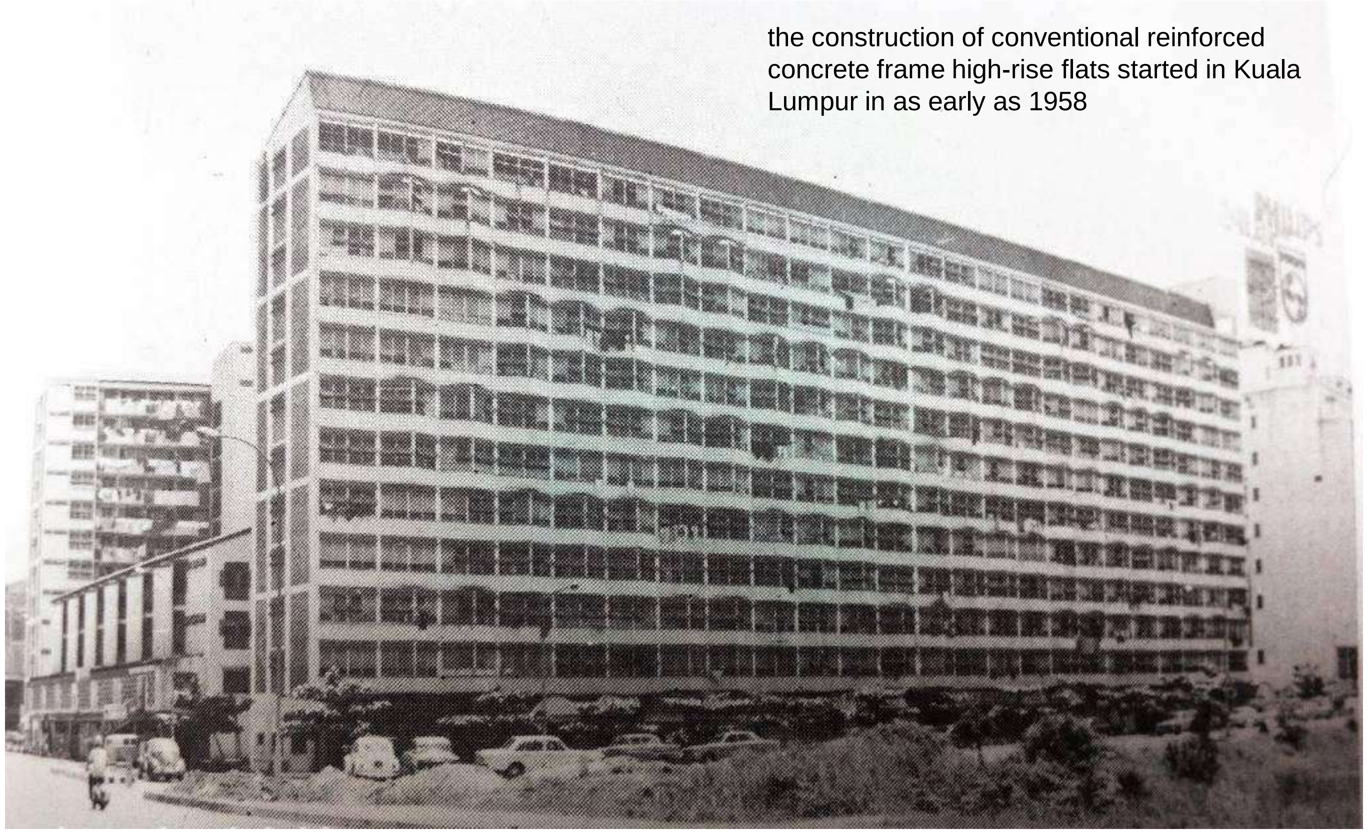

(*Source: NAPIC*)

**Figure 2.** Total number of landed and high rise residential development by state in 2015

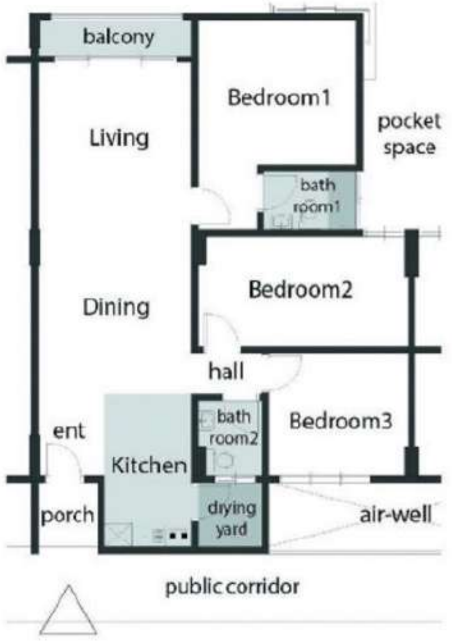
Built before 1957, the Sulaiman Courts was Malaysia's first-ever high-rise apartment. Tunku Abdul Rahman, the country's first prime minister, wanted the Courts to offer reasonably priced housing, so he ordered construction of the building.

# STRATA - FLATS


Accordingly, (NSW Planning, 2011) Low-rise medium density residential development is desirable because it: 1. is less expensive to build 2. Does not need major site amalgamation 3. Performs environmentally better than most high-rise housing (Holloway and Bunker, 2006; Pullen, 2007) 4. Can deliver a greater mix of more affordable housing types 5. Fits perfectly into existing streets and neighbourhoods 6. Suits a wide range of demographic groups



the construction of conventional reinforced concrete frame high-rise flats started in Kuala Lumpur in as early as 1958




Typical layout

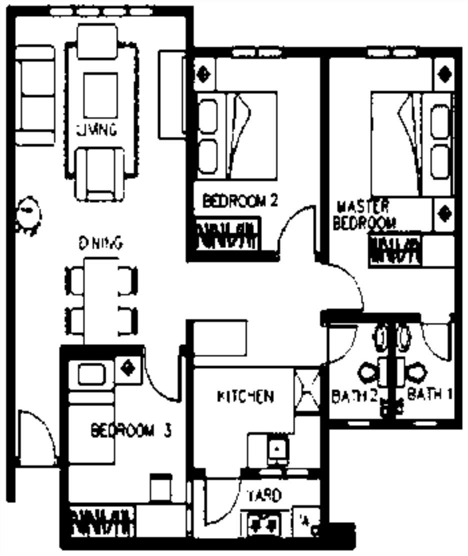
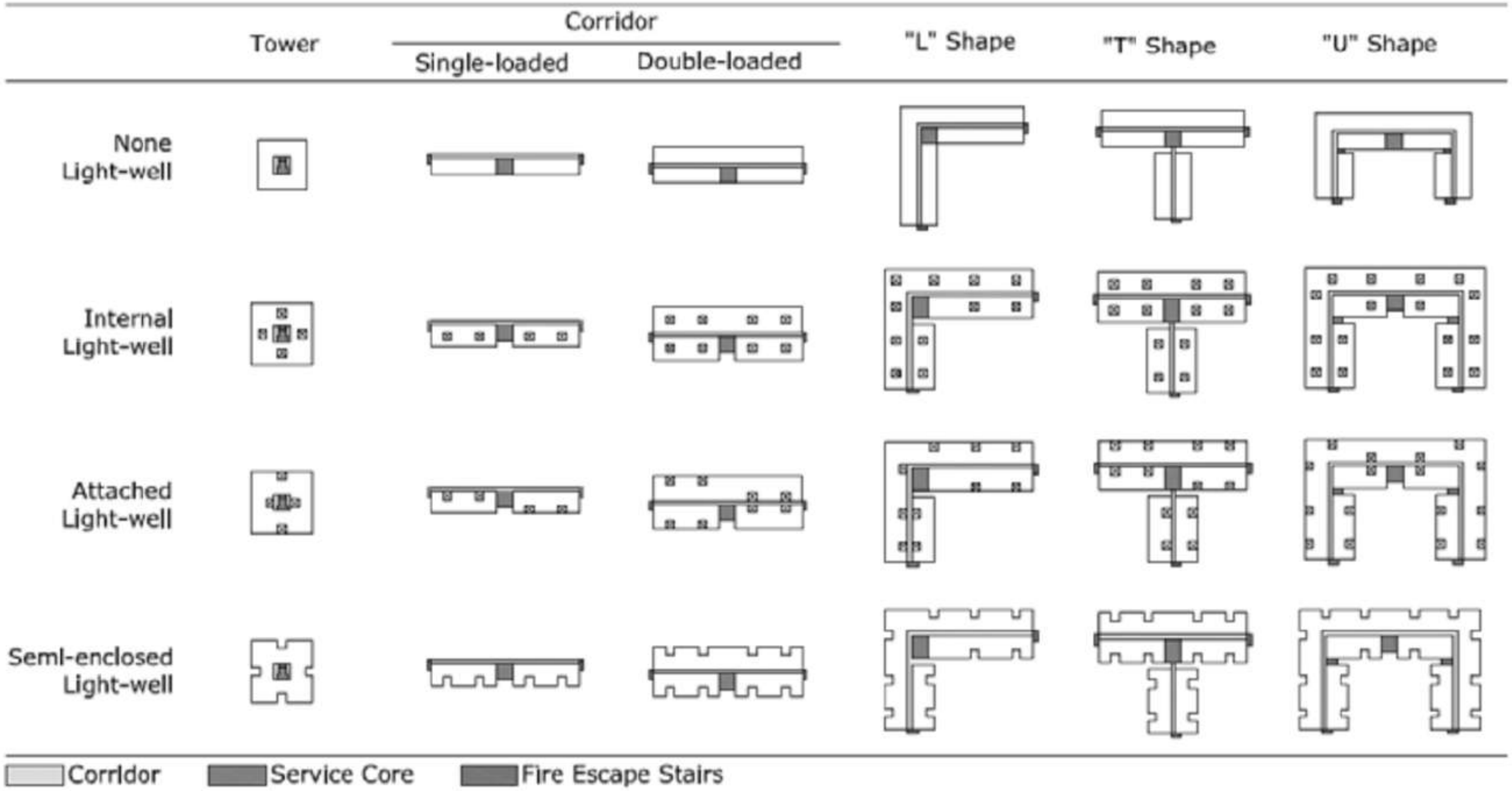
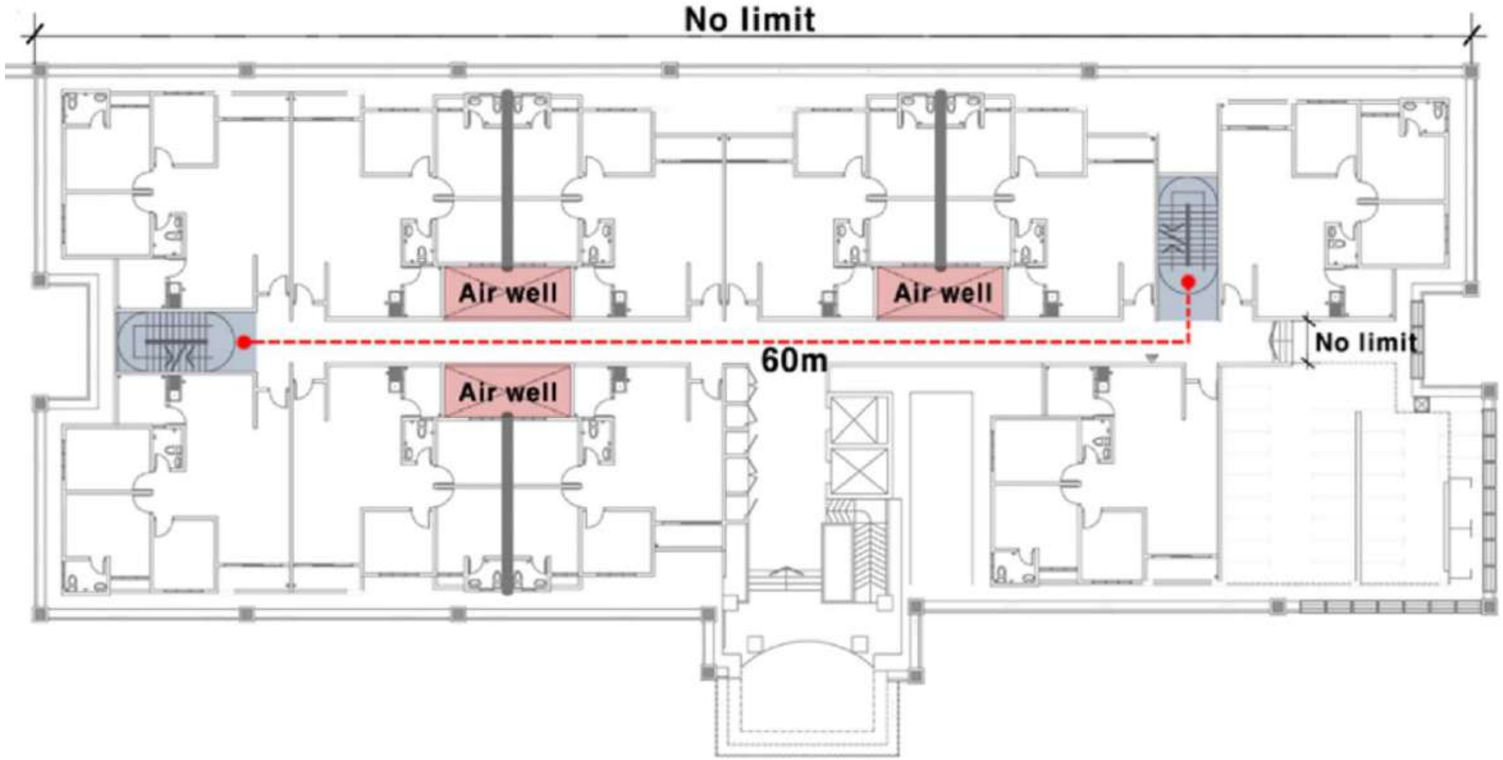



*Table 3 Common possible configurations of HRR buildings derived theoretically from literature review and developed according to the light-well types (plans).*





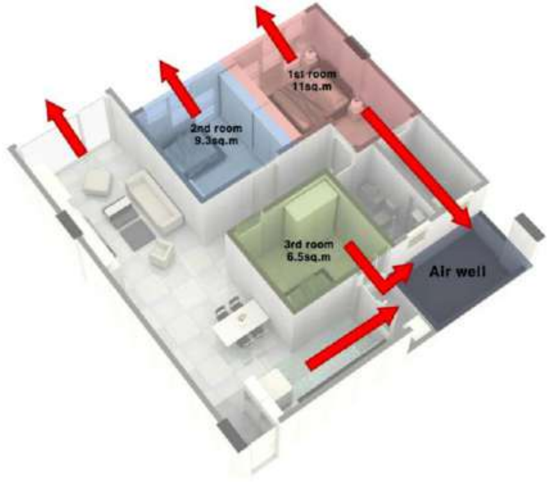




Figure 3. *Typical Unit Plan*

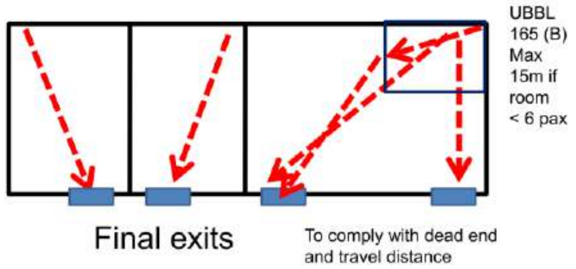
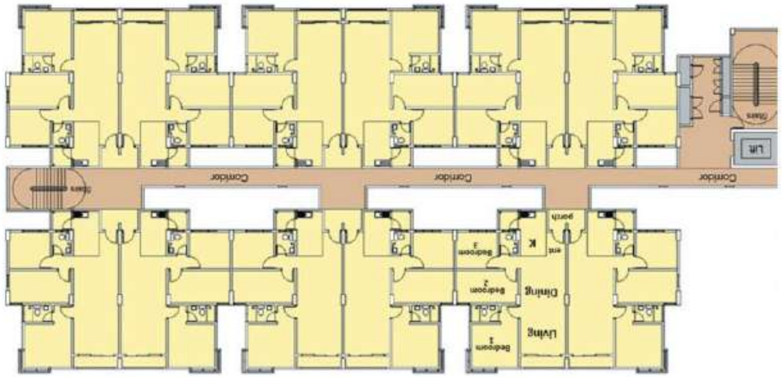
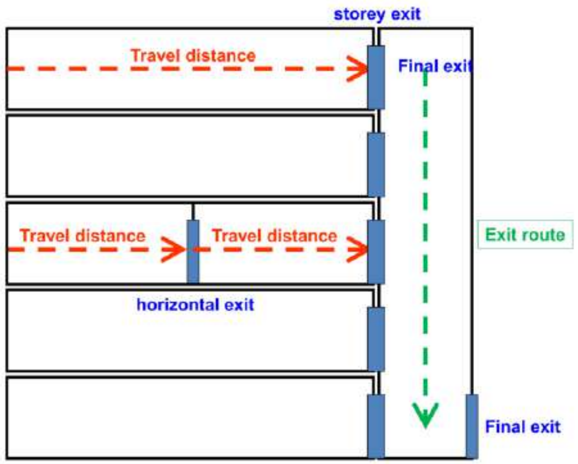



Figure 2. *Block Plan and Travel Distance*



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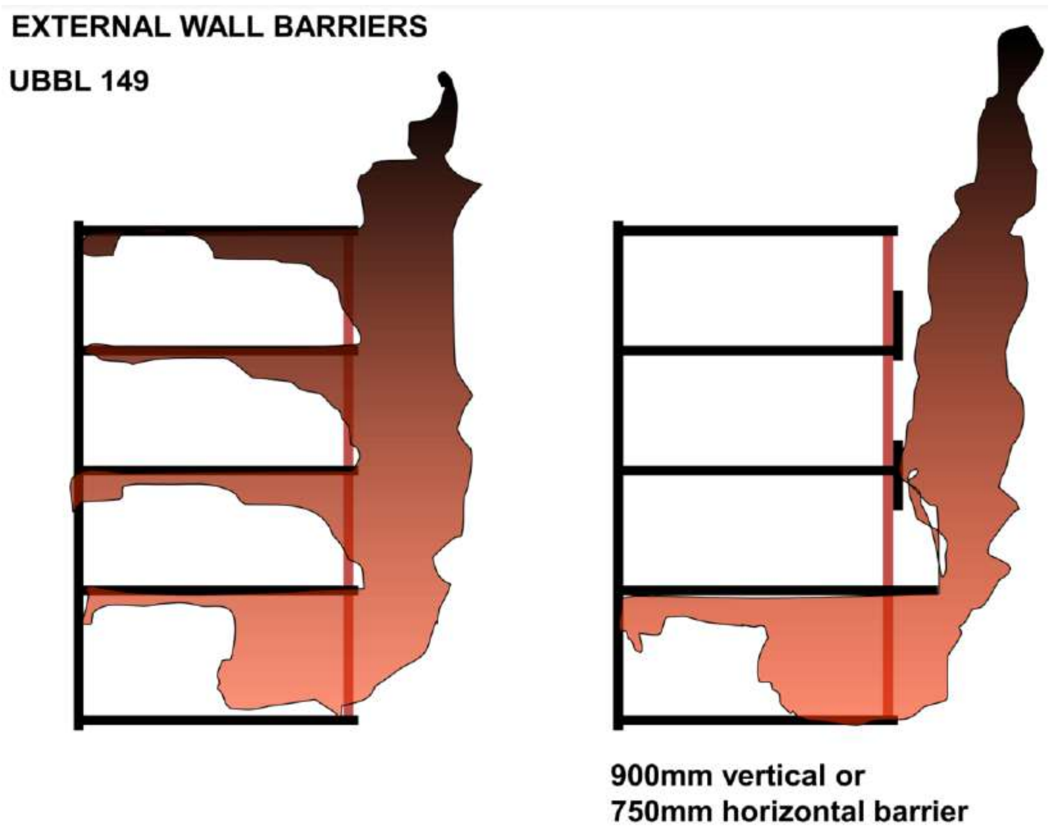


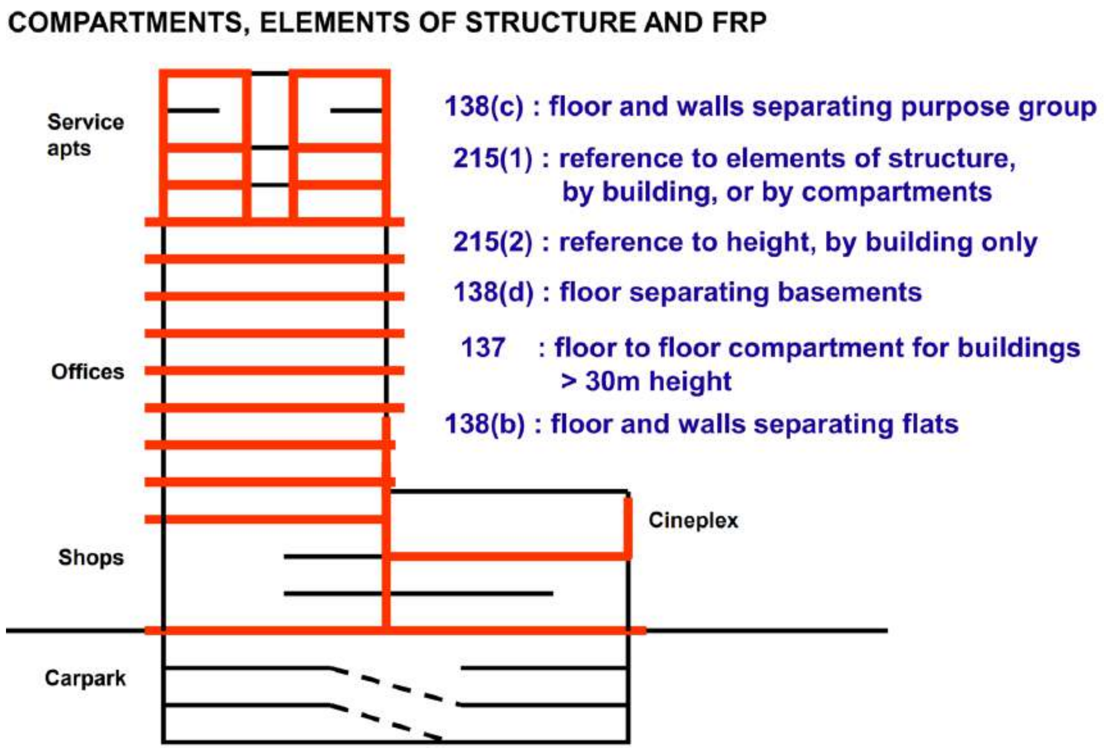

# CONDOMINIUM

The additional attributes which distinguish condominium to flat and apartment are better facilities offered and well planned design to suit the taste of high income people.

condominiums can be defined as a form of co ownership over a multi-unit property where the owner-occupier has individual ownership of one unit but common ownership of the grounds and common areas with other unit owners.


facilities that can be found in most of the condominium are swimming pool and children's pool, gymnasium, 24 -hour security service, launderette service, jogging track, squash and tennis court, function room, poolside barbecue pit, and landscaped.

# MASTER PLAN


**Block A** (High Rise Residences)

**Block B** (Low Rise Residences)

N

- A GUARD HOUSE
- B ENTRANCE STATEMENT
- C SWIMMING POOL
- D WADING POOL
- E OUTDOOR SHOWER
- F GYM & STUDIO / JACUZZI
- G CHILDREN'S PLAYGROUND
- H BBQ AREA
- I OPEN LAWN
- J NURSERY / SAUNA / HALF BASKETBALL COURT
- K FUNCTION HALL / SQUASH COURT

**Table 15:** Scenery From The Condominium Unit.

| View | Percentage |
|---|---|
| Facing swimming pool | 34 |
| Facing town area | 22 |
| Facing green scenery | 43 |
| Missing | 1 |

**Table 16: Number Of Bedrooms Demanded**

| Number of Bedrooms | Percentage |
|---|---|
| **2** | **1** |
| **3** | **38** |
| **4** | **61** |
| **Total** | **100** |

**Table 17:** Preferred Built-Up Area

| Built-Up Area | Percentage |
|---|---|
| 800 - 1,000 sq. ft. | 0.4 |
| 1,001 - 1,200 sq. ft. | 3.8 |
| 1,201 - 1,400 sq. ft. | 52.7 |
| 1,401 - 1,600 sq. ft. | 37.0 |
| 1,601 and above sq. ft. | 6.1 |
| **Total** | **100** |

**Table 20:** Main **Factors That Influence Demand**

| Main Factors | Ranking |
|---|---|
| Location of development | 1 |
| Characteristics of the building | 2 |
| Price | 3 |
| Recreational facilities | 4 |
| View | 5 |
| Marketing | 6 |
| Financing | 7 |
| Management | 8 |
| Surrounding environment | 9 |
| Investment | 10 |
| Good reputation of the developer | 11 |

**Table 18: Preferred Recreational Facilities.**

| Recreational Facilities | Rank |
|---|---|
| Parking Space | 1 |
| Swimming Pool | 2 |
| Function Hall | 3 |
| Tennis/Badminton Courts | 4 |
| Gymnasium | 5 |
| Sauna | 6 |
| Children Playground | 7 |
| Jogging Track | 8 |
| Gazebo | 9 |

**Table 19: Amenities Facilities Demanded**

| Amenities Facilities | Rank |
|---|---|
| 24 Hours Guard/Survelliance | 1 |
| Automatic Sprinkler | 2 |
| Nursery | 3 |
| Free Club Membership | 4 |
| High Speed Modern Lifts | 5 |
| Mini Markets | 6 |
| Cafeteria | 7 |
| Launderette | 8 |
| BBQ Area | 9 |

Unit Size
450 - 650 sqft  30% maximum
650 - 800 sqft  20-50% maximum
800 sqft  +++   50%  minimum

Parking Lot
1 lot per Unit Residential (low cost)
2 lot per Unit Residential (medium cost)
+ 10% visitor parking

1 lot per 500sqft commercial
1 lot per 500 sqft institutional / transport hub

+ 30% road circulation

<u>Parking Lot</u>

| | |
|---|---|
| 1 lot per 500 sqft | commercial |
| 1 lot per 500 sqft | transport hub |
| 1 lot per 2000 sqft | factory |
| 1 lot per 1000 sqft | educational |
| 1 lot per 2000 sqft | community center |
| 1 lot per 750 sqft | funeral home |
| 1 lot per 250 sqft | community pool |
| 1 lot per 8 seat | stadium |
| 2 lot per sports | court |
| 1 lot per 1000 sqft | Event hall |
| 1 lot per 5 student | University |
| 1 lot per 750 sqft | religious building |
| 1 lot per 750 sqft | hospital |
| 1 lot per 215 sqft | hypermarket |

+ 30% road circulation